PROGRAM DESIGN MANUAL

CHAD WESLEY SMITH

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Welcome to the Juggernaut Powerlifting Program Design Manual!

The objective of this book is to bridge the gap between the Scientific Principles of Strength Training and giving you pre-written programs or templates, we want to equip you to create programs for yourself and your athletes for a life-time of training.

Understanding the practical applications of Specificity, Overload, Fatigue Management, SRA, Variation, and Phase Potentiation is critical and being able to adjust how these principles are implemented based on an athlete's Sex, Age, Size, Strength, Experience, and Lifestyle is what separates good programs from great ones.

The Juggernaut Powerlifting Program Design Manual has created a systematic approach to athlete assessment helping you make the most informed decisions possible regarding:

- Volume Thresholds
- Training Frequency
- Phase Length and Proportion
- Overloading and Fatigue Management Strategies
- Exercise Selection

Coaching certainly has artistic and scientific aspects to it, but really many of those 'artistic' decisions are products of deeper understanding based upon seeing thousands of different athletes and even more coaching problems and this manual will help accelerate your understanding of those 'artistic' decisions so you can make more sound and scientific decisions about them.

ASSESSING
THE ATHLETE

The Scientific Principles of Strength Training will apply to athletes of all types, regardless of gender, age, experience, strength or any other factor, you must satisfy those principles. Since you are reading this book though and undoubtedly are a smart, strong, good-looking and conscientious person, you already know that though and you want to further refine your program design skills, so that you are optimizing programs for every athlete. To optimally implement the Scientific Principles you must understand your athlete, understand how different factors affect the application of the Principles and know that what is optimal for one, isn't optimal for another and what is optimal now for someone, likely won't still be optimal in the future for them.

Answer the following questions, these answers will help inform many decisions we make throughout the rest of this book..

Sex:

-Female

-Male

Bodyweight:

-Light, Female-Under 57. Male-Under 75kg.

-Middle, Female-57 to 75kg. Male-75 to 100kg.

-Heavy, Female-75 to 90kg. Male-100-125kg

-Super Heavy, Female-Over 90kg. Male-Over 125kg

Height:

-Short, Female-Under 160cm. Male-Under 170cm.

-Medium, Female-160 to 167cm. Male-170 to 182cm.

-Tall, Female-168 to 175cm. Male-183 to 195cm.

-Very Tall, Female-Over 175cm. Male-Over 195+cm.

Strength (Refer To Classification Charts on pages 11 & 12)

Squat:

-Low (Class VI-III)

-Moderate (Class II-I)

-High (Master-Elite)

-Very High (Intl Elite)

Bench:

-Low (Class VI-III)

-Moderate (Class II-I)

-High (Master-Elite)

-Very High (Intl Elite)

Deadlift:

-Low (Class VI-III)

-Moderate (Class II-I)

-High (Master-Elite)

-Very High (Intl Elite)

Experience:

-Beginner, Less Than 4 Years of Serious Strength Training
-Intermediate, 4-8 Years of Serious Strength Training
-Advanced, 8-12 Years of Serious Strength Training
-Very Advanced, Over 12 Years of Serious Strength Training

Age:

-19 Years Old and Younger
-20 to 29 Years Old
-30 to 39 Years Old
-40 to 49 Years Old
-50 Years Old and Older

Diet:

-Poor, Not Getting Sufficient Calories or Intentionally In A Deficit
-Average, Calories Are Equated but Macro Breakdown and Nutrient Timing Isn't Accounted For
-Good, Calories Are Equated with Good Macro Balance and Attention to Nutrient Timing

Sleep:

-Poor, 5 Hours or Less On Average
-Average, 5 to 7 Hours On Average
-Good, 7 or More Hours On Average

Stress Away from Training

-High. This could be related to training for other endeavors along with lifting, a manual labor job or other factors leading to a high-stress lifestyle.

-Medium. You may have some other physically demanding hobbies that you participate in 1-2x/week and have a mix of low and high-stress days in your job/ relationship/etc

-Low. You have a job that isn't physically demanding and lifting is your main physical activity.

Historical Workload

1. 1x/week frequency per lift with minimal accessory work.
2. 1 to 2x/week frequency per lift with moderate accessory work.
3. 2x/week frequency per lift with significant accessory work.
4. 2 to 3x/week frequency per lift with moderate to significant accessory work.
5. 3x/week or more frequency per lift with moderate to significant accessory work.

Historical Recovery Ability

1. Will often be sore 3 days after training and struggle to increase workloads.
2. Will often be sore 2 days after training.
3. Will often be sore 1-2 days after training and seem to respond to average workloads.
4. Rarely sore more than 1 day after training and am able to increase workloads.
5. Rarely sore more than 1 day after training, respond to high workloads and possibly have a background in endurance sports or CrossFit.

These answers will be important throughout the book and we will discuss their implications at length.

MEN'S CLASSIFICATION

OPEN	Raw 52	Raw 56	Raw 60	Raw 67.5	Raw 75	Raw 82.5	Raw 90	Raw 100	Raw 110	Raw 125	Raw 140	Raw 140+
Int'l Elite	472.5	515.0	555.0	620.0	680.0	730.0	770.0	812.5	840.0	865.0	905.0	932.5
Elite	435.0	472.5	507.5	570.0	622.5	670.0	707.5	745.0	772.5	795.0	830.0	855.0
Master	400.0	435.0	467.5	522.5	572.5	615.0	652.5	686.0	707.5	730.0	757.5	780.0
Class I	360.0	390.0	417.5	470.0	512.5	550.0	580.0	615.0	635.0	652.5	682.5	702.5
Class II	317.5	342.5	362.5	412.5	450.0	485.0	512.5	540.0	557.5	575.0	600.0	620.0
Class III	275.0	300.0	322.5	362.5	395.0	422.5	447.5	472.5	487.5	502.5	522.5	537.5
Class IV	240.0	260.0	280.0	315.0	342.5	370.0	390.0	410.0	427.5	440.0	457.5	470.0
	SQ 52	SQ 56	SQ 60	SQ 67.5	SQ 75	SQ 82.5	SQ 90	SQ 100	SQ 110	SQ 125	SQ 140	SQ 140+
Int'l Elite	165.0	182.5	200.0	227.5	252.5	270.0	290.0	305.0	320.0	332.5	352.5	362.5
Elite	152.5	167.5	182.5	207.5	230.0	247.5	265.0	280.0	292.5	305.0	322.5	332.5
Master	140.0	155.0	167.5	190.0	212.5	227.5	245.0	257.5	270.0	280.0	295.0	305.0
Class I	125.0	137.5	150.0	172.5	190.0	205.0	217.5	230.0	240.0	250.0	265.0	275.0
Class II	110.0	122.5	137.5	150.0	167.5	180.0	192.5	202.5	212.5	222.5	235.0	242.5
Class III	95.0	107.5	115.0	132.5	145.0	157.5	167.5	177.5	185.0	192.5	202.5	210.0
Class IV	85.0	92.5	100.0	115.0	127.5	137.5	145.0	152.5	162.5	170.0	177.5	182.5
	BP 52	BP 56	BP 60	BP 67.5	BP 75	BP 82.5	BP 90	BP 100	BP 110	BP 125	BP 140	BP 140+
Int'l Elite	117.5	127.5	137.5	155.0	170.0	182.5	192.5	202.5	210.0	217.5	225.0	232.5
Elite	107.5	117.5	127.5	142.5	155.0	167.5	177.5	185.0	192.5	197.5	207.5	215.0
Master	100.0	107.5	117.5	130.0	142.5	152.5	162.5	170.0	177.5	182.5	190.0	195.0
Class I	90.0	97.5	105.0	117.5	127.5	137.5	145.0	152.5	157.5	162.5	170.0	175.0
Class II	80.0	85.0	95.0	102.5	112.5	122.5	127.5	135.0	140.0	145.0	150.0	155.0
Class III	67.5	75.0	80.0	90.0	97.5	105.0	112.5	117.5	122.5	125.0	130.0	135.0
Class IV	60.0	65.0	70.0	80.0	85.0	92.5	97.5	102.5	107.5	110.0	115.0	117.5
	DL 52	DL 56	DL 60	DL 67.5	DL 75	DL 82.5	DL 90	DL 100	DL 110	DL 125	DL 140	DL 140+
Int'l Elite	190.0	202.5	215.0	240.0	257.5	277.5	290.0	305.0	310.0	315.0	325.0	335.0
Elite	175.0	187.5	197.5	220.0	237.5	255.0	265.0	280.0	285.0	290.0	300.0	307.5
Master	160.0	172.5	182.5	200.0	217.5	232.5	245.0	257.5	262.5	265.0	272.5	280.0
Class I	145.0	152.5	162.5	180.0	195.0	210.0	217.5	230.0	235.0	237.5	245.0	252.5
Class II	127.5	135.0	150.0	157.5	170.0	185.0	192.5	202.5	207.5	210.0	217.5	222.5
Class III	110.0	117.5	125.0	140.0	150.0	160.0	167.5	177.5	180.0	182.5	187.5	192.5
Class IV	97.5	102.5	110.0	122.5	130.0	140.0	145.0	152.5	157.5	160.0	165.0	170.0

Raw 52	Raw 56	Raw 60	Raw 67.5	Raw 75	Raw 82.5	Raw 90	Raw 100	Raw 110	Raw 125	Raw 140	Raw 140+	OPEN
487.5	530.0	572.5	637.5	700.0	752.5	792.5	837.5	865.0	890.0	932.5	960.0	Int'l Elite
447.5	487.5	522.5	587.5	640.0	690.0	727.5	767.5	795.0	820.0	855.0	880.0	Elite
412.5	447.5	482.5	537.5	590.0	632.5	672.5	705.0	727.5	752.5	780.0	802.5	Master
370.0	402.5	430.0	485.0	527.5	567.5	597.5	632.5	655.0	672.5	702.5	722.5	Class I
327.5	352.5	395.0	425.0	462.5	500.0	527.5	555.0	575.0	592.5	617.5	637.5	Class II
282.5	310.0	332.5	372.5	407.5	435.0	460.0	487.5	502.5	517.5	537.5	552.5	Class III
247.5	267.5	287.5	325.0	352.5	380.0	402.5	422.5	440.0	452.5	470.0	485.0	Class IV
52	56	60	67.5	75	82.5	90	100	110	125	140	140+	
170.0	187.5	205.0	235.0	260.0	277.5	297.5	315.0	330.0	342.5	362.5	372.5	Int'l Elite
157.5	172.5	187.5	212.5	237.5	255.0	272.5	287.5	302.5	315.0	332.5	342.5	Elite
145.0	160.0	172.5	195.0	220.0	235.0	252.5	265.0	277.5	287.5	305.0	315.0	Master
130.0	142.5	155.0	177.5	195.0	210.0	225.0	237.5	247.5	257.5	272.5	282.5	Class I
112.5	125.0	142.5	155.0	172.5	185.0	197.5	207.5	220.0	230.0	242.5	250.0	Class II
97.5	110.0	117.5	137.5	150.0	162.5	172.5	182.5	190.0	197.5	207.5	217.5	Class III
87.5	95.0	102.5	117.5	132.5	142.5	150.0	157.5	167.5	175.0	182.5	187.5	Class IV
52	56	60	67.5	75	82.5	90	100	110	125	140	140+	
120.0	132.5	142.5	160.0	175.0	187.5	197.5	207.5	217.5	225.0	232.5	240.0	Int'l Elite
110.0	120.0	132.5	147.5	160.0	172.5	182.5	190.0	197.5	202.5	212.5	222.5	Elite
102.5	110.0	120.0	135.0	147.5	157.5	167.5	175.0	182.5	187.5	195.0	200.0	Master
92.5	100.0	107.5	120.0	132.5	142.5	150.0	157.5	162.5	167.5	175.0	180.0	Class I
82.5	87.5	97.5	105.0	115.0	125.0	132.5	140.0	145.0	150.0	155.0	160.0	Class II
70.0	77.5	82.5	92.5	100.0	107.5	115.0	120.0	125.0	130.0	135.0	140.0	Class III
62.5	67.5	72.5	82.5	87.5	95.0	100.0	105.0	110.0	112.5	117.5	120.0	Class IV
52	56	60	67.5	75	82.5	90	100	110	125	140	140+	
195.0	207.5	222.5	247.5	265.0	285.0	297.5	315.0	320.0	325.0	335.0	345.0	Int'l Elite
180.0	192.5	202.5	227.5	245.0	262.5	272.5	287.5	292.5	297.5	310.0	317.5	Elite
165.0	177.5	187.5	205.0	225.0	240.0	252.5	265.0	270.0	272.5	280.0	287.5	Master
150.0	157.5	167.5	185.0	200.0	217.5	225.0	237.5	242.5	245.0	252.5	260.0	Class I
132.5	140.0	155.0	162.5	175.0	190.0	197.5	207.5	212.5	217.5	225.0	230.0	Class II
112.5	120.0	130.0	145.0	155.0	165.0	172.5	182.5	185.0	187.5	192.5	197.5	Class III
100.0	105.0	112.5	125.0	135.0	145.0	150.0	157.5	162.5	165.0	170.0	175.0	Class IV

OPEN	Raw 52	Raw 56	Raw 60	Raw 67.5	Raw 75	Raw 82.5	Raw 90	Raw 100	Raw 110	Raw 125	Raw 140	Raw 140+
Class II	127.5	135.0	150.0	157.5	170.0	185.0	192.5	202.5	207.5	210.0	217.5	222.5
Class III	110.0	117.5	125.0	140.0	150.0	160.0	167.5	177.5	180.0	182.5	187.5	192.5
Class IV	97.5	102.5	110.0	122.5	130.0	140.0	145.0	152.5	157.5	160.0	165.0	170.0

Raw 52	Raw 56	Raw 60	Raw 67.5	Raw 75	Raw 82.5	Raw 90	Raw 100	Raw 110	Raw 125	Raw 140	Raw 140+	OPEN
132.5	140.0	155.0	162.5	175.0	190.0	197.5	207.5	212.5	217.5	225.0	230.0	Class II
112.5	120.0	130.0	145.0	155.0	165.0	172.5	182.5	185.0	187.5	192.5	197.5	Class III
100.0	105.0	112.5	125.0	135.0	145.0	150.0	157.5	162.5	165.0	170.0	175.0	Class IV
0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	

OPEN	CI 52	CI 56	CI 60	CI 67.5	CI 75	CI 82.5	CI 90	CI 100	CI 110	CI 125	CI 140	CI 140+
Int'l Elite	482.5	525.0	565.0	632.5	692.5	745.0	785.0	827.5	857.5	882.5	922.5	950.0
Elite	442.5	480.0	517.5	580.0	635.0	682.5	720.0	760.0	785.0	810.0	847.5	872.5
Master	407.5	442.5	477.5	532.5	582.5	625.0	662.5	697.5	722.5	742.5	772.5	797.5
Class I	367.5	395.0	427.5	480.0	522.5	562.5	592.5	625.0	647.5	665.0	695.0	715.0
Class II	322.5	347.5	387.5	420.0	457.5	495.0	522.5	550.0	567.5	587.5	612.5	630.0
Class III	280.0	305.0	327.5	367.5	402.5	430.0	455.0	480.0	497.5	512.5	532.5	547.5
Class IV	245.0	265.0	287.5	320.0	347.5	375.0	397.5	417.5	435.0	447.5	465.0	480.0
	SQ 52	SQ 56	SQ 60	SQ 67.5	SQ 75	SQ 82.5	SQ 90	SQ 100	SQ 110	SQ 125	SQ 140	SQ 140+
Int'l Elite	175.0	192.5	210.0	237.5	265.0	285.0	302.5	320.0	335.0	350.0	370.0	382.5
Elite	160.0	175.0	192.5	217.5	242.5	260.0	277.5	292.5	307.5	320.0	340.0	350.0
Master	147.5	162.5	177.5	200.0	222.5	240.0	257.5	270.0	282.5	295.0	310.0	320.0
Class I	132.5	145.0	157.5	180.0	200.0	215.0	227.5	242.5	252.5	262.5	280.0	287.5
Class II	117.5	127.5	145.0	157.5	175.0	187.5	202.5	212.5	222.5	232.5	245.0	252.5
Class III	100.0	112.5	122.5	137.5	152.5	165.0	177.5	185.0	195.0	202.5	215.0	220.0
Class IV	87.5	97.5	107.5	120.0	132.5	142.5	152.5	162.5	170.0	177.5	187.5	192.5
	BP 52	BP 56	BP 60	BP 67.5	BP 75	BP 82.5	BP 90	BP 100	BP 110	BP 125	BP 140	BP 140+
Int'l Elite	117.5	127.5	137.5	155.0	170.0	182.5	192.5	202.5	210.0	217.5	225.0	232.5
Elite	107.5	117.5	127.5	142.5	155.0	167.5	177.5	185.0	192.5	197.5	207.5	215.0
Master	100.0	107.5	117.5	130.0	142.5	152.5	162.5	170.0	177.5	182.5	190.0	195.0
Class I	90.0	97.5	105.0	117.5	127.5	137.5	145.0	152.5	157.5	162.5	170.0	175.0
Class II	80.0	85.0	95.0	102.5	112.5	122.5	127.5	135.0	140.0	145.0	150.0	155.0
Class III	67.5	75.0	80.0	90.0	97.5	106.0	112.5	117.5	122.5	125.0	130.0	135.0
Class IV	60.0	65.0	70.0	80.0	85.0	92.5	97.5	102.5	107.5	110.0	115.0	117.5
	DL 52	DL 56	DL 60	DL 67.5	DL 75	DL 82.5	DL 90	DL 100	DL 110	DL 125	DL 140	DL 140+
Int'l Elite	190.0	202.5	215.0	240.0	257.5	277.5	290.0	305.0	310.0	315.0	325.0	335.0
Elite	175.0	187.5	197.5	220.0	237.5	255.0	265.0	280.0	285.0	290.0	300.0	307.5
Master	160.0	172.5	182.5	200.0	217.5	232.5	245.0	257.5	262.5	265.0	272.5	280.0
Class I	145.0	152.5	162.5	180.0	195.0	210.0	217.5	230.0	235.0	237.5	245.0	252.5
Class II	127.5	135.0	150.0	157.5	170.0	185.0	192.5	202.5	207.5	210.0	217.5	222.5
Class III	110.0	117.5	125.0	140.0	150.0	160.0	167.5	177.5	180.0	182.5	187.5	192.5
Class IV	97.5	102.5	110.0	122.5	130.0	140.0	145.0	152.5	157.5	160.0	165.0	170.0

Raw 52	Raw 56	Raw 60	Raw 67.5	Raw 75	Raw 82.5	Raw 90	Raw 100	Raw 110	Raw 125	Raw 140	Raw 140+	OPEN
497.5	540.0	582.5	652.5	712.5	767.5	807.5	852.5	882.5	910.0	950.0	977.5	Int'l Elite
455.0	495.0	532.5	597.5	655.0	702.5	742.5	782.5	807.5	835.0	872.5	897.5	Elite
420.0	455.0	492.5	547.5	600.0	645.0	682.5	717.5	745.0	765.0	795.0	822.5	Master
377.5	407.5	440.0	495.0	537.5	580.0	610.0	645.0	667.5	685.0	715.0	737.5	Class I
332.5	357.5	400.0	432.5	470.0	510.0	537.5	567.5	585.0	605.0	630.0	650.0	Class II
287.5	315.0	337.5	377.5	415.0	442.5	467.5	495.0	512.5	527.5	547.5	565.0	Class III
252.5	272.5	295.0	330.0	357.5	387.5	410.0	430.0	447.5	460.0	480.0	495.0	Class IV
52	56	60	67.5	75	82.5	90	100	110	125	140	140+	
180.0	197.5	217.5	245.0	272.5	292.5	312.5	330.0	345.0	360.0	380.0	395.0	Int'l Elite
165.0	180.0	197.5	225.0	250.0	267.5	285.0	302.5	317.5	330.0	350.0	360.0	Elite
152.5	167.5	182.5	205.0	230.0	247.5	265.0	277.5	290.0	305.0	320.0	330.0	Master
137.5	150.0	162.5	185.0	205.0	222.5	235.0	250.0	260.0	270.0	287.5	295.0	Class I
120.0	132.5	150.0	162.5	180.0	192.5	207.5	220.0	230.0	240.0	252.5	260.0	Class II
102.5	115.0	125.0	142.5	157.5	170.0	182.5	190.0	200.0	207.5	222.5	227.5	Class III
90.0	100.0	110.0	122.5	137.5	147.5	157.5	167.5	175.0	182.5	192.5	197.5	Class IV
52	56	60	67.5	75	82.5	90	100	110	125	140	140+	
120.0	132.5	142.5	160.0	175.0	187.5	197.5	207.5	217.5	225.0	232.5	240.0	Int'l Elite
110.0	120.0	132.5	147.5	160.0	172.5	182.5	190.0	197.5	202.5	212.5	222.5	Elite
102.5	110.0	120.0	135.0	147.5	157.5	167.5	175.0	182.5	187.5	195.0	200.0	Master
92.5	100.0	107.5	120.0	132.5	142.5	150.0	157.5	162.5	167.5	175.0	180.0	Class I
82.5	87.5	97.5	105.0	115.0	125.0	132.5	140.0	145.0	150.0	155.0	160.0	Class II
70.0	77.5	82.5	92.5	100.0	107.5	115.0	120.0	125.0	130.0	135.0	140.0	Class III
62.5	67.5	72.5	82.5	87.5	95.0	100.0	105.0	110.0	112.5	117.5	120.0	Class IV
52	56	60	67.5	75	82.5	90	100	110	125	140	140+	
195.0	207.5	222.5	247.5	265.0	285.0	297.5	315.0	320.0	325.0	335.0	345.0	Int'l Elite
180.0	192.5	202.5	227.5	245.0	262.5	272.5	287.5	292.5	297.5	310.0	317.5	Elite
165.0	177.5	187.5	205.0	225.0	240.0	252.5	265.0	270.0	272.5	280.0	287.5	Master
150.0	157.5	167.5	185.0	200.0	217.5	225.0	237.5	242.5	245.0	252.5	260.0	Class I
132.5	140.0	155.0	162.5	175.0	190.0	197.5	207.5	212.5	217.5	225.0	230.0	Class II
112.5	120.0	130.0	145.0	155.0	165.0	172.5	182.5	185.0	187.5	192.5	197.5	Class III
100.0	105.0	112.5	125.0	135.0	145.0	150.0	157.5	162.5	165.0	170.0	175.0	Class IV

WOMEN'S CLASSIFICATION

OPEN	Raw	Raw	Raw	Raw	Raw	Raw	Raw	Raw	Raw	Raw	Raw
	44	48	52	56	60	67.5	75	82.5	90	90+	
Int'l Elite	285.0	305.0	327.5	347.5	367.5	405.0	437.5	467.5	502.5	530.0	Int'l Elite
Elite	260.0	280.0	300.0	320.0	337.5	370.0	400.0	430.0	462.5	485.0	Elite
Master	237.5	255.0	272.5	290.0	305.0	337.5	365.0	392.5	417.5	440.0	Master
Class I	212.5	230.0	245.0	260.0	275.0	305.0	330.0	352.5	375.0	395.0	Class I
Class II	190.0	202.5	217.5	232.5	245.0	270.0	287.5	312.5	335.0	352.5	Class II
Class III	165.0	177.5	190.0	202.5	215.0	237.5	255.0	275.0	287.5	307.5	Class III
Class IV	142.5	152.5	165.0	172.5	185.0	202.5	217.5	235.0	252.5	265.0	Class IV
	SQ	SQ	SQ	SQ	SQ	SQ	SQ	SQ	SQ	SQ	
Int'l Elite	100.0	107.5	115.0	122.5	130.0	142.5	157.5	167.5	182.5	195.0	Int'l Elite
Elite	92.5	97.5	105.0	112.5	120.0	132.5	145.0	155.0	167.5	180.0	Elite
Master	82.5	90.0	95.0	102.5	107.5	120.0	132.5	140.0	152.5	162.5	Master
Class I	75.0	80.0	85.0	92.5	97.5	107.5	117.5	127.5	137.5	147.5	Class I
Class II	65.0	70.0	77.5	80.0	87.5	95.0	102.5	112.5	122.5	130.0	Class II
Class III	57.5	62.5	67.5	70.0	75.0	85.0	92.5	100.0	105.0	115.0	Class III
Class IV	50.0	52.5	57.5	60.0	65.0	72.5	77.5	85.0	92.5	97.5	Class IV
	BP	BP	BP	BP	BP	BP	BP	BP	BP	BP	
Int'l Elite	67.5	72.5	77.5	82.5	87.5	95.0	102.5	110.0	117.5	125.0	Int'l Elite
Elite	62.5	65.0	70.0	75.0	80.0	87.5	95.0	100.0	107.5	115.0	Elite
Master	55.0	60.0	65.0	67.5	72.5	80.0	85.0	92.5	97.5	102.5	Master
Class I	50.0	55.0	57.5	62.5	65.0	72.5	77.5	82.5	87.5	92.5	Class I
Class II	45.0	47.5	52.5	55.0	57.5	62.5	67.5	72.5	80.0	82.5	Class II
Class III	37.5	42.5	45.0	47.5	50.0	55.0	60.0	65.0	67.5	72.5	Class III
Class IV	32.5	35.0	37.5	40.0	42.5	47.5	52.5	55.0	60.0	62.5	Class IV
	DL	DL	DL	DL	DL	DL	DL	DL	DL	DL	
Int'l Elite	117.5	125.0	135.0	145.0	150.0	165.0	177.5	190.0	200.0	210.0	Int'l Elite
Elite	107.5	115.0	125.0	132.5	137.5	152.5	162.5	175.0	185.0	192.5	Elite
Master	97.5	105.0	112.5	120.0	125.0	137.5	147.5	157.5	167.5	175.0	Master
Class I	87.5	95.0	102.5	107.5	112.5	125.0	132.5	142.5	150.0	157.5	Class I
Class II	77.5	85.0	90.0	95.0	100.0	110.0	117.5	127.5	135.0	140.0	Class II
Class III	67.5	75.0	80.0	85.0	87.5	97.5	102.5	110.0	115.0	122.5	Class III
Class IV	57.5	62.5	67.5	72.5	75.0	82.5	87.5	95.0	100.0	105.0	Class IV

Raw	Raw	Raw	Raw	Raw	Raw	Raw	Raw	Raw	Raw	OPEN
44	48	52	56	60	67.5	75	82.5	90	90+	
312.5	335	360	382.5	405	445	482.5	515	552.5	582.5	Int'l Elite
285	307.5	330	352.5	372.5	407.5	440	472.5	510	532.5	Elite
262.5	280	300	320	335	372.5	402.5	432.5	460	485	Master
235	252.5	270	285	302.5	335	362.5	387.5	412.5	435	Class I
210	222.5	240	255	270	297.5	317.5	345	367.5	387.5	Class II
182.5	195	210	222.5	237.5	262.5	280	302.5	317.5	337.5	Class III
157.5	167.5	182.5	190	202.5	222.5	240	257.5	277.5	292.5	Class IV
44	48	52	56	60	67.5	75	82.5	90	90+	
110	117.5	127.5	135	142.5	157.5	172.5	185	200	215	Int'l Elite
102.5	107.5	115	125	132.5	145	160	170	185	197.5	Elite
90	100	105	112.5	117.5	132.5	145	155	167.5	180	Master
82.5	87.5	92.5	102.5	107.5	117.5	130	140	152.5	162.5	Class I
72.5	77.5	85	87.5	97.5	105	112.5	125	135	142.5	Class II
62.5	70	75	77.5	82.5	92.5	102.5	110	115	127.5	Class III
55	57.5	62.5	65	72.5	80	85	92.5	102.5	107.5	Class IV
44	48	52	56	60	67.5	75	82.5	90	90+	
75	80	85	90	97.5	105	112.5	120	130	137.5	Int'l Elite
70	72.5	77.5	82.5	87.5	97.5	105	110	117.5	127.5	Elite
60	65	72.5	75	80	87.5	92.5	102.5	107.5	112.5	Master
55	60	62.5	70	72.5	80	85	90	97.5	102.5	Class I
50	52.5	57.5	60	62.5	70	75	80	87.5	90	Class II
42.5	47.5	50	52.5	55	60	65	72.5	75	80	Class III
35	37.5	42.5	45	47.5	52.5	57.5	60	65	70	Class IV
44	48	52	56	60	67.5	75	82.5	90	90+	
130	137.5	147.5	160	165	182.5	195	210	220	230	Int'l Elite
117.5	127.5	137.5	145	152.5	167.5	180	192.5	202.5	212.5	Elite
107.5	115	125	132.5	137.5	152.5	162.5	172.5	185	192.5	Master
97.5	105	112.5	117.5	125	137.5	145	157.5	165	172.5	Class I
85	92.5	100	105	110	120	130	140	147.5	155	Class II
75	82.5	87.5	92.5	97.5	107.5	112.5	120	127.5	135	Class III
62.5	70	75	80	82.5	90	97.5	105	110	115	Class IV

OPEN	CI	CI	CI	CI	CI	CI	CI	CI	CI	CI
	44	48	52	56	60	67.5	75	82.5	90	90+
Int'l Elite	290.0	310.0	332.5	355.0	375.0	412.5	445.0	477.5	512.5	540.0
Elite	265.0	285.0	305.0	325.0	342.5	377.5	407.5	437.5	470.0	495.0
Master	240.0	260.0	277.5	295.0	312.5	342.5	372.5	400.0	427.5	450.0
Class I	215.0	232.5	250.0	265.0	280.0	310.0	335.0	360.0	382.5	402.5
Class II	192.5	205.0	222.5	235.0	250.0	275.0	292.5	317.5	342.5	360.0
Class III	167.5	180.0	195.0	205.0	217.5	240.0	260.0	280.0	292.5	315.0
Class IV	145.0	155.0	167.5	177.5	187.5	205.0	222.5	237.5	257.5	270.0
	SQ	SQ	SQ	SQ	SQ	SQ	SQ	SQ	SQ	SQ
Int'l Elite	105.0	112.5	120.0	127.5	137.5	150.0	165.0	177.5	192.5	205.0
Elite	95.0	102.5	110.0	117.5	125.0	137.5	152.5	162.5	177.5	187.5
Master	87.5	92.5	100.0	107.5	115.0	125.0	137.5	147.5	160.0	172.5
Class I	77.5	85.0	90.0	95.0	102.5	112.5	125.0	132.5	145.0	155.0
Class II	70.0	75.0	80.0	85.0	92.5	100.0	110.0	117.5	127.5	137.5
Class III	60.0	65.0	70.0	75.0	80.0	87.5	95.0	102.5	110.0	120.0
Class IV	52.5	57.5	60.0	62.5	70.0	75.0	82.5	87.5	97.5	102.5
	BP	BP	BP	BP	BP	BP	BP	BP	BP	BP
Int'l Elite	67.5	72.5	77.5	82.5	87.5	95.0	102.5	110.0	117.5	125.0
Elite	62.5	65.0	70.0	75.0	80.0	87.5	95.0	100.0	107.5	115.0
Master	55.0	60.0	65.0	67.5	72.5	80.0	85.0	92.5	97.5	102.5
Class I	50.0	55.0	57.5	62.5	65.0	72.5	77.5	82.5	87.5	92.5
Class II	45.0	47.5	52.5	55.0	57.5	62.5	67.5	72.5	80.0	82.5
Class III	37.5	42.5	45.0	47.5	50.0	55.0	60.0	65.0	67.5	72.5
Class IV	32.5	35.0	37.5	40.0	42.5	47.5	52.5	55.0	60.0	62.5
	DL	DL	DL	DL	DL	DL	DL	DL	DL	DL
Int'l Elite	117.5	125.0	135.0	145.0	150.0	165.0	177.5	190.0	200.0	210.0
Elite	107.5	115.0	125.0	132.5	137.5	152.5	162.5	175.0	185.0	192.5
Master	97.5	105.0	112.5	120.0	125.0	137.5	147.5	157.5	167.5	175.0
Class I	87.5	95.0	102.5	107.5	112.5	125.0	132.5	142.5	150.0	157.5
Class II	77.5	85.0	90.0	95.0	100.0	110.0	117.5	127.5	135.0	140.0
Class III	67.5	75.0	80.0	85.0	87.5	97.5	102.5	110.0	115.0	122.5
Class IV	57.5	62.5	67.5	72.5	75.0	82.5	87.5	95.0	100.0	105.0

CI Raw	CI Raw	CI Raw	CI Raw	CI Raw	CI Raw	CI Raw	CI Raw	CI Raw	CI Raw	OPEN
44	48	52	56	60	67.5	75	82.5	90	90+	
320	340	365	390	412.5	455	490	525	565	595	Int'l Elite
292.5	312.5	335	357.5	377.5	415	447.5	482.5	517.5	545	Elite
265	285	305	325	345	377.5	410	440	470	495	Master
237.5	255	275	292.5	307.5	340	367.5	395	420	442.5	Class I
212.5	225	245	257.5	275	302.5	322.5	350	377.5	395	Class II
185	197.5	215	225	240	265	285	307.5	322.5	347.5	Class III
160	170	185	195	207.5	225	245	262.5	282.5	297.5	Class IV
44	48	52	56	60	67.5	75	82.5	90	90+	
115	125	132.5	140	152.5	165	182.5	195	212.5	225	Int'l Elite
105	112.5	120	130	137.5	152.5	167.5	180	195	207.5	Elite
97.5	102.5	110	117.5	127.5	137.5	152.5	162.5	175	190	Master
85	92.5	100	105	112.5	125	137.5	145	160	170	Class I
77.5	82.5	87.5	92.5	102.5	110	120	130	140	152.5	Class II
65	72.5	77.5	82.5	87.5	97.5	105	112.5	120	132.5	Class III
57.5	62.5	65	70	77.5	82.5	90	97.5	107.5	112.5	Class IV
44	48	52	56	60	67.5	75	82.5	90	90+	
75	80	85	90	97.5	105	112.5	120	130	137.5	Int'l Elite
70	72.5	77.5	82.5	87.5	97.5	105	110	117.5	127.5	Elite
60	65	72.5	75	80	87.5	92.5	102.5	107.5	112.5	Master
55	60	62.5	70	72.5	80	85	90	97.5	102.5	Class I
50	52.5	57.5	60	62.5	70	75	80	87.5	90	Class II
42.5	47.5	50	52.5	55	60	65	72.5	75	80	Class III
35	37.5	42.5	45	47.5	52.5	57.5	60	65	70	Class IV
44	48	52	56	60	67.5	75	82.5	90	90+	
130	137.5	147.5	160	165	182.5	195	210	220	230	Int'l Elite
117.5	127.5	137.5	145	152.5	167.5	180	192.5	202.5	212.5	Elite
107.5	115	125	132.5	137.5	152.5	162.5	172.5	185	192.5	Master
97.5	105	112.5	117.5	125	137.5	145	157.5	165	172.5	Class I
85	92.5	100	105	110	120	130	140	147.5	155	Class II
75	82.5	87.5	92.5	97.5	107.5	112.5	120	127.5	135	Class III
62.5	70	75	80	82.5	90	97.5	105	110	115	Class IV

OVERVIEW OF THE PHASES

Hypertrophy
Strength
Peaking
Deload
Transitional Weeks
Bridge Phases

We implement a phasic system into our training to ensure that each necessary quality for the powerlifter is developed to its fullest abilities and that the athlete achieves their highest fitness levels for their competition, not before or after. A phasic system for powerlifting will primarily focus on training for Hypertrophy, General Strength and Peaking, organized strategically to bring the athlete to the competition platform with the most muscle, greatest strength, best technique, prepared neural qualities and lowest fatigue possible.

A phasic structure, when implemented correctly, allows each quality-hypertrophy, strength, and peaking/neural force production-to be developed to a higher level than if these qualities were trained simultaneously. While all of these qualities are important to a powerlifter's success, training simultaneously to improve qualities of the 10rm (hypertrophy) and 1rm (peaking) will compete against

each other, generating too much muscular fatigue through hypertrophy to allow for maximum performance in the 1rm and too much neural fatigue, as well as stress on the connective tissue, from peaking to push the volume to the necessary levels for optimal hypertrophy. A common refrain from critics of phasic training is that the athlete is constantly moving away from the last quality-most notably losing muscle as they transition away from hypertrophy and approach a meet, that would only be an issue if improper phase lengths and overloading parameters were utilized. Nobody can dispute that a block dedicated purely to Hypertrophy will be the best way to add muscle mass and when the subsequent General Strength phase is sufficiently voluminous, there will be no risk of losing muscle during this time. While volume will inevitably drop during a peaking phase, choosing an appropriate length peaking block for the athlete's qualification will allow you to avoid any muscle atrophy during this phase heading into a meet.

In this next section, I will briefly outline the goals and overloading parameters of each key phase of our training. We will expand on each phase in later sections

Hypertrophy

Training for Hypertrophy creates the foundation of success for a powerlifter. Simply put, the athlete with more muscle mass has a greater potential to lift heavier weights, as well as being more durable to hard training. Hypertrophic gains are driven by increases in volume, so your focus should be on increasing number of sets, the number of reps per set, weight or some combination of the 3 over time.

Goals

Build Muscle
Increase Work Capacity
Avoid Adaptive Resistance to Heavy Training
Avoid Adaptive Resistance to Technical Development in Competitive Lifts

Overloading Parameters

55-75% of 1rm
Sets of 6 to 12 Reps
Increases in total volume will lead to Hypertrophic gains.

Volume Ranges

Minimum Effective Volumes

-Squat: 5 to 10 Overloading Sets/Week
-Bench: 6 to 12 Overloading Sets/Week
-Deadlift: 4 to 7 Overloading Sets/Week

Maximum Recoverable Volumes

-Squat: 10 to 16 Overloading Sets/Week
-Bench: 14 to 20 Overloading Sets/Week
-Deadlift: 8 to 14 Overloading Sets/Week

Strength

Goals

Improve Force Production

Begin Technical Development of Competitive Exercises

Maintain Muscle Mass Developed During Hypertrophy

Overloading Parameters

70-90% of 1rm

Sets of 3 to 6 Reps

Strength adaptations necessitate increases in intensity over time

Volume Ranges

Minimum Effective Volumes

-Squat: 3 to 8 Overloading Sets/Week

-Bench: 5 to 11 Overloading Sets/Week

-Deadlift: 3 to 6 Overloading Sets/Week

Maximum Recoverable Volumes

-Squat: 6 to 12 Overloading Sets/Week

-Bench: 8 to 16 Overloading Sets/Week

-Deadlift: 4 to 10 Overloading Sets/Week

Peaking

Peaking training is a uniquely important phase of the competitive powerlifter's training. This phase is the culmination of a well designed and diligently executed training program and while it is unlikely that great peaking training can salvage poor efforts preceding it, it is very possible for a poorly organized peaking plan to cause your previous hard work to be squandered. Optimal peaking training hinges on striking a balance between developing technical precision, building neural force production and optimizing the fitness/fatigue relationship for the day of competition.

Goals

Develop Technical Prowess
Enhance Neural Force Production
Maintain Fitness from Previous Blocks
Decay Fatigue for Day of Competition

Overloading Parameters

85-100% of 1rm
Sets of 1 to 3 Reps
High-intensity lifts are needed to practice competition technique and improve neural qualities.

Volume Ranges

Minimum Effective Volumes

-Squat: 2 to 7 Overloading Sets/Week
-Bench: 4 to 9 Overloading Sets/Week
-Deadlift: 1 to 4 Overloading Sets/Week

Maximum Recoverable Volumes

-Squat: 3 to 9 Overloading Sets/Week
-Bench: 5 to 12 Overloading Sets/Week
-Deadlift: 2 to 7 Overloading Sets/Week

Deload

The Deload is a very important though often overlooked aspect of effective and sustainable training. Some lifters and coaches want to lead you to believe that deloads aren't necessary or that feeling you need a deload is an indication that you haven't prioritized recovery properly, to those people, I would respond that their normal training isn't as overloading and thus, effective, as it could be. Good training is very challenging and should be accompanied by planned deloads organized around times of functional overreaching.

Goals
Decay Fatigue
Create Smooth Transition from One Phase to the Next

Overloading Parameters
80-90% of Normal Training Intensity
50-75% of Normal Training Volume

For example, if you did 5x8 Squats with 150kg during the last week of a Hypertrophy Phase, an appropriate Deload could be 3x8 at 130kg, 4x6 at 130kg would also be effective.

Transitional Weeks

Transitional Weeks are a strategy that we use to create a smooth transition after competition back into higher volume training. A common situation that arises is an athlete goes through a 4 week peaking block and taper where they perform sets of 1-3 reps almost exclusively, they compete in the meet where they do 9 singles, they likely take the week off after the meet or do very light training, now they have 5 weeks where they haven't performed an overloading set of more than 3 reps and they try and jump back into Hypertrophy training for sets of 8 and are so limited by their temporary reduction in fitness and work capacity that they can barely survive the session and are sore for a week, sound familiar? We mitigate this issue with the use of Transitional Week(s) which allow a slower, smoother ramp-up back into higher volume work.

One to Three weeks of transitional work is typical, the longer the athlete has been performing low volume training, the longer they will need during the transitional period. The first step of designing a transitional program is to identify where you want to begin your hypertrophy work at, establish that target and then work backward from there. Let's say that you want to begin your Hypertrophy Block with a session of 4x8 at 300#, transitional weeks could look like:

1. Week Transition
    Week 1-4x5 at 300#
2. Week Transition
    Week 1-4x4 at 300#
    Week 2-4x6 at 300#
3. Week Transition
    Week 1-4x4 at 280#
    Week 2-4x4 at 300#
    Week 3-4x6 at 300#

These transitional weeks should make the re-introduction of higher volume Hypertrophy work much smoother, helping your athletes be more successful by avoiding excessive fatigue. Even when an athlete is very eager about getting back into hard training, I still highly suggest this type of work, they may seem bored in the short term but over the course of a training cycle and more so a (hopefully) long lifter career, those 1-3 weeks of boredom will seem insignificant to them. During the week after a competition, I usually have lifters avoid doing many if any, barbell lifts but rather they will benefit from focusing on performing 3-5 lifting sessions mostly on machines (if available) to promote blood flow into the muscles with little stress on the joints and nervous system. Performing LISS aerobic work like walking, biking or swimming can also be useful to promote recovery. I would also encourage an athlete to do activities away from lifting that they enjoy, like pick-up basketball or something similar, in the immediate time following a meet, simply to create a mental break for themselves.

Bridge Phases

A Bridge Phase is a term I use to describe a period of training that creates a 'bridge' between the end of training for one meet and the start of focused preparation for the next meet. While it may be possible and perhaps more 'scientifically sound' for an athlete with a to go from meet, to Transitional Weeks into Hypertrophy for an extended time, followed by strength and peaking again or to have a short time between competitions so that they are only performing intensive meet training, I do not find this to be a long-term sustainable plan. This is where Bridge Phases can become critical; to either provide a time of greater variation for a mental and physical break for the athlete, helping them avoid adaptive resistance and staleness or for an athlete who may be detrained to build up work capacity in preparation for harder training to follow. These type of Bridge Phases are something that in my own career, I failed to pro-actively include, so I had to reactively include them when small injuries, frustration, and boredom built up.

The first way to effectively utilize a Bridge Phase is to help an athlete who is de-trained either due to injury or just a period of inactivity to build up work capacity in preparation for more overloading and specific meet preparation. This period should focus on variation and frequency while slowly and smoothly increasing fitness levels. Variation is of great importance because it will help the detrained athlete build a broader base of fitness, correcting potential imbalances they may have while improving coordination. Higher frequency training will be useful to these athletes because they are operating at lower levels than they have previously been capable of, it will be difficult for them to get significant stimulus from a single session, so they will need greater

frequency to build their fitness back. Along with training at higher frequencies, utilizing submaximal work is critical here, as it will allow the athlete to execute every lift with great technique while staying far enough away from failure so fatigue doesn't accumulate too quickly and any risk of injury is mitigated.

My preferred way to organize a Bridge Phase for an athlete who needs to regain fitness to begin harder, more focused training is to give them a fix volume and intensity but allow them to adjust their reps per set by feel and utilize controlled rest periods. This helps to ensure that they are performing quality reps and moving quickly to improve special work capacity. Here is a sample of a Bridge Phase for this purpose:

Day 1	Week 1	Week 2	Week 3
Pause Squat	70%x24 Reps-90 sec rests	55%x30 Reps-60 sec rests	75%x18 Reps, 105 sec rests
Bench Press	70%x24 Reps-90 sec rests	55%x30 Reps-60 sec rests	75%x18 Reps, 105 sec rests
Opposite Stance Deadlift	70%x24 Reps-90 sec rests	55%x30 Reps-60 sec rests	75%x18 Reps, 105 sec rests
Day 2			
Low Bar Squat	55%x30 Reps-60 sec rests	70%x24 Reps-90 sec rests	60%x24 Reps-75 sec rests
Widegrip Bench	55%x30 Reps-60 sec rests	70%x24 Reps-90 sec rests	60%x24 Reps-75 sec rests
Bentover Rows	5x10	5x10	5x8
Shoulders/Arms	3x15 each	3x15 each	3x12 each
Day 3			
Closegrip Bench	70%x24 Reps-90 sec rests	55%x30 Reps-60 sec rests	75%x18 Reps, 105 sec rests
Comp Stance Block Pull	70%x24 Reps-90 sec rests	55%x30 Reps-60 sec rests	75%x18 Reps, 105 sec rests
Back Raises	5x10	5x10	5x8
Arms	3x15 each	3x15 each	3x12 each
Day 4			
High Bar Squat	55%x30 Reps-60 sec rests	70%x24 Reps-90 sec rests	60%x24 Reps-75 sec rests
Feet Up Bench	55%x30 Reps-60 sec rests	70%x24 Reps-90 sec rests	60%x24 Reps-75 sec rests
Lat Pulldowns	5x10	5x10	5x8
Shoulders/Arms	3x15 each	3x15 each	3x12 each

Day 1	Week 4	Week 5	Week 6
Pause Squat	60%x24 Reps-75 sec rests	80%x12 Reps-120 sec rests	65%x18 Reps, 90 sec rests
Bench Press	60%x24 Reps-75 sec rests	80%x12 Reps-120 sec rests	65%x18 Reps, 90 sec rests
Opposite Stance Deadlift	60%x24 Reps-75 sec rests	80%x12 Reps-120 sec rests	65%x18 Reps, 90 sec rests
Day 2			
Low Bar Squat	75%x18 Reps, 105 sec rests	65%x18 Reps, 90 sec rests	80%x12 Reps-120 sec rests
Widegrip Bench	75%x18 Reps, 105 sec rests	65%x18 Reps, 90 sec rests	80%x12 Reps-120 sec rests
Bentover Rows	5x8	5x6	5x6
Shoulders/Arms	3x12 each	3x10 each	3x10 each
Day 3			
Closegrip Bench	60%x24 Reps-75 sec rests	80%x12 Reps-120 sec rests	65%x18 Reps, 90 sec rests
Comp Stance Block Pull	60%x24 Reps-75 sec rests	80%x12 Reps-120 sec rests	65%x18 Reps, 90 sec rests
Back Raises	5x8	5x6	5x6
Arms	3x12 each	3x10 each	3x10 each
Day 4			
High Bar Squat	75%x18 Reps, 105 sec rests	65%x18 Reps, 90 sec rests	80%x12 Reps-120 sec rests
Feet Up Bench	75%x18 Reps, 105 sec rests	65%x18 Reps, 90 sec rests	80%x12 Reps-120 sec rests
Lat Pulldowns	5x8	5x6	5x6
Shoulders/Arms	3x12 each	3x10 each	3x10 each

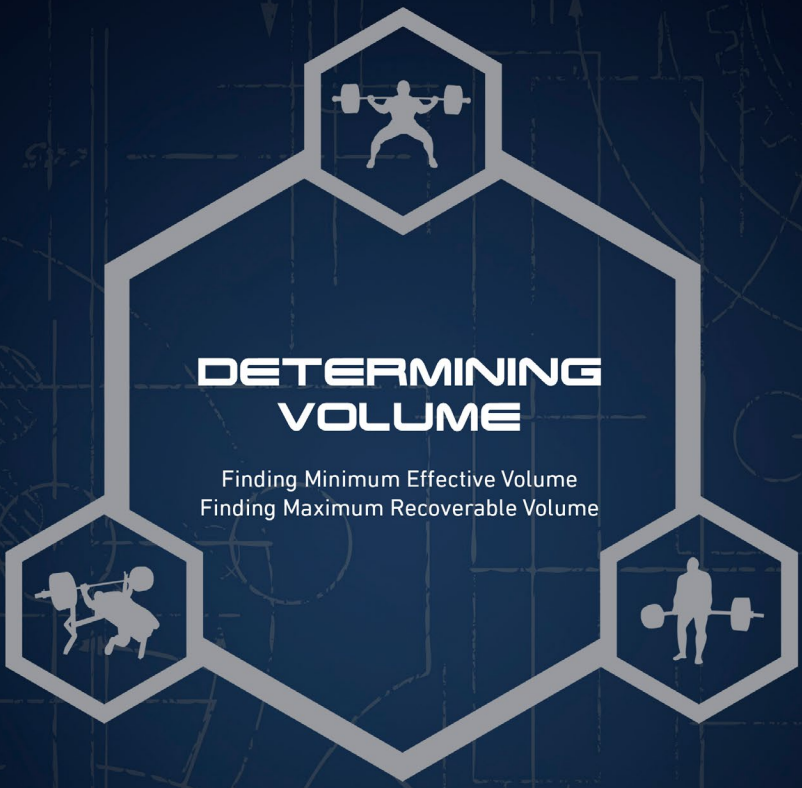
All of the exercises that are only give a % x Reps should be performed in sets of 3-6 reps based on feel, only crisp, explosive reps with good technique should be done. As soon as the bar slows down or technique falters, rack the bar and rest. This may mean that 70%x24 Reps actually breaks down as Sets of 5, 5, 4, 4, 3, 3 reps and that is fine. The program above can be adjusted in many ways dependent on the athlete's qualification and needs.

For example:

- Rest periods can be lengthened or shortened based on fitness levels.
- Total reps can be changed +/-3. I suggest decreasing reps for bigger/stronger athletes and increasing them for smaller/weaker and female athletes.
- Intensity can increase 2.5-5% for smaller/weaker and female athletes.
- Frequency could increase. This program could be performed up to 6 days/week. In its current 4 day iteration, I would train on Mon/Wed/Fri/Sat or Mon/Tues/Thurs/Fri. You could also add aerobic capacity work on Off Days.
- Amount of accessory exercises can increase.

A Bridge Phase that is being used to add variation to an athlete's training, giving them a mental break from more specific and structured training while allowing them to rehab small nagging injuries, explore new movement patterns and avoid staleness/adaptive resistance can take many different forms. This could be the best time that Strongman, Athletic Performance or CrossFit strategies could be very useful. There are so many potential ways that this program could be constructed and be useful in this context, so I will just give you a few guidelines to keep in mind when creating it.

- Let your athletes have fun! This can be accomplished by doing new activities, old activities that they haven't done in a while or just setting new temporary short-term goals.
- Be an athlete. Get your lifters moving, this could be sprinting (I'd suggest hills or sleds), throwing, jumping and carrying.
- Create progress. The Principle of Overload should still be adhered to in this phase but it doesn't just mean that you do more volume or more weight in this looser structure. You can overload in those manners, you can do more reps, you can go faster, you can rest less or any combination of the above.
- Don't forget the details. Since so much of the training you are doing is going to be new and fun, there still needs to be attention put towards some things that are often tedious like breathing drills and corrective exercises. This probably means 10-20 minutes of small details before you get to play.



DETERMINING VOLUME

Finding Minimum Effective Volume
Finding Maximum Recoverable Volume

One of the most critical aspects of optimizing training to the individual is finding the right volume to work with. We must know how much training the athlete needs to make progress (Minimum Effective Volume-MEV) and how much they can potentially tolerate (Maximum Recoverable Volume-MRV). Finding the precise numbers is an inexact science, as it will change throughout the lifter's career, and will require a process of trial and error to more finely tune, but by critically analyzing several factors about the athlete, we can make a very well informed estimate, to begin with. Let's take a deeper look at these Individual Differences and how they affect MEV/MRV.

Sex

Female lifters are generally smaller and less muscular than their male counterparts of similar qualification, this coupled with lower testosterone, causes each session to be less fatiguing and necessitates them to train at higher volumes. Male lifters will have lower MEVs and MRVs than female lifters.

Bodyweight

A lifter's bodyweight is highly correlated (hopefully) to their muscle mass. More muscle mass means more muscle to be damaged and this greater amount of damaged muscle will take longer to recover, lessening the amount of training the athlete can do. Greater muscle mass also allows the athlete to better hold fitness. Higher body weights will lower both MEV and MRV. One potential caveat to this would be an athlete who carries very high levels of body fat, which could raise MEV.

Height

A taller lifter will move the bar a greater distance on each rep causing more work to be done on each rep, this will make each set more fatiguing than it would be for a shorter lifter. Greater height causes both MEV and MRV to lower.

Strength

The stronger a lifter becomes the more stimulating each set of work they do is and the more fatigue a given set will generate. Strength is hard earned and also hard to maintain. A lifter's strength will cause their MEV to raise and their MRV to lower, this type of factor makes the training of very strong lifters' a unique challenge.

Experience

A more experienced lifter will have a greater special work capacity than a beginner lifter, and more than they had when they were beginners or intermediates. This greater special work capacity should allow them to train with more volume throughout the week. When a lifter becomes Very Experienced (ie. Consistent hard training for over 12 years), it is likely that they will not be able to tolerate as many sets of overloading work as they previously could because each of these sets will be so stimulating. Similarly to strength, an experienced lifter has done so much training in their past, that small amounts of training have a very little effect, so as a lifter's experience increases, their MEV will also increase. MRV will also increase with experience, likely until the athlete becomes Very Experienced, at which time there will be a small downturn in MRV.

Age

I'm often asked questions to the effect of, "I'm 45 years old, how do I need to change my training?" But biological age on its own does not give us that much information in regards to how much training an athlete needs or can tolerate. There can be a massive difference in training needs from one 45 years old to another but many of the other Individual Differences we are considering here will account for those differences. What Biological Age does largely tell us is that all things being equal (which they rarely are), a 25-year-old can do more training than a 35-year-old, who can do more than a 45-year-old and so on. A younger athlete can benefit from less training (Lower MEV), largely due to hormonal reason and can generally tolerate higher training (Raise MRV).

Diet

The quality of an athlete's diet has a unique impact on their training volume requirements. From the perspective of MEV, a Poor Diet (insufficient calories) will also necessitate that the athlete does more training to overcome this catabolic effect. On the other hand, a Poor Diet will significantly limit the total amount of training the athlete can recover from, causing MRV to be lowered. A Good Diet (Caloric Surplus with Attention to Timing and Nutrient Balance) will allow the athlete to benefit from less training (Lower MEV), while also enabling them to tolerate slightly higher volume (Raise MRV).

Sleep

Along with Diet, Sleep is the foundation of recovery and affects MEV/MRV in a similar manner. Poor Sleep (Less Than 5 Hours/Night on Average) will actually require you to do slightly more training to make progress because of the diminished recovery capacity, you'll need to give your body more stimulus. Conversely, Poor Sleep will significantly diminish the total amount of training you can recover from, Sleep is very important and has a significant impact on your training. Getting Good Sleep (Over 7 Hours/Night on Average) will allow you to benefit from slightly less training because you'll be better deriving the benefits of the work you are doing but will also allow you to push your total volume to higher levels.

Stress

Your body can not differentiate between stress being imposed on it, it does not know the difference between squats, manual labor or arguments, it only understands that stress is being imposed upon it.

If you have High-Stress levels outside of your training, whether that is due to physical demands of your job, long hours studying for exams or other reasons, you will face similar effects as having a Poor Diet or Poor Sleep, you will need to do slightly more training to see benefit (Raise MEV), while ultimately not being able to train as hard as you potentially could (Lower MRV). Having Low Stress outside of training will benefit you in being able to train slightly less and still benefit (Lower MEV) while allowing you to train more and harder and still recover (Raise MRV).

Historical Workload

The amount of training you have utilized in the past will influence the amount of training you need to do in the present and future. If you have performed high volumes of training previously, you'll like have a higher MEV than counterparts who have not trained in this manner. These high training volumes should have also raised your special work capacity and with that increased your MRV.

Historical Recovery Ability

The term Historical Recovery Ability is our attempt to quantify your genetic ability to recover from training. Genetics will be the biggest influence on how much training an athlete needs and is able to do but is also the most difficult to measure. We all know somebody who seems like they get stronger from just looking at weights and somebody who seems to be able to train endlessly and continue improving. Think critically about the training you have done in your life and how you tend to feel from it and respond to it, if you feel like you have an Exceptional Historical Recovery Ability, both your MEV and MRV will increase, while if you traditionally have a Poor Historical Recovery Ability, both volume landmarks will decrease.

Now that we understand the implications of each of these factors on your Minimum Effective Volume (MEV) and Maximum Recoverable Volume (MRV), we can use this information and your answers from the Assessing The Athlete section to understand the magnitudes of impact of each of these factors and find your specific MEV and MRV for each phase of training, Hypertrophy, Strength and Peaking.

Finding MEV

Step 1: Collect Your Answers from the Assessing the Athlete Section

Step 2: Establish Baseline MEV. For this, we will use the Median of our established ranges.

	Hypertrophy	**Strength**	**Peaking**
Squat:	7.5 Sets	5.5 Sets	4.5 Sets
Bench:	9 Sets	8 Sets	6.5 Sets
Deadlift:	5.5 Sets	4.5 Sets	2.5 Sets

Step 3: Find your answers on the following chart and move the MEV up/down according to the corresponding number. For example, in Squat Hypertrophy we will begin with 7.5 Sets, if you are a Female, increase by 1.5 Sets, Middleweight, increase by .5 Sets and so on.

Lower Volume						Higher Volume
-1.5	-1	-.5	0 Male	+.5	+1	+1.5 Female
Super		Heavy		Middle		Light
	Very Tall	Tall		Medium	Short	
		Low	Moderate	High	Very High	
	Beginner	Intermediate		Advanced	Very Advanced	
	<19	20s	30s	40s	50+	
		Good	Avg	Poor		
		Good	Avg	Poor		
		Low	Avg	High		
	1	2	3	4	5	
	1	2	3	4	5	

Your MEVs:

Hypertrophy	Squat:	Bench:	Deadlift:
Strength	Squat:	Bench:	Deadlift:
Peaking	Squat:	Bench:	Deadlift:

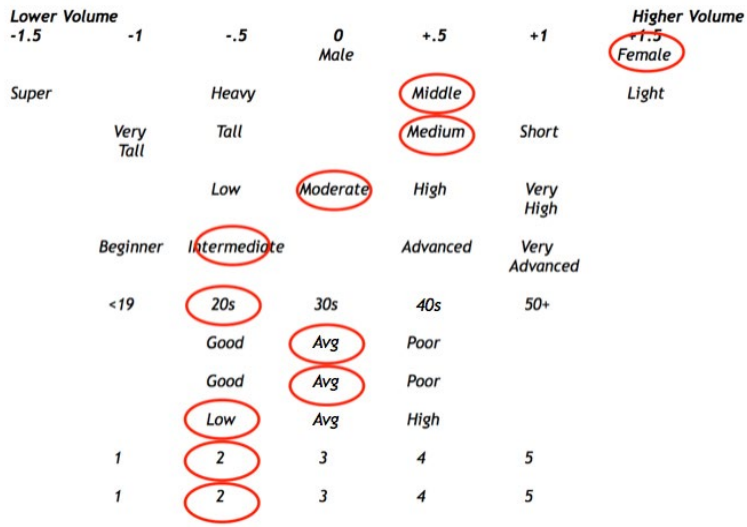
If you received a number that ends in .5, I would suggest you round that up for MEV.

For example, let's run through this process for an athlete with the following characteristics:

Female, Middle Weight, Medium Height, Moderate Strength, Intermediate Experience, 27 Years Old, Average Diet, Average Sleep, Low Away from Training Stress, 2 Historical Workload and 2 Historical Recovery Ability.

Starting with MEV Baseline Numbers in Hypertrophy of 7.5 Sets/Week in the Squat, 9 Sets/Week in the Bench and 5.5 Sets/Week in the Deadlift, our example athlete's numbers are adjusted as follows...

Characteristic	Adjustment	New Totals
Gender	Female +1.5 Sets/Week	Squat 9 Sets/Week Bench 10.5 Sets/Week Deadlift 7 Sets/Week
Weight	Middle +.5 Sets/Week	Squat 9.5 Sets/Week Bench 11 Sets/Week Deadlift 7.5 Sets/Week
Height	Medium +.5 Sets/Week	Squat 10 Sets/Week Bench 11.5 Sets/Week Deadlift 8 Sets/Week
Strength	Moderate No Adjustment	Squat 10 Sets/Week Bench 11.5 Sets/Week Deadlift 8 Sets/Week
Experience	Intermediate -.5 Sets/Week	Squat 9.5 Sets/Week Bench 11 Sets/Week Deadlift 7.5 Sets/Week
Age	20s -.5 Sets/Week	Squat 9 Sets/Week Bench 10.5 Sets/Week Deadlift 7 Sets/Week
Diet	Average No Adjustment	Squat 9 Sets/Week Bench 10.5 Sets/Week Deadlift 7 Sets/Week
Sleep Quality	Average No Adjustment	Squat 9 Sets/Week Bench 10.5 Sets/Week Deadlift 7 Sets/Week
Stress Away from Training	Low -.5 Sets/Week	Squat 8.5 Sets/Week Bench 10 Sets/Week Deadlift 6.5 Sets/Week
Historical Workload	2 -.5 Sets/Week	Squat 8 Sets/Week Bench 9.5 Sets/Week Deadlift 6 Sets/Week
Historical Recovery Ability	2 -.5 Sets/Week	Squat 7.5 Sets/Week Bench 9 Sets/Week Deadlift 5.5 Sets/Week

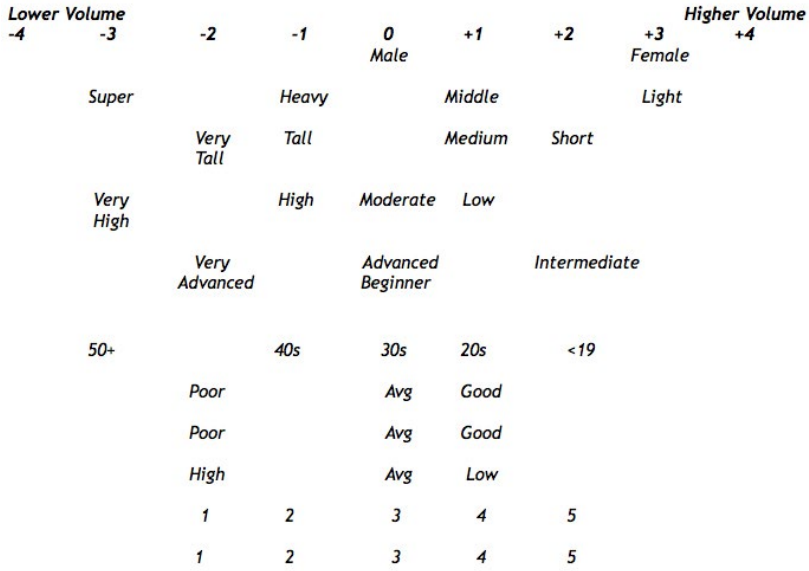


For this example athlete, their MEVs during a Hypertrophy Block are 7.5 Sets/Week for the Squat, 9 Sets/Week for the Bench Press and 5.5 Sets/Week for the Deadlift. You can extrapolate the results for Strength and Peaking here. Remember for Strength and Peaking, use their corresponding starting values in the table at the beginning of this section.

Finding Your MRV

Now let's repeat this process for MRV. For MRV we will use a new starting point, find the Median of our known MRV ranges below.

	Hypertrophy	Strength	Peaking
Squat:	14 Sets	9 Sets	6 Sets
Bench:	17 Sets	11 Sets	8.5 Sets
Deadlift:	11 Sets	7 Sets	4.5 Sets



Your MEVs:

Hypertrophy	Squat:	Bench:	Deadlift:
Strength	Squat:	Bench:	Deadlift:
Peaking	Squat:	Bench:	Deadlift:

If you find yourself with a number ending in .5, I'd suggest rounding down for MRV. There is the possibility that very big, very strong, Male athletes could finish with an MRV of .5 or lower, especially in the deadlift, if that is the case, you could treat .5 as a heavy deadlift every other week, or just raise the number to 1.

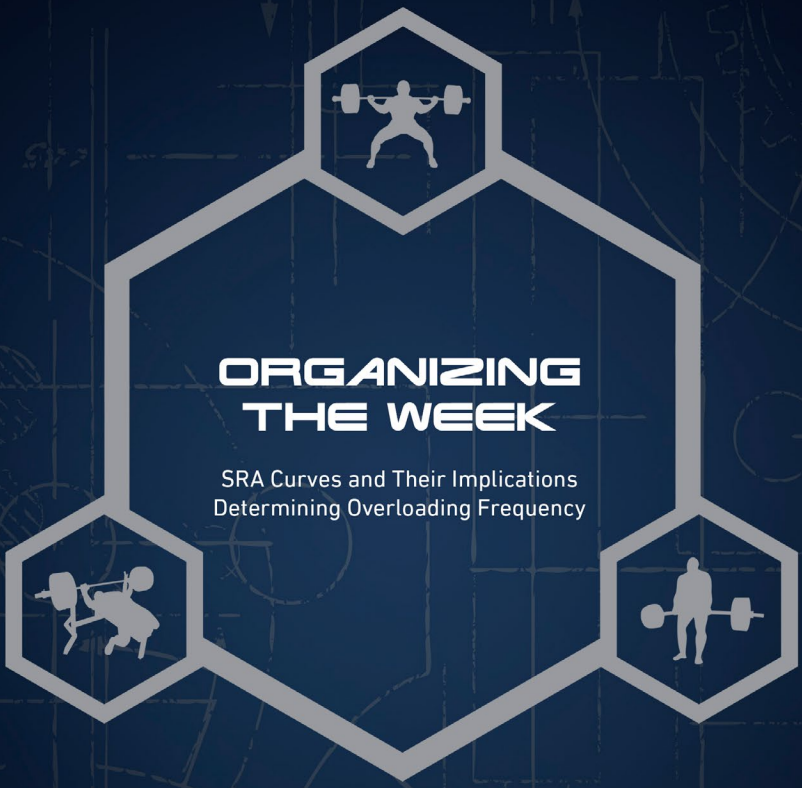
Let's use our same example athlete from MEV to illustrate this process for MRV.

Lower Volume								Higher Volume
-4	-3	-2	-1	0	+1	+2	+3	+4
				Male			Female	
	Super		Heavy		Middle		Light	
		Very Tall	Tall		Medium	Short		
	Very High		High	Moderate	Low			
		Very Advanced		Advanced Beginner		Intermediate		
	50+		40s	30s	20s	<19		
		Poor		Avg	Good			
		Poor		Avg	Good			
		High		Avg	Low			
		1	2	3	4	5		
		1	2	3	4	5		

Starting with MRV Baseline Numbers in Hypertrophy of 14 Sets/Week in the Squat, 17 Sets/Week in the Bench and 11 Sets/Week in the Deadlift, our example athlete's numbers are adjusted as follows...

Characteristic	Adjustment	New Totals
Gender	Female +3 Sets/Week	Squat 17 Sets/Week Bench 20 Sets/Week Deadlift 14 Sets/Week
Weight	Middle +1 Set/Week	Squat 18 Sets/Week Bench 21 Sets/Week Deadlift 15 Sets/Week
Height	Medium +1 Set/Week	Squat 19 Sets/Week Bench 22 Sets/Week Deadlift 16 Sets/Week
Strength	Moderate No Adjustment	Squat 19 Sets/Week Bench 22 Sets/Week Deadlift 16 Sets/Week
Experience	Intermediate +2 Sets/Week	Squat 21 Sets/Week Bench 24 Sets/Week Deadlift 18 Sets/Week
Age	20s +1 Set/Week	Squat 22 Sets/Week Bench 25 Sets/Week Deadlift 19 Sets/Week
Diet	Average No Adjustment	Squat 22 Sets/Week Bench 25 Sets/Week Deadlift 19 Sets/Week
Sleep Quality	Average No Adjustment	Squat 22 Sets/Week Bench 25 Sets/Week Deadlift 19 Sets/Week
Stress Away from Training	Low +1 Set/Week	Squat 23 Sets/Week Bench 26 Sets/Week Deadlift 20 Sets/Week
Historical Workload	2 -1 Set/Week	Squat 22 Sets/Week Bench 25 Sets/Week Deadlift 19 Sets/Week
Historical Recovery Ability	2 -1 Set/Week	Squat 21 Sets/Week Bench 24 Sets/Week Deadlift 18 Sets/Week

For this example athlete, their MRVs during a Hypertrophy Block are 21 Sets/Week for the Squat, 24 Sets/Week for the Bench Press and 18 Sets/Week for the Deadlift.
You can extrapolate the results for Strength and Peaking here. Remember for Strength and Peaking, use their corresponding starting values in the table at the beginning of this section.



ORGANIZING THE WEEK

SRA Curves and Their Implications
Determining Overloading Frequency

When looking to organize the training week, which entails determining our optimal frequency, deciding how to spread your volume across that frequency and then how to fill in that volume with the right exercises, we must understand the Principle of Stimulus Recovery Adaptation. The main issue that we are going to be dealing with SRA is how frequently we're going to introduce overload and hard training. Every different lift and different physical system is going to have different length SRA curve, and finding the optimal frequency of overload training is going to allow you to train as hard as possible, as frequently as possible, and with doing that, make gains as quickly as possible. The three different powerlifts, squat, bench and deadlift all have different SRA lengths.

The deadlift is the longest SRA curve. It is the most stressful to the body from both the neural standpoint and a physical standpoint, it will be trained the least frequently. The squat is going to be the next most fatiguing and the bench can probably be trained the most frequently. There will be some small considerations to be made for Individual Differences within that, such as if a sumo deadlifter is going to be able to train more frequently than a conventional deadlifter. A high bar squat or a front squat can be trained more frequently than a low bar back squat. Close grip bench probably more frequently than a wide grip bench, and people's unique leverages will come into account as well. Short lifters are going to be able to train more frequently than taller lifters. The bar is not moving as far and is not as much work. Different physical qualities also have different SRA lengths. Technique, hypertrophy, force output, and tissue integrity all need to be considered separately when organizing your training frequency based on SRA curves. Technical abilities, have the shortest SRA curve, meaning that you can and should train them the most frequently, and technique can be trained with relatively low intensity. To use a bit of an extreme example from another sport, think about a sport like golf. Golf is extremely technical, much more precise movement than powerlifting is, but obviously, the intensity is very low because a golf club is so much lighter than a squat, bench or deadlift. Golfers play 18 holes in the morning, 18 holes in the afternoon, repeat that day after day after day because the only thing that they are really dealing with is Technical SRA, and it recovers very, very quickly so they can and need to train it very frequently to maximize their abilities. From the strength sports world, consider weightlifting.

It is not uncommon for weightlifters to train six, nine, maybe even 12 sessions per week, and the weightlifting movements, the snatch and clean, are definitely more technical than the squat bench and deadlift. The weights also are not as heavy and there is no eccentric portion to the lifts, so they do not create as much muscular damage, they can and need to train them more frequently to develop their techniques to the necessary levels.
Hypertrophy is going to be the second shortest SRA length of these different physical qualities. Muscle size is one of the most important underlying factors to strength. The bigger a muscle is the better chance that athletes will have to be exceptionally strong. We need to frequently train for hypertrophy, we need to frequently introduce that stimulus to the muscles, but different size muscles, different insertion angles are going to dictate how frequently we can train those muscles. In short, small muscles can and should be trained more frequently than large muscles. Training shoulders, triceps, and biceps much more frequently than your lats or erectors, hamstrings, will be a good idea.

Force production is also a paramount indicator of strength, the better the athlete can produce force, the better they can recruit muscle fibers and apply that into the barbell, the better powerlifter they are going to be, and it has been shown that advanced lifters are more capable of higher force production than novice lifters, even with the same amount of muscle mass. The more advanced a lifter becomes, the longer their Force Production SRA curve is going to be and the longer duration they are going to need between high-intensity training sessions where they have to call on those abilities.

Finally, connective tissue integrity, the health of your tendons and ligaments, has to be considered as well. And those are going to have the longest SRA curb of the four physical factors, really heavy peaking training, is going to be the most stressful to your joints and ligaments, you will need the longest amount of time between those sessions. Properly applied SRA means a well thought out arrangement of training days. Frequency in and of itself does not matter if you are not organizing the days properly. Squatting 3x/week is good and benching 2x/week and deadlifting 1.5x/week, but if the days are organized as squat, squat, squat, bench, deadlift, bench, that is not going to be a good organization and will violate the Principle of SRA. Of course, you are reading this so you are smart, I'm not too worried about you doing that, but make sure that you are spreading out those similar days, particularly squat and deadlift days, probably not putting them back to back, or at least if you are putting them back to back, considering why you're doing that within the context of these principles.

Ways that we tend to violate this principle, an under application of SRA is too much frequency, training hard too often. This is becoming more of an issue with the advent of high volume, high-frequency type programs and those can be great, but too much frequency is not going to be a positive thing. Very high-frequency training could be good for beginner lifters because they are not able to induce very much stress or very much stimulus in each training session, and the longer they are away from the technique of the lift, the more likely it is for that technique to begin to decay. Even if it is very light exposure to the lifts, maybe even on a daily basis, beginner lifters will benefit from very high frequency, but as you become more and more advanced, the frequency is probably going to need to go down.

Another under application of SRA would be the other side of the coin, too low of frequency. Very low-frequency training, like doing lifts only once a week or maybe every other week could lead to diminishing fitness. Even though you are very, very well recovered, you do not have enough stimulus to keep the fitness high between training sessions. There are some very high-level lifters who have been extremely successful with very low-frequency training, maybe squatting every other week, deadlifting every 10 to 14 days but you must consider with that type of training that if an athlete is exceptionally strong and they are squatting 900+ pounds in training, benching well into the 500s and deadlifting in the 800s, they are able to produce such significant stimulus per training session that it may be necessary for them to have an exceptionally long time between hard training sessions to be recovered enough to have the next transition. Also, it's important to consider that the use of Soviet sport supplements can help an athlete hold on to muscle mass, even with very low-frequency training.

An important consideration when determining training frequency and the implications of SRA on this is how to adjust your training when moving into a Peaking Phase. Peaking is unique from Hypertrophy and Strength, in that due to the higher intensity loads, neural fatigue will be very prominent which means that overloading sessions cannot be as frequent BUT since technical prowess is among the primary goals of a peaking phase and Technical SRA is very short compared to Neural SRA, we have a problem to solve. The solution to that problem is the Technical Session. A Technical Session will provide the opportunity for technical practice in the competition lifts performed at a low enough intensity to actually help decay fatigue through improved blood flow.

Typically, reducing your number of overloading sessions by 1 from your Strength Phase and replacing that with a Technical Session is appropriate. So if you had been squatting 3x/week during a Strength Phase, now during Peaking you would have 2 Overloading sessions and 1 Technical Session. The Technical Session should be anywhere from 10-50% lighter than an Overloading session, the bigger, stronger and more advanced an athlete, the greater reduction in intensity is appropriate. As an athlete becomes more advanced and experienced, they are able to get good technical practice from very light weights because they are more in tune with very subtle changes in the lift. If you have an athlete who can squat 225# and has been training for 1 year, it is very unlikely that they would be able to notice small shifts in their bodyweight forward or backward or change in knee position with even 75% of their 1rm (170#). While a very experienced lifter who squats 700# and has been lifting for 15 years should be adept enough to notice technical faults in their own lifting even at 315# (45% of 1rm). All lifters should strive to focus on the details of their lifts at light weights so they can move toward this skill and be able to get more quality work out of less load.

Determining Frequency

In addition to the Individual Differences we must consider when determining the frequency, you must also take into account Time, Special Work Capacity and Phase. Time is the first consideration because while we can design the most scientifically sound program possible, it will never work if the athlete isn't able to adhere to it.

When an athlete's training time is limited to only short training sessions, more frequent sessions, which are less overloading by nature, may be necessary. If the athlete's time is limited in how many days per microcycle they may train, they will need bigger sessions to satisfy overload and train with their optimal frequency (or close to it).

An athlete's Special Work Capacity is their ability to perform more training in a given session, so while we may find that based on the athlete's MRV and Optimal Frequency, they should be performing 8-10 overloading sets in a session, they may lack the Special Work Capacity to complete this much work, so they'll need to spread their Weekly Volume out over more sessions.

Which Phase an athlete is in must be considered when determining how many Overloading sessions they perform in a microcycle. Each Hypertrophy, Strength and Peaking sessions have their own unique SRA Curves and rates at which they will accumulate Fatigue, so they may require different training frequencies, or different organizations of this frequency to ensure that the necessary Overloading parameters are met. This is particularly unique during a Peaking phase where Neural Fatigue is higher due to greater Relative Intensity, this leads to less frequent overloading but that works against one of the primary goals of Peaking, Technical Prowess. To ensure that this is goal is still achieved, it is necessary that we introduce Technical Sessions.

Squat:	**1x/Week**	**2x/Week**	**3x/Week**	**4x/Week**
Bench Press:	**2x/Week**	**3x/Week**	**4x/Week**	**5x/Week**
Deadlift:	**1x/Week**	**2x/Week**		**3x/Week**

	Male		*Female*	
Super	*Heavy*	*Medium*		*Light*
V Tall	*Tall*	*Medium*		*Short*
V High	*High*	*Moderate*		*Low*

V Advanced	*Advanced*	*Intermediate*	*Beginner*	
50s	*40s*	*30s*	*20s*	*<19*
Poor	*Avg*	*Good*		
Poor	*Avg*	*Good*		
High	*Avg*	*Low*		
1	*2*	*3*	*4*	*5*
1	*2*	*3*	*4*	*5*

As you go about determine your training frequency, it is likely that you will be selecting a range, such as 2-3x/week Squatting, 3-4x/week Benching or 1-2x/week Deadlifting, this will likely result in what we will call a Half or Small Training Session. For example, if you have an MRV of 16 sets in the Squat, to be spread out between 2-3 sessions/ week, you may have 2 bigger sessions, one of 7-8 sets, another of 5-6 sets and a 3rd small session of 2-3 sets, this smaller session of 2-3 sets would be our 'Half' Session. You can also manipulate this Frequency and how it impacts Fatigue Accumulation through strategic exercise selection, which will discuss more later.

Here are some example scenarios and how we would solve them.

3 Training Days/Week

MRVs: Squat-15 Sets, Bench-18 Sets, Deadlift 10 Sets

Frequency: Squat-2 Sessions/Week, Bench-3 Sessions/Week, Deadlift 1.5 Sessions/Week

Day 1	Day 2	Day 3	Day 4	Day 5	Day 6	Day 7
Squat 9 Sets	Off	Bench 10 Sets	Off	Deadlift 7 Sets	Off	Off
Bench 5 Sets				Squat 6 Sets		
Deadlift 3 Sets				Bench 3 Sets		

4 Training Days/Week

MRVs: Squat-16 Sets, Bench-20 Sets, Deadlift 11 Sets

Frequency: Squat-2.5 Sessions/Week, Bench-3 Sessions/Week, Deadlift 1.5 Sessions/Week

Day 1	Day 2	Day 3	Day 4	Day 5	Day 6	Day 7
Squat 8 Sets	Off	Bench 6 Sets	Off	Squat 5 Sets	Bench 10 Sets	Off
Bench 4 Sets		Deadlift 7 Sets		Deadlift 4 Sets		
		Squat 3 Sets				

5 Training Days/Week

MRVs: Squat-13 Sets, Bench-17 Sets, Deadlift 8 Sets

Frequency: Squat-2 Sessions/Week, Bench-3 Sessions/Week, Deadlift 1.5 Sessions/Week

Day 1	Day 2	Day 3	Day 4	Day 5	Day 6	Day 7
Squat 8 Sets	Bench 9 Sets	Deadlift 5 Sets	Off	Squat 5 Sets	Bench 6 Sets	Off
Bench 2 Sets				Deadlift 3 Sets		

6 Training Days/Week

MRVs: Squat-20 Sets, Bench-24 Sets, Deadlift 16 Sets

Frequency: Squat-3.5 Sessions/Week, Bench-4 Sessions/Week, Deadlift 2.5 Sessions/Week

Day 1	Day 2	Day 3	Day 4	Day 5	Day 6	Day 7
Squat 8 Sets	Bench 10 Sets	Deadlift 8 Sets	Bench 3 Sets	Squat 6 Sets	Bench 7 Sets	Off
Bench 4 Sets		Squat 4 Sets		Deadlift 5 Sets	Squat 2 Sets	
Deadlift 3 Sets						

There are many potential ways to effectively organize these training weeks. The primary things to keep in mind are:

- Training does not happen in a vacuum, so what you do Day 1 will affect Day 2 and so on.
- Having bigger (more sets) and smaller (less sets) sessions can help manage fatigue throughout the week.
- Exercise selection can also help mitigate fatigue throughout the week. For example, if you are going to Bench on Day 1 and Day 2 with Day 2 as the bigger session, choosing something like Feet Up Bench on Day 1, which does not for as much weight to use but is still overloading, would be a good option.

INDIVIDUALIZING TRAINING LECTURE

PASSWORD: **PDM2018**

CREATING THE INDIVIDUAL SESSION

Exercise Selection
Considerations by Phase

Exercise Selection
Considerations by Weak Point

Spreading Volume
Across Exercise Variations

Exercise Selection

When selecting exercises for a program, one of the foremost considerations should be what phase is being performed. Each phase has a different objective and there are nuanced differences in regards to exercise choices for each one. Here are some things to keep in mind for each phase:

Hypertrophy

Hypertrophic gains are driven by increased volume, so choosing exercises that lend themselves to higher volume being performed are a good choice. Hypertrophy occurs the furthest away from competition and because of the higher reps being done, is inherently less specific than other phases, further implementing the Principle of Variation at this time will help avoid Adaptive Resistance in the athlete. Here are some of my favorite exercises to use to fulfill your MEV/MRV in each lift:

Squat

High Bar Squat

The Low Bar Squat isn't preferable in Hypertrophy because it will fatigue the lower back too much and interfere with deadlift training. The High Bar Squat is typically the foundational lift for the Squat during Hypertrophy.

Front Squat

The Front Squat lends itself to high volumes and is an excellent choice to develop the strength and size of the quads.

Belt Squat

The Belt Squat is a staple in Juggernaut programming and is particularly effective during a Hypertrophy Block.

Leg Press/Hack Squat-Machine

work is often demonized for the strength athlete but during Hypertrophy training, when volume is king, these are a great way to add up volume without too much fatigue.

Unilateral Work

Step Up, Split Squat and Lunge variations are all simple ways to add volume and grow your legs.

Bench Press

There is not a notable difference in the amount of fatigue generated by Competition Bench Press, Closegrip and Widegrip Bench Press so it is not as paramount to adjust exercise selection for the Bench Press, the same way that the differences in fatigue between High and Low Bar Squat dictated that choice. Staying away from your competition technique in the Bench Press during Hypertrophy (or not performing more than 1/3 of your reps this way) is still advisable to help avoid Adaptive Resistance, so to that end, here are some other good options for your overloading work for the Bench Press to fulfill your MEV/MRV.

Widegrip Bench Press

These can be very stressful to the pecs/shoulders so some lifters may not be able to train with them for high volumes but if you can, they are a great exercise option to build Pec strength and size and help athletes who struggle off the chest.

Closegrip Bench Press

Some lifters may struggle with bicep tendonitis when performing high volumes of Closegrip Bench but they are a very useful movement to build muscle in the triceps.

Feet Up Bench Press

This variation will eliminate leg drive and make the athletes pecs and shoulders work harder to move the weight. You also cannot lift as much weight in this variation so it lends itself to more volume being performed.

Incline Bench Press

This is an extra level of variation from the flat competition Bench Press so it will be useful in this phase to avoid Adaptive Resistance over the course of a program.

Military Press Variations

Seated, standing, dumbbell or barbell, during Hypertrophy is the time that the Military Press and its variations will be most useful to a powerlifter 's development.

Dumbbell Press Variations

Flat, Incline, Alternating or Floor Press variations with dumbbells lend themselves to very high reps being performed and are also a good option for Hypertrophy training.

Deadlift

Conventional Deadlifts

Whether you pull Conventional or Sumo in competition, we use the Conventional and its variations almost exclusively during Hypertrophy training. These will pack on muscle but also come at a high fatigue cost, so they likely won't make up the bulk of your MEV/MRV. Training touch-and-go and using straps can be a useful strategy during Hypertrophy as well.

Block Pulls

Pulling with the bar elevated 1-4" will give you many of the same benefits as Deadlifts from the floor but at a slightly lower fatigue cost, so you can likely perform more volume.

RDLs

Make sure to keep your technique very strict so that you can maximize your stimulus with the lowest weight possible. Utilizing the Box Deadlift/Vert Pull pull with the RDL is very helpful.

Good Mornings

Treat these similarly to the RDL. Using a SSB or Cambered Bar is acceptable if your shoulders need them.

Back Raises/Reverse Hypers/GHRs/Hip Thrusts

Choosing 2-3 of these movements to perform each week will help add targeted volume to the muscles that drive heavy deadlifts.

Other Lifts

Training the Lats, as well as direct training for the Shoulders, Triceps and Biceps, should be secondary in your focus during all phases of preparation for Powerlifting. Neither the Squat, Bench or Deadlift are a true tester of Lat strength and though strength in the Lats is useful for success in all of the lifts, they are not the primary mover that many would like you to believe. Doing enough training for the Lats to improve stability, keep your shoulders healthy and create some mechanical advantage in the Bench setup is good but training to have a huge Barbell Row or treating back training as a top priority, will be energy wasted that you could have better spent on training more specific exercises or recovering.

Strength

Strength adaptations are driven by increased intensity, so choosing exercises that lend themselves to greater overload should be prioritized. When an athlete moves from Hypertrophy to Strength, they are getting closer to a meet so there should be an increase in Specificity toward the competition variations, they do not need to be used exclusively but should be a higher priority. For example, if an athlete is going to use the Low Bar Squat in competition, they should do at least 1/3 of their Squats during a strength phase in this manner (this would also be the time to introduce knee wraps if they are going to compete in them). It is not necessary, or even prudent, to do exclusively low bar squatting during this time though. Low Bar squatting is stressful to the arms/shoulders for many lifters, so performing them too frequently will disrupt your bench press training. Low bar squatting is also more fatiguing to the lower back, so training it too frequently can disrupt your deadlift training. Finally, because the Low Bar Squat allows your to move bigger weights, they will be more neurally fatiguing and rotating in High Bar Squats, Pause Squats or Front Squats can help manage that fatigue.

For the Bench Press, as with the Squat, you should now be shifting more of your training toward your competition technique. Because the Bench Press isn't as structurally or neurally fatiguing as any squat variation, you could likely perform it as the primary movement for all your sessions and follow it up with variations to address specific weak areas or you could do rotate your primary movement from session to session with competition technique and variations, either strategy will be effective.

In the Deadlift, if you compete with the Sumo Deadlift, you should introduce it into your program in the Strength Phase. You may not want to include it from the floor every week, as this can be very stressful to the hips, but using it every other week, alternating with either Sumo Block Pulls or Conventional variations is a useful strategy.

Peaking

The Peaking Phase is the most specific phase of meet preparation and should focus on almost exclusively the competition lifts with some accessory work to maintain hypertrophic gains. Some variations of the Squat, Bench, and Deadlift could still be included to address a specific weak point or to help manage fatigue but should make up less than 1/4 of your training. During Peaking could be a time when Mechanical Overload exercises are useful to help improve neural force production and get the lifter more confident under heavy weights but they should be used sparingly.

Directed Adaptation vs Adaptive Resistance

Two important concepts for you to understand in regards to program design for long-term training are directed adaptation and adaptive resistance. These two ideas are, in many ways, two sides of the same coin, and in many ways work against each other, but when you can make them work strategically with each other, that's when you're going to really get your best long-term training results.

So what is directed adaptation? It is a sub-principle of specificity and is looking at the logical sequencing of training towards a specific goal. So whether we are increasing volume for hypertrophic adaptations, increasing intensity for strength adaptations, or increasing specificity for technical adaptations. It must occur in a timely manner, so doing it over the course of several weeks or several months, not once this week, and once a month from now, and another month after that, it has to have some focused time dedicated towards it.

An example of this would be if your goal is to squat heavier for sets of 5, to satisfy directed adaptation we would want to be doing that once or twice a week. Do you want to get better at squatting sets of five? You squat for sets of five. If we were to introduce other goals to that, like maybe squatting for sets of 15, now instead of doing sets of 5 twice a week, you're doing sets of 5 once and sets of 15 later in the week. You could get better at both of those things, but neither of them will get as good as they could have been if all you were doing was directing your focus towards that singular goal of sets of 5. The more goals you introduce to that, maybe 5, 15s, and 1s, all three of them could improve, but they will not improve as much as they potentially could

have if you just focused on one goal. So as we introduce those other goals, we interfere with directed adaptation because the less time you train one of those singular goals, the quicker the qualities are going to decay. So even though you may make some improvement in 5s because you did not have as much dedicated time focused on just training the 5s, your best ability to squat those 5s is going to go away rather quickly.

That brings us to adaptive resistance, the other side of the coin of directed adaptation. To understand adaptive resistance, we must understand the concepts of positive and negative feedback loops. A positive feedback loop is one in which the product of the action produces more of the same. An example of this would be blood clotting. You get a cut, there is a small blood clot that occurs to stop the bleeding there and then more and more and more blood clotting happens to avoid that bleeding continuing. Adaptive resistance happens on a negative feedback loop in which the product of the system means that you get less of what you had before. So when you are hungry and then you eat, you do not get hungrier and hungrier from eating more and more, the way that when you have blood clotting, some blood clotting occurs, more and more occurs because of it. Rather a negative feedback loop means that as you eat, you get less and less hungry and the way that is going to result in training is the more and more you do a specific task, the less and less profound the results are going to become because of it.
As we work to balance directed adaptation versus adaptive resistance, we have directed adaptation that you have to have sequential, strategic, timely direction towards the goal. You have to train the thing that you want to get better at and you have to do it over and over, but as we do it over and

over, the result of it becomes diminished because of adaptive resistance's negative feedback loop. Finding that sweet spot between sufficient directed adaptation before adaptive resistance becomes too much, is a bit of the art of coaching.

How does directed adaptation and adaptive resistance relate to strength training success and success in powerlifting? There are four main categories that are going to be affected by these two concepts. First, the development of technique, next, hypertrophy, as you do more and more volume of work, that volume is going to become less and less impactful on you growing muscle. Your neural strength production is going to benefit from directed adaptation, but then as you get too much of it, you're going to run into adaptive resistance. Finally, your connective tissue, if all you are doing is the same thing over and over and over, everyone is familiar with the concept of an overuse injury, that would be the connective tissue's response to adaptive resistance.

In regards to the development of technique, we have the four stages of motor learning. We have unconscious incompetence, basically, you are bad at it and you don't even know how bad you are, you can't tell what you're doing wrong. Next is conscious incompetence, that you know you have technical flaws and you are thinking about them and sometimes you fix them and sometimes you are not able to. Third, is conscious competence where you are able to do things correctly, but you are really having to think your way through every step. Finally, our goal would be unconscious competence, where your technique has become automatic. It is your default and you can shut your brain off, get under the bar, and just move powerfully, and that is how you are going to be able to move the biggest weights.

Table 1

Weight Class	Weak Legs	Expected Squat/ Deadlift	Weak Back
Light (Male 60kg and Under, Female 52kg and Under)	81%	88%	95%
Medium (Male 67.5-82.5kg, Female 56-67.5kg)	80%	86%	92%
Heavy (Male 90-110kg, Female 75-82.5kg)	87%	95%	103%
Super (Male 125kg and Above, Female 90kg and Above)	94%	104%	114%

Technical learning tends to happen in this slow to rapid to slow process. So a beginner is going to have a hard time improving their technique because they do not have the motor control and the understanding of what they are doing. They are in either that unconscious incompetence or conscious incompetence, but as you become more advanced and make the shift towards conscious competence, their technique is able to improve very quickly because they can feel things that they are doing wrong and make those corrections. Then as the athlete becomes very, very advanced, that technical progress is going to slow down once again, because things are so fine-tuned with such small room for improvement.

An example of this outside of the lifting world would be sprinters. If you are familiar with track & field training, you have seen high-level sprinters train, they do not just sprint. They do not just do the most specific thing possible, but they break down their technique into parts and address it with different drills. Whether it's a-skip, b-skip, c-skip, different fast leg drills, running at different paces, pulling sleds, running hills, tempo runs, they are doing things that allow them to address different aspects and different parts of their technique so they can improve those specific parts.

If they are just sprinting, there is really no way for them, at such a high velocity, to make the technical adjustments that they would need to improve their sprinting technique the same way that lifters are going to need to use different variations to improve aspects of their technique. Those variations could come through different exercises, through different loading strategies, or tempos of the exercises.

Finding that variation sweet spot, how much variation is enough to avoid adaptive resistance while allowing you to keep making technical progress versus how much variation is too much to not allow you to develop the skill of the exercises and satisfy directed adaptation, that is sort of the magic that we are looking for. You have your competition exercise, you have its variations of it. So we have the squat as our competition exercise and then the variations would be front squat, high bar squat, pause squat, changing tempos and changing loading strategies as well. So different set rep schemes and then you have your smaller accessory work. We want to do is introduce variations that are specific enough to have a high transfer to the athlete's abilities and their technique, but they have to be different enough to create some new neural pathways. This is something that I have had to do with Marissa Inda's training and I think something we have really honed in on for her bench press training. She had had very, very good bench press results and then about a month out from the 2016 Nationals, her bench just started to look off and we were doing a lot of benching. I was like, "Why is her technique getting worse from doing the most specific exercise?" and we had definitely run into adaptive resistance.

After a poor bench result there, less than she was capable of doing, I decided no benching for the next six weeks after

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the meet, only bench variations, with the thought behind that being that she knew how to bench, she had good technique, but we were just running into these same neural pathways through adaptive resistance so that introducing more variations was going to allow a re-sensitizing of the technique and we have stuck with that now after every meet. She is having a shorter time of competition bench pressing with this break, four, six, eight weeks after the meet of little to no competition benching to re-sensitize and that has produced consistent improvements in the bench press. Where she had done 82.5kg at that disappointing National meet and then went on after, with this strategy, and did 92.5kg, 95kg, 95.5kg, 97.5kg, 97.5kg and 98kg at the subsequent meets. So it was actually taking away the competition bench press and allowing some neural re-sensitizing time to that, that helped her improve her technique.

So how long should you keep the competition variations in? You want to do the competition variation for 8 to 16 weeks leading into the competition. It does not mean that it has to be the only thing you do, but it should be the primary focus of your training. You will need to be on the longer side of that the less experienced you are. The more experienced you are, the more time you can take away from the competition technique because you are more skilled, you have done it more over the course of your training career, so you will remember how to do it more quickly.

You want to be complementing that with one to three different variations that are going to address different deficiencies that you may have. Those variations could be rotated every 4 to 8 weeks. Maybe you are doing the competition bench press, and then for a month you have close grip in with it, and the next month wide grip, and the next month

Spoto press. If you're just going to do competition bench for eight weeks going into a meet, you could have the same one or two or three variations along with that for the entire eight weeks training cycle. And then the next category of exercises, smaller stuff, whether that's dumbbell flies, front raises, overhead pressing variations, things that are a little bit less specific to the bench press. Those could be changed out every month. You do not want to change them too frequently, like every week because then you'll violate directed adaptation which supersedes adaptive resistance, but to keep your training interesting and to make sure that we're staying healthy, you could change them every four weeks.

Exercise Variation Strategy To Satisfy Directed Adaptation and Avoid Adaptive Resistance

Competition Exercise - Perform for 8 to 16 Weeks

Squat, Bench, Deadlift

Variations Specific to Weakpoint(s)
Choose 1 to 3 Movements and Perform for 4 to 12 Weeks

SSB Squat, Front Squat, Pause Squat. Close/Widegrip Bench, Incline Bench, Spoto Press. Halting Deadlift, Deficit Dead, Block Pull.

Smaller Assistance Movements-Choose 1 to 2 Movements Per Bodypart and Perform for 4 to 8 Weeks

Split Squats, Back Raises. DB Flies, DB Lateral Raises. Reverse Hypers, Dumbbell RDLs.

Adaptive resistance is going to play a role in hypertrophy training. In hypertrophy training as we look to overload volume, whether that is increasing every week or over the course of several mesocycles, that is what is going to drive your adaptations. As you are doing these exercises, not every muscle fiber is going to receive the same amount of stimulus and fatigue from it, some movements may tax the upper pec more rather than the lower pec and that is going to lead some muscles to become fatigued and overreached potentially sooner than others. With muscles being taxed to varying degrees, we can not just do the same exercise and continually load it more and more and more, and effectively grow the way that we are looking to. What we need to do is introduce different exercises at slightly different angles to avoid this issue, and that could be changing the exercise every mesocycle. Or it could be introducing two exercises within the same mesocycle and keeping them for several consecutive mesocycles.

We do not want to be just changing, changing, changing, and not coming back to the same exercise at all, because that's going to violate directed adaptation, which supersedes the importance of adaptive resistance, but you could pair two exercises together like the wide grip bench and incline bench as your primary exercises during a hypertrophy phase, and maybe that lasts 2-4 months. I would not go longer than four months with hypertrophy, you are potentially going to run into some fatigue management issues, some adaptive resistance issues, and you might just get really bored doing that.

Neural strength production is also something where people often make a mistake of pushing too long and running into adaptive resistance problems.

Max Aita, is the classic example of too much directed adaptation towards neural force production. Max did the Bulgarian system or similar training to that with daily maximums, multiple daily maximums in the front squat, snatch, and clean and jerk for about 13 consecutive years. His front squat, right off the bat, had a huge result improving 52kg on an already pretty strong front squat in just 6 weeks. It then took him to the remainder of the year, other 45ish weeks to improve 5kg, and then three years after that to improve another 5kg, and five years after that to improve another 5kg. So that is, at its most extreme level, adaptive resistance to neural force production because that is the only quality they were training by just training the one rep max. What is a one rep max? The most neural force production type of training; it is peaking training, very low volume, very high intensity, and high specificity. I would limit this type of training to at absolute maximum three months of neural force production focused training, but more likely it's going to be one to two months.

You saw Max had this huge result in six weeks, that is going to tend to be a good time and a lot of times you might want to err on a little bit shorter rather than a little bit too long. Most people will be one to two months. The biggest, strongest athletes might be able to do it for three months, but other smaller athletes are going to run into hypertrophic decay if it goes on too long. So while that type of training can have an extremely profound effect, like Max's 52kg PR in the front squat, you do not want to do it too long. It is all about balancing directed adaptation versus adaptive resistance.

"But Chad, can't I just change exercises and avoid this problem of adaptive resistance and just keep pushing my neural strength production week after week after month after month?" Yes and no, you can diminish the adaptive resistance that will occur, but if that's all you are focused on, you are going to run into a couple of problems of directed adaptation and specificity.

Say you are changing exercises week after week and always maxing out, whatever number max it is. If you are not very, very advanced and probably on a lot of performance enhancing drugs as well, you are going to run into hypertrophic decay, which will of course ultimately limit how strong you can become. As you continue to change exercises, you are going to run into that same kind of problem we talked about with trying to squat for 5 rep maxes and 15 rep maxes. You might get better at all of them, but you will never get as good at any of them as you potentially could and since we know what lifts are going to happen in a powerlifting meet, so those are the ones we need to make sure we are getting better at.

Yes, you can limit some adaptive resistance by changing exercises all the time, you might lose some hypertrophy and you are going to run into specificity problems and phase potentiation problems where you are not building one phase on the next to the ultimate highest peak possible. Rather than improving to up the highest level for one day that matters, you are maybe just going to to a lower level for a more extended period of time.

Finally, connective tissue can have a problem with adaptive resistance. If we are always training the same movements and that is occurring along the same force lines, you will run into overuse injuries. If all you ever do are the same movements, you are just going to get more and more beat up, so small changes to the angles of exercise are going to help keep you healthier longer.

How can we satisfy directed adaptation, which is the focused improvement of a specific quality while avoiding adaptive resistance, which is the lessening adaptations that will occur the longer you train for one specific goal. You can keep the competition exercise in for eight to 16 weeks at a time. You want to have one to three variations that compliment different weak points specific to you that are done after that exercise, and those can be rotated every four to eight weeks, and then finally you can have smaller accessory movements that you rotate every month. That will allow you to develop enough skill in the exercises to satisfy directed adaptation without doing them too long where you run into big adaptive resistance problems. In doing that, you want to limit hypertrophy and strength phases to four months at the longest. Peaking phases could be potentially as long as three months where you're really pushing for that neural strength production, but most likely more people will be in the one to two month range with the smaller you are and female athletes on the shorter end of things, and bigger, stronger male athletes, and particularly if performance-enhancing drugs are involved, can push towards the longer end of things.

Weakpoint Guide

One of the key aspects of optimizing a program to be help an individual athlete excel, requires that you strategically select exercises based upon weaknesses that athlete needs to address. Choosing exercise variations that best compliment the competitive exercises by bringing up lagging muscle groups or technical deficiencies in a lifter. Here are my favorite exercises to address various weak points for the Powerlifts.

Squat

Rounding Over

Being rounded over at the upper back in the squat is most commonly due to lack of tightness in the setup. To correct this, refer back to the Technique section of this manual and follow the steps for a rigid upper back position. If you feel you are doing a good job with the technical aspects of the setup but are still being rounded over, then we can look at exercises to address this problem.

Safety Squat Bar Squats

The Safety Squat Bar (SSB) is the most useful specialty bar for the squat and due to its forward facing camber places the weight slightly in front of the athlete's body. This will try to pull the athlete forward and force them to develop strength pushing back against the weight and fighting to stay upright.

Front Squats

Similarly to the SSB, Front Squats place the weight in front of the athlete and will cause the athlete to have to fight to stay upright and build up the postural strength in the upper back. The athlete's ability to overload front squats though is often limited because of mobility restrictions in holding the bar, so I don't find it to be quite as effective as the SSB for this purpose.

General Upper Back Training

Adding muscle and general strength to the Lats, Rhomboids and Traps will all benefit you in not losing your upper back position in the squat. Pullups, Chinups, Lat Pulldowns, DB Rows, Chest Supported Rows, Cable Rows, Upright Rows and Shrugs can all be useful for Hypertrophy and General Strength training for these muscle groups.

Sticking Above Parallel

The most common area for an athlete to miss a squat is a few inches above parallel on the way up. There are many reason that this can occur but the most common reason reason is a relative weakness in either the low back compared to the quads or the quads compared to the low back.

Weak Legs and Strong Back

I find this to be the most common deficiency in an athlete's squat. You can identify this discrepancy in strength when the athlete's knee shift backward as the first movement out of the hole, this causes their hips to rise faster than their shoulders. Because the athlete has a relatively strong back, they'll be able to fight against the weight for a few seconds in this position and occasionally good morning their way through the sticking point. A very heavy conventional deadlift compared to high bar squat would be another indicator that an athlete has this weakpoint. We use the following guidelines to help inform our decisions about Relative Leg vs Back Strength...

Certainly there will be athletes who may not follow these guidelines, usually ones with very short or long arms, but for most athletes this chart should be very useful.

To correct this deficiency, you need to be able to develop the legs, without loading the back. While exercises like High Bar Squats and Front Squats are thought to be great quad builders, and they are, for the athlete whose back tends to take over in their squats, they'll likely perform High Bar Squats or Front Squats with their hips rising faster than their shoulders. When this occurs, their back will be doing more work then their legs (quads) and the imbalance of strength will still exist.

For this reason, High Bar Squats and Front Squats aren't my favorite option for correct this problem, though they can still be used, as long as you are very mindful to maintain an upright torso position throughout the lift.

The most effective way to bring up leg strength relative to the back, will be to train the legs directly with movements that don't involve the back. This is best done with:

Belt Squats
Performed for sets of 10-20 reps following your primary squat training. Make sure that the belt hangs low off the the athlete's hips so that they can keep their torso upright and feel all the pressure in their quads.

Leg Press/Hack Squats
Performed for sets of 10-20 reps following your primary squat training. Adjust your foot position to ensure that you're maximizing pressure in the quads without aggravating the knees.

Unilateral Training
Lunges, Step Ups and Split Squat variations are all appropriate here for sets of 8 to 20 reps per leg.

All of these exercises will develop the legs, in absence of training the back and help you quickly and simply correct this strength imbalance.

From a technical standpoint, it is particularly important for athletes with relatively strong backs, to emphasize keeping their knees forward out of the hole. Because of their strong backs, there is a tendency for the knees to shift back early and allow the back to do more work but in the deeper positions of the squat, the back isn't in a great position to move the weight, so keeping the knees forward will allow the quads to be utilized where they are best suited and then the back to take over later in the movement.

For athletes who fall into the opposite category, weaker low back compared to relatively strong quads, we need to focus on technical cues and extra low back training to improve the squat. To identify an athlete with this issue, you should look for:

- *Relatively heavy squat when compared to conventional deadlift*
- *Knees shifting back out of the hole*
- *Explosive first few inches of ascent and then it's as if the lift hits a wall and it is quickly missed.*

Both the Strong Back and Weak Back groups will have a backward shift of the knees moving out of the hole, so it can be tricky to distinguish which category the athlete falls into. Strong backed athletes' knees shift back because their backs are trying to take over, while weak backed athletes' knees shift back because their back isn't strong enough to hold the angle as they push with their relatively strong legs. Strong backed athletes will usually be able to fight against the weight at their sticking point for a few seconds but weak backed athletes, while explosive out of the hole, will miss the lift quickly when they hit the sticking point. Relative deadlift strength should also be used when identifying leg vs back strength.

Once you've identified an athlete to be in the Weak Back/ Strong Legs group, you can go about fixing this issue by placing more emphasis on their training to these exercises:

Conventional Deadlifts

This is pretty straightforward and is the best low back exercise you can do.

RDLs

As squatting volumes increase, it may be hard to do too much deadlifting from the floor, so RDLs are a good option for low back training with less overall fatigue.

Good Mornings

I find these to be best suited in the 6-15 rep range. Keep your technique strict and try to get maximum effect from minimum weight.

Reverse Hypers/Back Raises

Strict reverse hypers and back raises, either done at 45 degrees or on a GHR, are simply but profoundly effective exercises for the lower back.

From a technical standpoint, it is important for athletes with this weakpoint to make sure they are maximizing the low back strength they do have. This will be best achieved by really focusing on creating pressure with the upper back into the bar. Actively driving the upper back into the bar, as you hit the hole-not after, will help your knees stay forward out of the hole and allow the hips and shoulders to rise together.

Other Exercises To Improve This Area

If you have a balance between back and leg strength, you can consider other exercises to improve your strength in the low end range of the squat. Developing more starting strength and the ability accelerate a dead weight at these joint angles will help you improve through this range of motion. Some of our favorite exercises to improve these qualities are:

Pause Squats - Pauses are a simple and specific way to improve strength in weak positions in the squat. Pauses can be performed at any point during the lift, on the way down, in the hole or on the way up. Spending more time in a weak position will help you improve your strength there and pauses will inherently help you spend more time in those positions. If you are missing squats at 3" above parallel, consider doing pauses at 2" above parallel. Above parallel pause squats can be done by stopping the squat with the pause as the lowest position, or you can squat to full depth and then pause on the way up (or pause on the way down, then squat to full depth after). Pausing for 2-5 seconds is sufficient for any pause squat variation, as pauses longer than that will likely reduce the weight that can be used too much. When doing pauses in the hole, be mindful to keep tension in your legs and actively hold yourself in position, rather than resting your hamstrings on your calves.

Dead Squats

Also know as Anderson Squats, these are performed by setting the bar on pins in a power rack, on blocks or hanging in straps, getting under the weight and standing up from the 'dead' position. These are great for building explosive strength and should only be performed as singles. Here is an example of how to progress Dead Squats in your program...

Week 1-6x1 at 60% w/ 30 sec rest
Week 2-8x1 at 60% w/ 30 sec rest
Week 3-10x1 at 60% w/ 30 sec rest
Week 4-5x1 at 55% w/ 30 sec rest
Week 5-8x1 at 65% w/ 45 sec rest
Week 6-7x1 at 70% w/ 60 sec rest
Week 7-6x1 at 75% w/ 75 sec rest
Week 8-4x1 at 65% w/ 60 sec rest
Week 9-5x1 at 80% w/ 90 sec rest
Week 10-4x1 at 85% w/ 120 sec rest
Week 11-3x1 at 90% w/ Rest As Needed
Week 12-None

Mechanical Overload

The other aspect of squat training that usually needs addressing is the athlete's ability to handle heavy weights, this can be from Neural adaptations or confidence. To enhance either of these, utilizing Mechanical Overload or exercises that allow the lifter to lift supramaximal weights can be useful. It is important to keep in mind with these exercises that they carry with them a high level of neural fatigue, which takes a long time to decay, so they should be used strategically and sparingly. To mitigate the fatigue generated by these exercises while still deriving benefit from them, focus on using 105-125% of your maximum squat, the stronger you are, the lower % will be necessary. A lifter who squats 800# will get plenty of stimulus from a Mechanical Overload exercise with 105% of his max (840#), while a more novice lifter who squat 300# likely wouldn't notice too much of a difference if they only used 315#. For this reason, the more notice lifter could utilize higher %s of overload, up to 20-25%.

Walkout and Hold - This is my most often prescribed form of Mechanical Overload and while causing some neural adaptations to prepare you for heavier weights, it is largely psychological. It is simple to perform, simply load the bar to 105-125% of your max, walk it out of the rack as if you're setting up to squat and hold it for 10-15 seconds. While you are holding the weight, you should be breathing as if you are about to initiate the squat, this may mean that you go through a few rounds of breathing and holding a brace for a few seconds during the exercise.

Reverse Band Squats - Attaching bands to the top of the rack or monolift to provide assistance out of the hole is also referred to as the Future Method. This is a way to go through a full range of motion squat with a weight above what you can actually squat with the goal of preparing the body and mind for heavier loads. Using a band setup that puts you at 90-97% of the total weight in the hole and 105-125% of your maximum at the top is ideal. While it may be tempting to load the bar to 150% of your max and utilize very strong bands to assist you, this strategy will bring your more Instagram likes and fatigue (both neural and physical) than it will actually improved results. Excessive overload in a movement like this will likely have a detrimental effect on other aspects of your training.

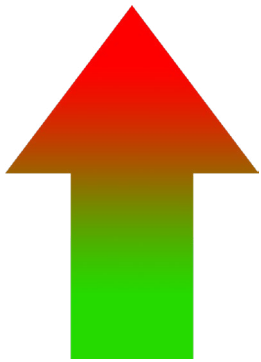
High Pin/High Box Squats - Utilizing these variations allow you to handle more weight because you're reducing the Range of Motion. These can be helpful to drive neural adaptations and improve confidence under heavy weights but you should be mindful of them leading to some technical disruption of your normal squat. Using a pin or box height that has you moving 25-75% of your normal ROM is sufficient.

All of these Mechanical Overload methods are highly stressful to the nervous system and body, so they should be used very sparingly. It isn't uncommon for me to program a single walkout and hold for an athlete in an entire training cycle, at most, I would utilize 1-2 sets of these type of exercises every 4 weeks.

The amount of fatigue generated by different variations is also important to consider when selecting exercises. Here is a look at how they compare relatively to each other:

Reverse Band Squat,
Walkout and Hold,
High Pin/Box Squats
Low Bar Squat w/ Wraps
Low Bar Squat
High Bar Squat
Front Squat
Leg Press, Hack Squat, Belt Squats
Unilateral Variations
(Split Squats, Lunges, Step Ups)

Most Fatiguing



Least Fatiguing

Bench Press

Being able to correctly identify your weakpoints in the bench press and then strategically select exercises to address to fix those problems will make your training more effective and efficient.

Weak Off the Chest

The ability to initiate the lift from the chest is largely influenced by the strength of your pecs, as well as the mechanical advantage you can create through your technique and muscularity.

Paused Benching

If you train mostly touch n go pressing in training and then struggle when you're required to pause at a meet, you're missing the lowest hanging fruit of improving your bench, more pausing. The paused bench press is the competition movement and should be your top training priority. A simple rule of thumb is to pause the first rep of all your sets, but if you really struggle with pausing, then pause all your reps. Training a slightly longer pause than what will be required in competition can be useful. Pauses from 2-5 seconds are appropriate for training.

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If you train mostly touch n go pressing in training and then struggle when you're required to pause at a meet, you're missing the lowest hanging fruit of improving your bench, more pausing. The paused bench press is the competition movement and should be your top training priority. A simple rule of thumb is to pause the first rep of all your sets, but if you really struggle with pausing, then pause all your reps. Training a slightly longer pause than what will be required in competition can be useful. Pauses from 2-5 seconds are appropriate for training.

Widegrip Benching

Using a wider grip will be more stressful on the pecs. Using a grip 2-3" wider than your competition grip is appropriate for widegrip training. If your competition grip is already at the maximum allowable width, then using widegrip in your training likely isn't useful. While Widegrip is great for building pec strength, it can be stressful to the shoulders, so be mindful of your scapular control and volume.

Extended ROM Benching

Benching through a larger ROM by using a Cambered Bar or Dumbbells is a useful technique to improve bottom end strength in the bench press. It is only necessary to extend the ROM .5-1.5" for a good carryover in the bench but this can be very stressful to the shoulders, so be mindful when including this in your training. It may be easier on the shoulders to use Dumbbells for a deep stretch when pressing, particularly with your palms in.

General Hypertrophy Training

Building size in the pecs and upper back will create a more mechanically advantageous position for you to press from. In general, the bench press will respond most to hypertrophy training compared to the squat and deadlift, this is why you will find more 'gym bros' with big benches, than you will with huge squats or deadlifts.

Explosive Pushup Variations

Explosive Pushup Variations are the Upper Body equivalent of jumps and are a great developer of Rate of Force Development. Depending on your fitness level and phase of training pushup variations could include Clapping Pushups, Pushups Onto Boxes, Drop Pushups or Rebound Pushups. Band assistance for athletes of lower preparation levels can be appropriate too.

Midrange Strength

The midrange of the bench press, generally 2-8" off the chest depending on setup and arm length, is most attributed to the strength of your shoulders. Exercises to improve strength here include:

Incline Bench Press

Incline Bench Press is one of the best overloading exercises for the upper pecs and shoulders. It is a great choice during Hypertrophy and Strength Phases but should not be included in Peaking Phases due to the technical disruption they may create. Both Barbell and Dumbbell variations can be useful but Barbell will allow for greater overloading stimulus, though Dumbbells may be less stressful to the shoulder joints.

Military Press

The Military Press is an excellent developer of shoulder and tricep strength but because of the direction of the press it will be slightly less specific to the bench press. It is a great choice for Hypertrophy and Strength Phases but do not allow yourself to become overly concerned with your maximum in the Military/Overhead Press, rather treat is as a builder of your bench press. This building will best occur in the 5-12 rep range, training it for lower reps at a higher RPE may be detract energy to train and recover from the bench press training and should be avoided close to meets for athlete's of higher qualification.

Spoto Press

The Spoto Press, popularized by the phenomenal bench presser Eric Spoto, is a bench rep where you pause the bar slightly off our your chest, never allowing the bar to rest on your chest. This variation will build strength in the shoulders, stability in your technique and improved control of the bar. Make sure that you are lowering the bar quickly to challenge yourself in controlling the bar during the pause.

Dead Bench

The Dead Bench is an exercise shown to me by Josh Bryant. This is performed by placing the bar 1-2" off the chest sitting on pins in a rack. You will press up from there and this will build greater starting strength because it eliminates the stretch reflex. This is a challenging variation and definitely helped build my bench when I included it. Here is an example of how to progress Dead Bench in your program...

Week 1-8x1 at 60% w/ 30 sec rest
Week 2-10x1 at 60% w/ 30 sec rest
Week 3-12x1 at 60% w/ 30 sec rest
Week 4-6x1 at 55% w/ 30 sec rest
Week 5-10x1 at 65% w/ 45 sec rest
Week 6-9x1 at 70% w/ 60 sec rest
Week 7-8x1 at 75% w/ 75 sec rest
Week 8-5x1 at 65% w/ 60 sec rest
Week 9-5x1 at 80% w/ 90 sec rest
Week 10-4x1 at 85% w/ 120 sec rest
Week 11-3x1 at 90% w/ Rest As Needed
Week 12-None

Front/Lateral Raises

Directly training the delts through front and lateral raises are a great option during Hypertrophy Phases to add muscle mass and can be useful to maintain muscle mass during lower volume strength and peaking phases. Whether performed with Dumbbells, Barbells or Plates, these exercises don't lend themselves well to maximal loading and should be train primarily in the 8-20 rep range.

Lockout Strength

The lockout of the bench press is largely driven by the strength of the triceps. One of geared powerlifting's greatest contributions to training ideas is their numerous exercises to improve the bench press lockout, since shirted bench pressing is so reliant upon tricep strength.

Closegrip Bench

This is the gold standard of tricep development, you can go heavy and it is highly specific to the bench press. Some people may suffer from shoulder pain from closegrip pressing because it will cause them to lose some scapular retraction when the bar is on their chest, so adding 1-2 boards can be appropriate.

Skullcrushers

Whether performed with a straight bar, EZ curl bar or dumbbells, the skullcrusher is a great option for overloading the triceps for growth and strength. Elbow pain may often accompany hard training on this exercise, so feel free to adjust hand position and arm angle by feel.

Dips

Dips are often a love/hate exercise for those looking to bench big. Those who love them find great strength development and growth through the shoulders and triceps by performing them. Those who hate them will be plagued with elbow, shoulder and/or sternum pain from just a few reps. If Dips feel good for you, then by all means, go for it and train them hard in the 5-15 rep range.

Other Tricep Work

Training the triceps becomes largely individualized based on how the lifter's elbows tolerate the training. There aren't necessarily good or bad exercises, but rather exercises that work for you. Pushdowns, JM Press, Tate Press, Dicks Press and many others can all be useful, so hit up YouTube and find some new variations that can work for you.

Mechanical Overload

The other aspect of bench training that usually needs addressing is the athlete's ability to handle heavy weights, this can be from Neural adaptations or confidence. To enhance either of these, utilizing Mechanical Overload or exercises that allow the lifter to lift supramaximal weights can be useful. It is important to keep in mind with these exercises that they carry with them a high level of neural fatigue, which takes a long time to decay, so they should be used strategically and sparingly. To mitigate the fatigue generated by these exercises while still deriving benefit from them, focus on using 105-125% of your maximum bench, the stronger you are, the lower % will be necessary. A lifter who benches 500# will get plenty of stimulus from a Mechanical Overload exercise with 110% of his max (525#), while a more novice lifter who benches 250# likely wouldn't notice too much of a difference if they only used 265#, so a 10-15% overload may be more appropriate. Female lifters, who are benching <200# will definitely need to utilized overloads of 20%+ to drive neural adaptation. For this reason, the more novice lifter could utilize higher %s of overload, up to 20-25%.

Slingshot

The Slingshot can be a useful tool for many aspects of training the Bench Press. First and foremost, it is designed as a mechanical overload tool and can help you achieve the 5-25% overload we referred to above. Be mindful that some people will experience technical disruption from using the Slingshot, so it may not be for everybody and you need to ensure that it is sized properly. The primary way that I prescribe the Slingshot for Mechanical Overload is to have the athlete add an extra set or two after their heaviest raw work. For example, during a training session that you work up to 315x3 raw, you could then add then Slingshot for 2x3 at 335. Utilizing this type of mechnical overload carries with it greater fatigue, so it likely should not be used every week. Every 2-4 weeks would be appropriate for mechanical overload training.

The Slingshot can also be a useful tool for injury work-arounds. If you are experiencing elbow or shoulder pain, using the Slingshot to reduce stress in the more vulnerable positions of the bench can be helpful. Just add it to your regularly planned bench work if your pain is light, while using it with reduced weight could be needed if pain is more significant.

Reverse Band Bench Pressing

Similarly to the Slingshot, Reverse Band Presses will assist you off the chest and help you handle a heavier weight. The type of band you utilize will vary widely by your strength and rack setup but the same 105-125% overload rule should be applied. Overloading beyond this point may be cool for an Instagram video but will likely generate more fatigue, elbow pain, shoulder pain and risk than its worth.

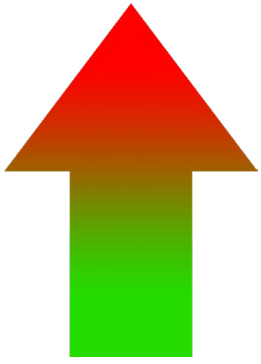
High Board/Rack Lockouts

Benching with a significantly reduced ROM like a 4-5 Board Press or presses from High Pins could be a potential over-load tool but in my opinion can cause too much technical disruption to be a viable option.

Here is a look at the relative fatigue generated by different Bench Press variations.

Reverse Band Bench,
Slingshot, High Pin Press,
Negatives Bench, Closegrip Bench,
Widegrip Bench, Standing Overhead
Incline Bench, Spoto Press,
Floor Press, Seated Overhead
Dumbbell Press Variations
Machine Pressing Variations
Single Joint Exercises
(Skullcrushers, Shoulder Raises, etc)

Most Fatiguing



Least Fatiguing

Deadlift

Being able to correctly identify your weakpoints in the deadlift and then strategically select exercises to address to fix those problems will make your training more effective and efficient.

Weak From the Floor

Weakness of the floor in the deadlift can be attributed to a lack of tension throughout the body or inability to effectively push with the legs from this position. To address this issue from a technical standpoint, better breathing and bracing is key. From a strength standpoint, stronger quads and lats will be beneficial. The Deadlift is unique in that, it doesn't have an eccentric phase in competition, so there isn't a stretch reflex to begin moving the bar upward, for this reason, including explosive work can be beneficial.

Deficit Deads

Deficit Deads are usually the most popular movement when discussing strength from the floor. They can be very useful but need to be treated strategically, as a bigger deficit, doesn't necessarily mean a better deficit. When the deficit becomes too large, it will alter the position too much for high transfer to the competitive exercise. For most lifters, .5-1.5" should be a sufficient deficit, if you are very tall/thin or very mobile, up to a 3" deficit could become useful. Sumo pullers can also use deficits and will need to err on the lower side of things because it will be a very stressful position for the hips.

Low Pause Deadlifts

The Low Pause Deadlift is a very challenging movement that will build great strength and stability in the starting position of the movement. When performing a Low Pause Deadlift, you want to only pull the bar barely off the ground, no more than 1", pause for a 2-3 count and then explosively finish the lift. These will be useful for the Conventional and Sumo puller alike.

Halting Deadlift
1.
2.
3.
4.

Halting Deadlift

The simplest way to improve at something is to do it more, so if you are struggling at the start of a deadlift, The Halting Deadlift will give you twice as much practice in this part of the movement. In the Halting Deadlift, pull the bar to the bottom of the knees, lower it back to the ground and then perform a full repetition. Both styles of Deadlift will benefit from Halting Deadlifts.

Coan Style Deadlift

Ed Coan, one of the greatest deadlifters of all-time, performed his reps of deadlift in a very challenging, very beneficial manner. Pull your deadlift up, as you typically would and then lower the bar under control, setting it down as if you're trying to stay silent, let the tension out of the bar, but keep the tension throughout your body and pull again. Coan Style Deadlifts will be great for sumo or conventional pullers.

High Bar/Front/SSB Squats

Improved quad strength will help you generate greater leg drive to start the lift and all of these squatting variations will help build that. Front and SSB Squats will also have the added benefit of challenging your upper back strength. Adding a pause to these squats can also be helpful when trying to improve the start of the Deadlift.

Box Squats

While I don't find the Box Squat to be useful in building raw squatting strength, it can be a useful tool for the Sumo Deadlift to improve hip strength, as well as hip and groin flexibility. Use a stance approximating your deadlift start position and a parallel, or slightly above box.

Weighted Jumps

Improving Rate of Force Development can benefit a lifter who is slow from the floor. Jumping while holding weights, either on a trap bar, dumbbells or kettlebells in each hand, is a simple but effective way to improve your explosive strength. Keep the quality high and volume low. Three to 10 sets of 1 to 3 reps should be sufficient.

Overhead Backwards MB Throw

Starting the ball on the ground can add a slight degree of extra benefit to this movement which will teach you to quickly generate force through the ground while also demanding that you improve your hip hinging pattern so you can effectively load the hips and glute for a throw of maximum distance.

Pendlay Rows

A lack of lat tension can cause you to struggle off the floor and lose your position, dooming the lift when it has barely begun. Pendlay Rows will help teach you how to quickly generate tension in your lats. Maximize the effectiveness of this movement by demonstrating control in the eccentric before resetting the weight on the floor, similar to a Coan Style Deadlift.

Weak Around the Knees

Weakness around the knees, either below or above, in the deadlift is most often caused by a lack of hamstring strength and/or a lack of lat tension/strength which allows the bar to drift away from your body, making the lift more challenging than it should be.

RDL/Stiff Leg/Box Deadlift

The RDL or Stiff Leg Deadlift (I consider these to be the same exercise, don't actually do Stiff Leg Deadlifts with your legs locked straight) is the best option for building highly transferable hamstring strength for the Deadlift. These can be done with tremendous overload and mimic the positions of the deadlift but with higher engagement of the hamstrings because of the high hip position.

The Box Deadlift, is an RDL variation first shown to me by Boris Sheiko and Mikhail Koklyaev at a seminar I hosted with them in Summer 2015 and since then I have implemented into my own and my athletes' training with great success. The Box Deadlift, especially when done with a Vert Pull device, allows the athlete to maintain very strict RDL technique with totally vertical shin throughout the movement or possibly even a negative shin angle for greater hamstring engagement.

Pause Deadlifts

If you want to get stronger in a specific position, spend more time in that position. Paused work, for any of the lifts, will inherently cause you to spend more time in a position that you may be weak in. If you are struggling in the deadlift at a given point, performing a 2-3 count pause, just below that position is a simple and effective fix. It is important to pause .5-1.5" below the point where you are actually

missing the lift, because that is the area where the bar is slowing down before it actually stops at the 'sticking point'. When performing Pause Deadlifts, it is common that athletes don't come to a complete stop but rather perform a 'California roll' through their pause point or just pull the bar slowly from the floor to the pause and then accelerate from there. To maximize the effectiveness of Pause Deadlifts, you need pull with maximal force to the pause, have a distinct time of isometric contraction while you hold the weight in place and then accelerate the bar to lockout with maximal force.

Touch and Go Deadlifts

There is often debate among coaches and lifters about performing Deadlifts from a 'dead stop' or touch and go. While I think that the Dead Stop starts are the most valuable for building technique and strength off the floor in the Deadlift, Touch n Go reps can be very useful to build up your hamstring strength. Basically everybody can go heavier in the Touch n Go than they can from a dead stop, so if you're doing a back down set or straight across sets or are in a higher volume phase, then going a bit heavier and performing Touch n Go reps is a great way to overload the hamstrings. Adding straps may be appropriate here.

Bent Over Rows

General Lat Strength is important to ensure that the bar stays close to your body throughout the Deadlift and while many exercises like pull-ups and cable rows can help build this, Bent Over Rows, particularly with a double overhand grip, will help strengthen the midrange position of the dead-lift best. These will be a heavy and taxing exercise, some momentum generate from the body is acceptable but keep it within reason. Try to approximate your torso angle at the midrange of a deadlift when setting up for this exercise. Wearing straps for this movement is probably worthwhile.

Weak At Lockout

The Deadlift Lockout is driven by glute and low back strength, as well as the upper back. Oftentimes a poor lockout can be related to problems in the starting position or a loss of hip position throughout the lift but for this section we will assume that your technique is stabilized (if not refer back to the Technique section) and that you just need to improve strength in specific areas.

Block Pulls

Elevating the bar onto Blocks, typically 2-4" high, can be a useful way to strengthen the lockout, though it seems to work differently for Conventional vs Sumo Deadlifters. For the Conventional Deadlift, Block Pulls are often weaker then pulls from the floor. This is due to the lifter's inability to generate leg drive with the bar in the elevated position, causing the back and hips to have to work harder to move the weight, overloading these areas will help strengthen the lockout for the conventional puller.

For the Sumo Deadlifter, Block Pulls are almost always stronger than pulls from the floor but training them for maximum weights in low rep ranges is so neurally taxing that you can't perform enough volume to build the hypertrophy and general strength needed for carryover to an improve lockout. If you are a Sumo Puller, looking to use Block Pulls to improve your Lockout, I'd suggest training them mostly in the 4-8 rep range for multiple sets and reserving any maximal Block Pulls in the 1-3 rep range for a Mechanical Overload tool, which we will discuss more in a bit.

BB Hip Thrusts/Glute Bridges

The Glutes are a very powerful hip extensor and can be critical to a strong lockout, as well as maintaining low back health. Both the Barbell Hip Thrust and Barbell Glute Bridge can be useful exercises to build Hypertrophy and Strength. These are not exercises to test your maximum on, but rather use them in the 5-12 rep range to build general strength and hypertrophy, ensuring that you are achieving full HIP extension (no Thoracic or Lumbar extension) on every rep. Try to get more out of less weight in these exercises by being strict in your technique and ensuring a strong contraction at the top of every rep.

Good Mornings

The Good Morning is a great exercise for building strength throughout the posterior chain, particularly in the low back. Many conventional deadlifters, particularly those with larger midsections, will tend to have their hips lockout too early, causing their low back to have to finish the majority of the work when locking out a deadlift. The Good Morning is an excellent choice to build up low back stability and strength. As with the BB Hip Thrust, I'd suggest training the Good Morning for moderate reps in the 5-12 rep range, focusing on strict technique and maximizing the effectiveness of the exercise with as light of weight as you can. If you struggle with the proper hip hinging pattern for the Good Morning, using a setup similar to a Box Deadlift/Vert Pull can be helpful. Anchoring a band around your waist that resists your hips can also be a way to improve hip extensor strength in this movement.

Back Raises

Back Raises or Back Hypers (Hyperextensions) are a very simple but effective strength builder for the low back. The technique is simple and unlikely to be screwed up and they can be loaded either with a bar across the back or by holding weights across the chest. When performing back raises, ensure that you stay in a neutral rib position, not allowing your ribs to rise and flair, this will help ensure that your low back and glutes are working effectively. These are a great exercise to train for sets of 8-20 reps and adding isometric holds at the top of the movement and controlled eccentrics are options to increase training effect without adding load.

Reverse Hypers

Similarly to the Back Raise, the Reverse Hyper is a very effective exercise to build up low back and glute strength. I do not suggest performing them in a 'swinging' manner, as this can lead to spinal flexion under load which can be injurious, rather, perform your Reverse Hypers with strict control for maximum effect. These are a great exercise to train for sets of 8-20 reps and adding isometric holds at the top of the movement and controlled eccentrics are options to increase training effect without adding load.

Shrugs

The final piece of the Deadlift Lockout is the ability to push the chest through to stand tall with the shoulders back and a strong upper back from shrugs at various angles can help ensure you aren't getting stuck with rounded shoulders. Training heavy Barbell Shrugs and Dumbbell Shrugs with an upright or slightly leaned over posture will build strength in the upper and mid traps to ensure you can finish a heavy pull.

Mechanical Overload

Mechanical Overload for the Deadlift isn't as significant an asset in training as it is for the Squat and Bench because 1-heavy deadlifts already generate such great neural fatigue and 2-the deadlift doesn't carry the same psychological intimidation effect of feeling heavy as the squat or bench because it doesn't have an eccentric phase preceding its concentric phase. For these reasons, I don't advocate for much Mechanical Overload training in the Deadlift but if you feel it is useful for you, here are my suggestions.

Block Pulls

For the conventional deadlifer, the Block Pull isn't likely to be a mechanical overload tool but if you are able to pull more from blocks than you are from the floor, it can be used to drive Neural adaptations with 105-125% of your maximum. Sumo Deadlifts can almost definitely pull more with the bar elevated 2-4" than they can from the floor and can use this as a Mechanical Overload tool for a max set of 1-3 reps. Keep in mind though that if you are pulling 105-125% of your best from from the floor, from blocks, that there will be tremendous neural fatigue generated. If you use Block Pulls as part of your peaking block, they should be done far enough away from competition to allow the neural fatigue to decay and the stronger you are, the further from the meet that needs to be.

Reverse Band Deadlifts

Reverse Band Deadlifts can be used to help an athlete accommodate to a heavier load or build a stronger lockout. Utilize Bands from the safety rails or top of a rack so that you're able to lift 105-125% of your maximum, the stronger you are the lower in that percentage range will be appropriate. As with any Mechanical Overload strategy, Reverse Band Deadlifts carry with them a substantial amount of Neural Fatigue that must be properly managed heading into a competition.

Grip Training

For the competitive powerlifter, your Deadlift can only be as strong as your grip allows it to be and there is nothing more frustrating than dropping a heavy pull at lockout, so lets ensure that doesn't happen.

Holding Your Last Rep

The simplest strategy to improve your grip strength for the Deadlift, doesn't require any extra exercises, just simply hold the last rep of each set at lockout for 10-15 seconds.

Timed Holds

Set the bar on the safeties in a power rack so that you don't have to lift it very far and hold for 30 seconds, then rest for 90 seconds, repeat this 3x. Begin with 50% of your max and build up as you can.

Double Overhand Deadlifts

Another simple strategy to add some extra grip work in, though I don't feel it as effective/specific as the two previous strategies, is to begin your deadlift training with a double overhand grip and only switching to over/under or a hook grip once you can no longer hold on.

Farmers Walks

Farmers Walks are a fun way to build up some grip strength as well as improve trunk stability and GPP. You don't need to train these for maximum weight if you aren't preparing for a strongman contest but challenging yourself for 15-50yd walks can be beneficial in a general training phase.

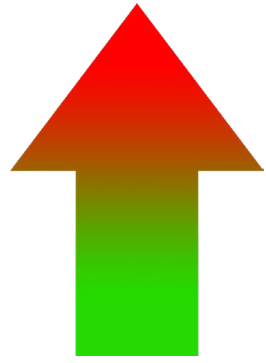
Other Grip Training Ideas

There are lots of grip training tools available such as Rolling Thunder, Pinch Grip Blocks, Grenade Grips and Grippers but you must remember that the ability to hold onto a heavy deadlift is dictated by your Supporting Grip, so while closing a #3 Captains of Crush or picking up a heavy Blob may mean you have a strong crushing or pinch grip, it won't necessarily translate to holding onto a heavy deadlift. These tools are useful for general grip strength through and chances are unless you are a VERY strong deadlifter, that if you have good general grip strength, you will be able to hold onto your pulls.

Here is a look at the relative fatigue generated by different Deadlift variations.

Reverse Band Deadlift,
Deficit Pulls,
Sumo Block Pulls
(If done for Mechanical Overload)
Conventional Deadlift
Conventional Block Pulls
(Assuming You Can Do More from Floor)
Sumo Deadlift, RDLs,
Good Mornings
Back Raises, Reverse Hypers, GHRs

Most Fatiguing



Least Fatiguing

Now that we know how to organize your Volume Landmarks across your optimal Frequency and understand what exercises are most useful for what purposes, we can decide how to actually select which exercises to fill up that volume with. Here are some example scenarios:

4 Training Days/Week

MRVs: Squat-16 Sets, Bench-20 Sets, Deadlift 11 Sets

Frequency: Squat-2.5 Sessions/Week, Bench-3 Sessions/Week, Deadlift 1.5 Sessions/Week

Weaknesses: Squat-Weak Legs/Strong Back, Bench-Weak Off Chest, Deadlift-Weak Off Floor

Day 1	Day 2	Day 3	Day 4	Day 5	Day 6	Day 7
Low Bar Squat 4 Sets	Off	Bench Press 3 Sets	Off	High Bar Squat 5 Sets	Bench Press 7 Sets	Off
Front Squat 4 Sets		Widegrip Bench 3 Sets		Deficit Deadlift 4 Sets	DB Bench w/ Deep Stretch 3 Sets	
Feet Up Bench 4 Sets		Deadlift 4 Sets				
		Pendlay Rows 3 Sets				
		Belt Squat 3 Sets				

5 Training Days/Week

MRVs: Squat-13 Sets, Bench-17 Sets, Deadlift 8 Sets

Frequency: Squat-2 Sessions/Week, Bench-3 Sessions/Week, Deadlift 1.5 Sessions/Week

Weaknesses: Squat-Rounding Over, Bench-Midrange, Deadlift-Lockout

Day 1	Day 2	Day 3	Day 4	Day 5	Day 6	Day 7
Low Bar Squat 5 Sets	Bench 6 Sets	Deadlift 5 Sets	Off	Low Bar Squat 3 Sets	Bench 3 Sets	Off
SSB Squat 3 Sets	Seated Military Press 3 Sets			Front Squat 2 Sets	Incline Bench 3 Sets	
Spoto Press 2 Sets				3" Block Pull 3 Sets		

6 Training Days/Week

MRVs: Squat-20 Sets, Bench-24 Sets, Deadlift 16 Sets

Frequency: Squat-3.5 Sessions/Week, Bench-4 Sessions/Week, Deadlift 2.5 Sessions/Week

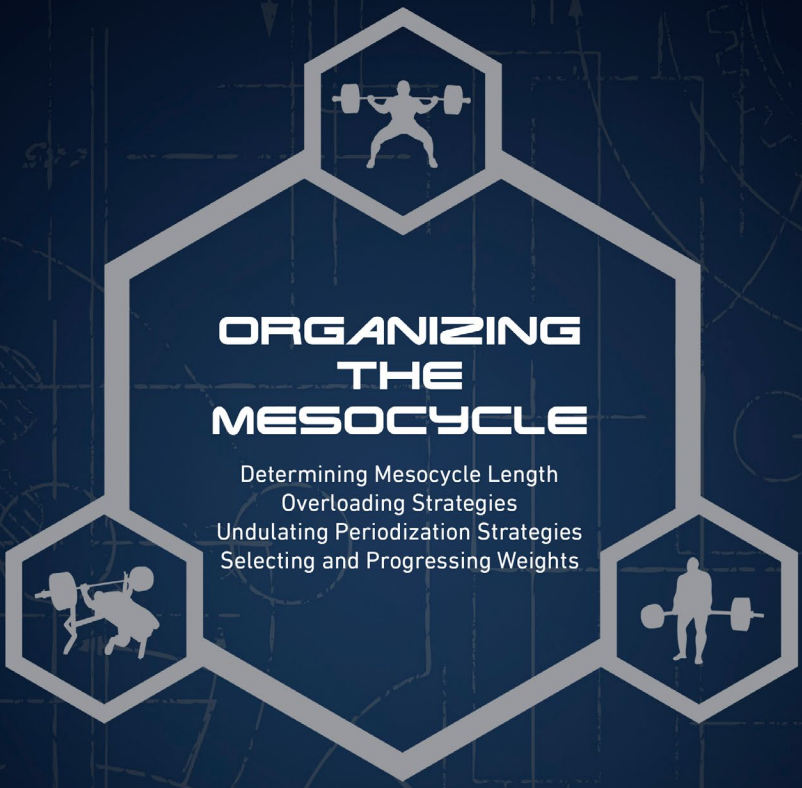
Weaknesses: Squat-Strong Legs/Weak Back, Bench-Lockout, Deadlift-Midrange

Day 1	Day 2	Day 3	Day 4	Day 5	Day 6	Day 7
Low Bar Squat 5 Sets	Bench 6 Sets	Deadlift 5 Sets	Dips 3 Sets	Low Bar Squat 3 Sets	Bench 4 Sets	Off
RDLs 3 Sets	Closegrip Bench 4 Sets	Pause Deadlift 3 Sets		Good Mornings 3 Sets	Closegrip Bench 3 Sets	
Skullcrushers 4 Sets		Back Raises 4 Sets		RDLs 5 Sets	Back Raises 2 Sets	
Bentover Rows 4 Sets						

There are many potential ways to effectively select exercises to satisfy your volume landmarks. The primary things to keep in mind are:

- *An athlete's special work capacity, may necessitate them splitting their volume landmarks between more movements. For example, hey may fatigue too much to continue Closegrip Benching and need to finish their volume with Skullcrushers or Tricep Pushdowns.*

- *Not all volume is created equal, if your MRV is 16 Sets of Squats, 16 sets of Low Bar Squats and 4 Sets of Squats, 4 Sets of Leg Presses, 4 Sets of Good Mornings and 4 Sets of Lunges is not equal, so if you are selecting more exercises that are less fatiguing, keep in mind that you will likely need to do a few more sets total.*



ORGANIZING THE MESOCYCLE

Determining Mesocycle Length
Overloading Strategies
Undulating Periodization Strategies
Selecting and Progressing Weights

Now we understand how much work you need to do, how to organize that work throughout the training week and how to select exercises to satisfy that volume through your optimal frequency, so we are ready to look at longer term strategies to progress the athlete.

Determining Mescocycle Length

Mesocycle length is an important consideration as it relates to an athlete's ability to accumulate fatigue. Too long of a mesocycle may lead to too much fatigue being accumulated and non-functional overreaching to occur. Too short of a mesocycle can lead to an athlete to waste time deloading when it is not necessary. Some people would make an argument to deload only when you feel it become necessary but I find two potential issues with this. First, if you are preparing for a meet, you have a set amount of time to train and the proportion of Hypertrophy, Strength and Peaking you employ is important, so separating those phases with planned deloads is best practice. Second, if a deload is

The relationship between MEV and MRV is a very useful tool to use when assessing Mesocycle Length, if we use our example athlete from the Establishing MEV/MRV section, let's look at their MEVs and MRVs for Hypertrophy:

Lift	MEV	MRV	Difference
Squat	7.5 Sets/Week	21 Sets/Week	13.5 Sets
Bench	9 Sets/Week	24 Sets/Week	15 Sets
Deadlift	5.5 Sets/Week	18 Sets/Week	12.5 Sets

In an case such as this, with a substantial spread between MEV and MRV, it is likely that using 5 or 6 weeks of Overloading Training before a planned deload is prudent. We would refer to this as a 5+1 or 6+1 paradigm.

A male lifter who is bigger and stronger is likely to have a closer relationship between MEV and MRV, something like the following...

Lift	MEV	MRV	Difference
Squat	11 Sets/Week	16 Sets/Week	5 Sets
Bench	13 Sets/Week	19 Sets/Week	6 Sets
Deadlift	7.5 Sets/Week	12 Sets/Week	4.5 Sets

When the spread between MEV/MRV is smaller, shorter mesocycles will be necessary and training on a 3+1 or 4+1 paradigm is likely. Typically, 2 to 6 weeks of overloading is used followed by a 1 week deload. When there is a wide gap between MEV/MRV, the athlete can make significant jumps from week to week either in volume or intensity in a linear fashion. As an athlete gains experience and strength, their MEV and MRV will trend closer together and more strategic fatigue management ideas may be necessary. The section on Undulating Periodization Strategies will come in handy for those athletes.

Overloading Strategies

Now that we have decided how long a Mesocycle should be, we can determine how we should satisfy the Principle of Overload over the course of those Mesocycles. The first method we will look at is the simplest, a Linear Progression.

Linear Progression of Overloading will be extremely effective for Beginner lifter, many intermediates and even potentially some Advanced lifters with exceptional recovery ability. In Hypertrophy, we will look to add sets each week, while also making small increases in intensity. In Strength Phases, we will progress intensity on a weekly basis while making small reductions in Volume weekly.

During Peaking, we may not be able to increase intensity weekly because of the neural fatigue that will accompany the higher loads, but increasing weight on a near weekly basis along with strategic reductions in volume to manage fatigue will be necessary.

Here is a look at one way we could apply Overloading in Hypertrophy for an athlete with the following volume landmarks.

Lift	MEV	MRV	Difference
Squat	8 Sets/Week	18 Sets/Week	10 Sets
Bench	12 Sets/Week	22 Sets/Week	10 Sets
Deadlift	7 Sets/Week	15 Sets/Week	8 Sets

Lift	Week 1	Week 2	Week 3	Week 4
Squat 1	Up to 8rm at 7RPE Drop 8-12%x4x8	Up to 8rm at 8RPE Drop 8-12%x5x8	Up to 8rm at 9RPE Drop 8-12%x6x8	Up to 8rm at 10RPE Drop 8-12%x7x8
Squat 2	3x8 at 85% of 8rm	5x8 at 85% of 8rm	7x8 at 85% of 8rm	8x8 at 85% of 8rm
Total # of Overloading Sets of Squat	8 Sets	11 Sets	14 Sets	17 Sets
Bench 1	Up to 8rm at 7RPE Drop 8-12%x5x8	Up to 8rm at 8RPE Drop 8-12%x6x8	Up to 8rm at 9RPE Drop 8-12%x7x8	Up to 8rm at 10RPE Drop 8-12%x8x8
Bench 2	2x8 at 85% of 8rm	3x8 at 85% of 8rm	4x8 at 85% of 8rm	5x8 at 85% of 8rm
Bench 3	4x8 at 85% of 8rm	5x8 at 85% of 8rm	6x8 at 85% of 8rm	7x8 at 85% of 8rm
Total # of Overloading Sets of Bench	12 Sets	15 Sets	18 Sets	21 Sets
Deadlift 1	Up to 6rm at 7RPE Drop 8-12%x3x6	Up to 6rm at 8RPE Drop 8-12%x4x6	Up to 6rm at 9RPE Drop 8-12%x5x6	Up to 6rm at 10RPE Drop 8-12%x6x6
Deadlift 2	3x6 at 85% of 6rm	4x6 at 85% of 6rm	5x6 at 85% of 6rm	6x6 at 85% of 6rm
Total # of Overloading Sets of Deadlift	7 Sets	9 Sets	11 Sets	13 Sets

You can see in this progression, the lifter starts their training at their MEV and progresses up towards their MRV over the course of the mesocycle. This mesocycle, for an athlete who has a relatively large gap between their MEV/MRV you can be fairly aggressive in adding volume each week but depending on the amount of time the athlete has to prepare for competition, you could strategize this differently. For example, if you wanted to prescribe 2 subsequent Hypertrophy Cycles, during the first one you may want to begin at MEV and then increase volume up to 80% of the lifter's MRV.

Then in the second cycle, you can stay at the MEV + 2 or 3 sets and then ramp up to 100% of MRV before deloading. Your Hypertrophy MRV and where you finish the final Hypertrophy Phase before transitioning to a Strength Phase will help inform your starting point for Strength.

While Hypertrophic gains are driven largely by overloading volume, Strength improvements will be dependent upon overloading intensity. With this increasing intensity and the greater neural, joint/tendon integrity fatigue, it will likely be necessary to decrease the number of Overloading sets per week, while keeping in mind to stay above the athlete's Strength MEV. Let's take a look a a possible Strength Phase that will build upon the above Hypertrophy Phase.

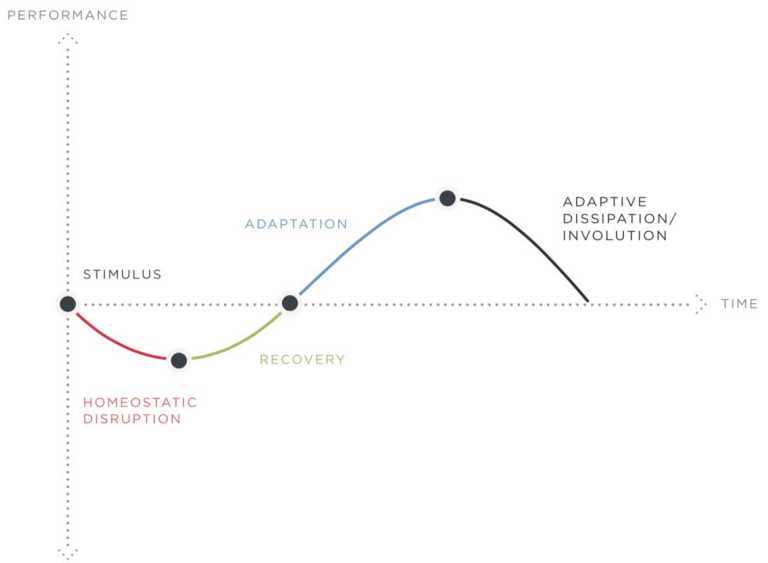
Lift	**Week 1**	**Week 2**	**Week 3**	**Week 4**
Squat 1	Up to 5rm at 7RPE Drop 8-12%x6x5	Up to 5rm at 8RPE Drop 8-12%x5x5	Up to 5rm at 9RPE Drop 8-12%x4x5	Up to 5rm at 10RPE Drop 8-12%x3x5
Squat 2	5x5 at 85% of 5rm	4-5x5 at 85% of 5rm	4x5 at 85% of 5rm	3-4x5 at 85% of 5rm
Total # of Overloading Sets of Squat	12 Sets	10-11 Sets	9 Sets	7-8 Sets
Bench 1	Up to 5rm at 7RPE Drop 8-12%x7x5	Up to 5rm at 8RPE Drop 8-12%x6x5	Up to 5rm at 9RPE Drop 8-12%x5x5	Up to 5rm at 10RPE Drop 8-12%x4x5
Bench 2	5x5 at 85% of 5rm	4-5x5 at 85% of 5rm	4x5 at 85% of 5rm	3-4x5 at 85% of 5rm
Bench 3	6x5 at 85% of 5rm	5-6x5 at 85% of 5rm	5x5 at 85% of 5rm	4-5x5 at 85% of 5rm
Total # of Overloading Sets of Bench	19 Sets	16-18 Sets	15 Sets	12-14 Sets
Deadlift 1	Up to 4rm at 7RPE Drop 8-12%x5x4	Up to 4rm at 8RPE Drop 8-12%x4x4	Up to 4rm at 9RPE Drop 8-12%x3-4x4	Up to 4rm at 10RPE Drop 8-12%x3x4
Deadlift 2	4x4 at 85% of 5rm	3-4x4 at 85% of 4rm	3x4 at 85% of 4rm	2-3x4 at 85% of 4rm
Total # of Overloading Sets of Deadlift	10 Sets	8-9 Sets	7-8 Sets	6-7 Sets

Undulating Periodization Strategies

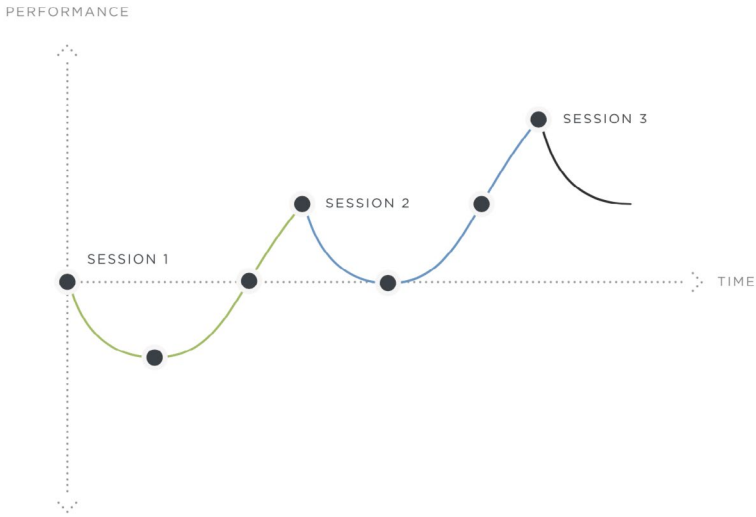
There are many different periodization strategies out there and a lot of times you will hear about Daily Undulating Periodization, or what is referred to as DUP. The way that I program involves undulating periodization, but I would not consider it to be DUP, as it's not variable based on any type of daily feedback or autoregulation in that sense, in the broader sense of that term. Rather, it's a more longterm planning of undulations that's really going to help us in three main areas of the Scientific Principles of Strength Training. Those three principles that this undulating periodization strategy is really going to help bolster and improve are Fatigue Management, Overload, and the Principle of Stimulus Recovery Adaptation (SRA).

The ability to affect fatigue management and be able to improve our fatigue management when designing programs will occur through being able to avoid accumulating too much fatigue because training does not happen in a vacuum and fatigue that is generated is not just from each singular workout. The squat workout does not just generate squat fatigue, and a bench workout does not just generate bench fatigue and fatigue from a deadlift workout, does not just affect the deadlift. No, they all generate fatigue on the entire system. The body does not differentiate between squatting, benching, and deadlifting, so there is cumulative fatigue. If you have a very hard squat work out on Monday, very taxing to the nervous system and muscular system, that is going to affect your Tuesday bench workout and maybe your Wednesday deadlift workout. Tuesday's bench is going to affect Wednesday's deadlift, Wednesday's deadlift is going to affect Friday's squat which is going to affect Saturday's bench. There are numerous different ways that you could structure the week, but understand that the fatigue being generated from one lift or one focused

training day carries over to subsequent days.
Undulating the periodization and varying the levels of intensity and volume through the different lifts is going to help avoid too much fatigue being accumulated and maybe one lift suffering at the benefit of another.
Undulating periodization is a strategic application of SRA, which means when I introduce a stimulus which generates fatigue and we have to generate fatigue to build fitness, the length of the SRA curves is going to vary based on how much fatigue is being generated.



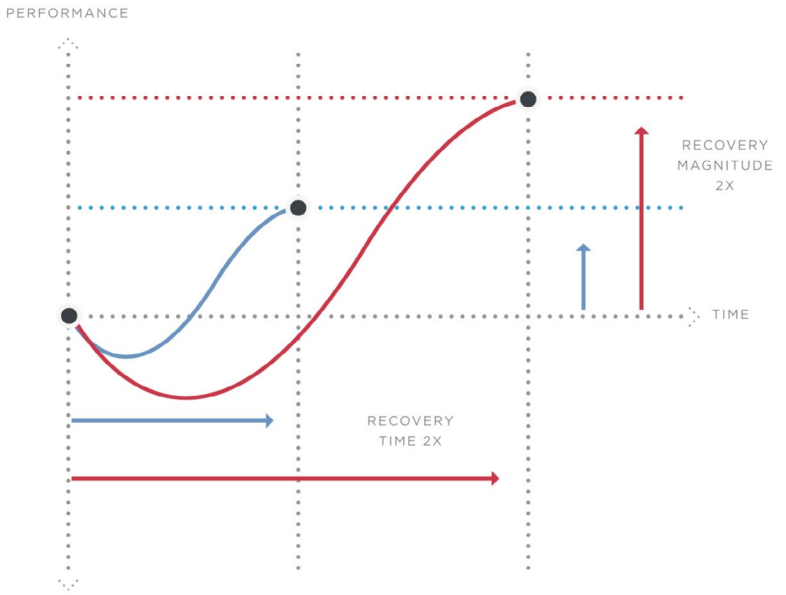
We could have a very hard workout, which has a longer SRA curve, we could have an easier workout that has a shorter SRA curve and strategically sequencing these will allow for the most overloading training in a specified time period.



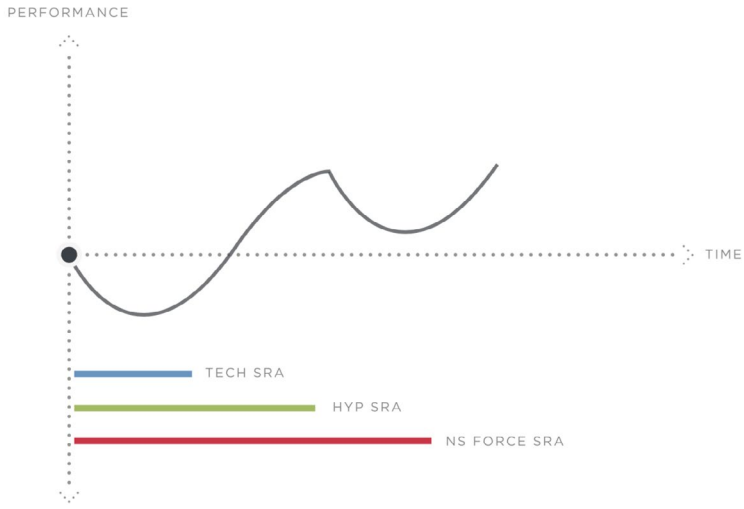
If we are training on a three up, one down schedule, three weeks of hard training, followed by a one week deload, we want do as much hard training as you possibly can in that time or let's even say you have 12 weeks before a meet, the more hard training you can do in those 12 weeks, the stronger you can potentially get by the meet. If you are not applying some level of strategy to the way you set up your frequency and the way that fatigue is managed within that frequency, you're not going to get to do quite as much hard training.

The undulating strategies are going to allow you to fit more hard training into a compressed time period and still manage fatigue, so you can have proper recovery, which in turn is going to allow you to potentially become stronger.

Finally, these strategies will affect Overload because the stronger an athlete becomes, the heavier weights they are lifting, the more experienced they are, the bigger, more muscular they are, the more stimulus they will derive from a single training session.



When you get an athlete who is very strong and the weights they are lifting become very heavy, a lot of fatigue is going to be generated from that. A longer SRA curve is going to be developed, but you could run into a problem there, because as they generate more fatigue and would require a longer SRA curve before being able to do another comparable workout, they are now potentially going to run into an issue of decaying technique as different physical qualities; strength, hypertrophy, tendon and ligament integrity, technique, neural fatigue, all have different length SRA curves, technique is very short.



If you have this big training session and then you wait a full week, or two weeks in some programs cases, between those you could actually experience technical decay. As the athlete becomes capable of greater and greater overload, they need to be more strategic in the way that they organize their training so they can make sure that technique is still developing while they are not beating up their joints, tendons, and ligaments too much, ensuring the nervous system is able to recover between these very hard overload sessions.

The final factor that is going to make this undulating periodization strategy so valuable to you as an athlete, or for the athletes that you coach is that it's going to create an improved predictability of training.

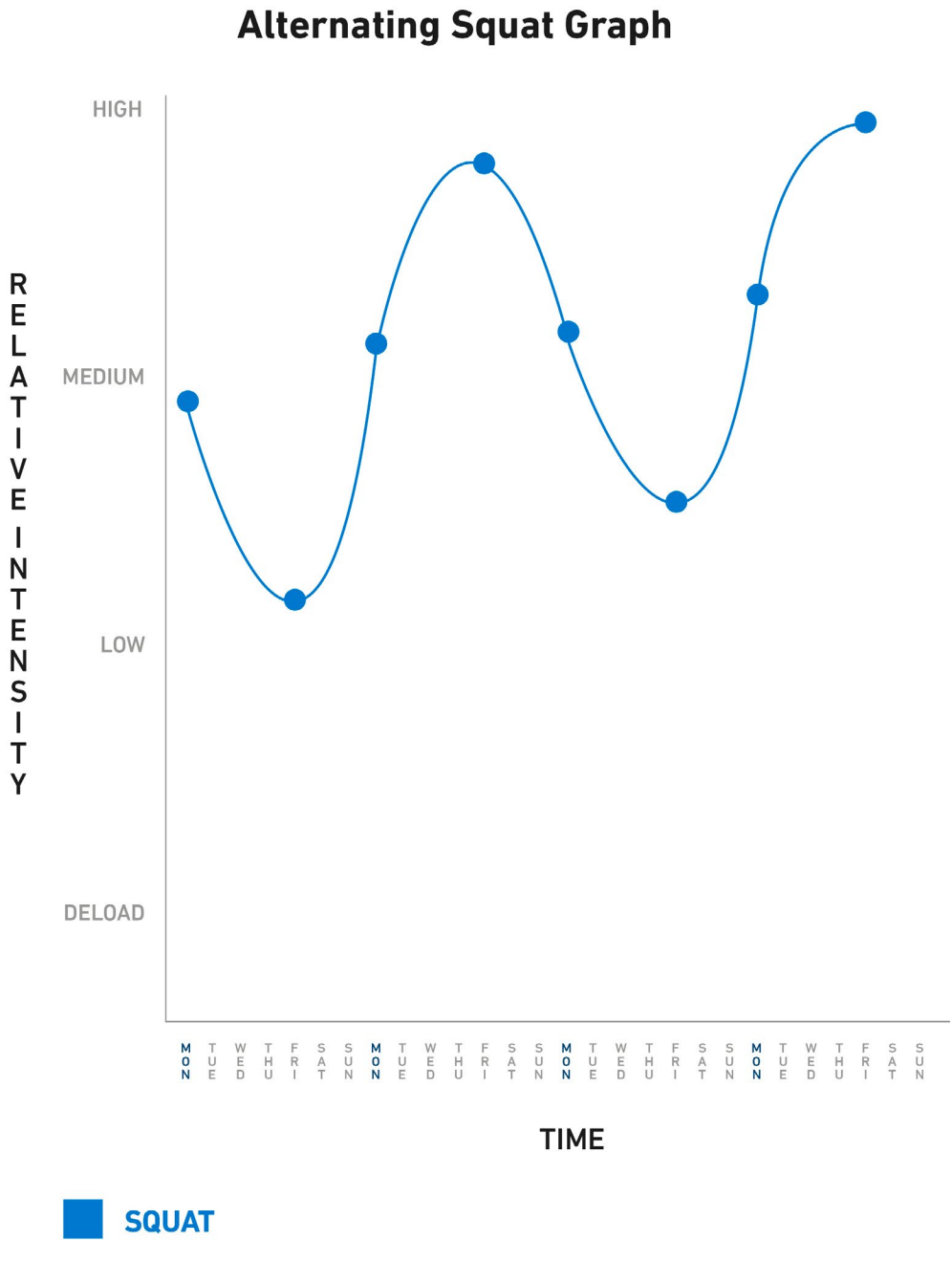
Training that has a more predictable direction to it is going to allow, for you as an athlete or coach, to better say, "This is the day that I'm going to feel strong." As a competitive powerlifter, the better that you can say, "This is the day, the competition day, where my athlete will be the strongest," the better coach you will be, the more successful athlete

you will be. This is something that I take a great amount of pride in, that I have peaked extremely well over the duration of my powerlifting career. I'm very proud of Marisa every time that she puts her best lifts of the training cycle on the platform. The day that it matters. It goes beyond just peaking because being ready for that one competition day is the most important thing. Within the course of a training cycle, you are going to have days where you feel great, you are going to have days where you feel rundown, days where you are supposed to go heavy and days that are not designed to be as heavy.

Creating this undulating strategy is going to give you a better understanding and control of which days you are going to feel good on, so you can plan for those to be heavy days. Which days you are going to feel rundown on, so you can plan for those to be lighter days. Usually with every big, high, heavy, hard day that you have, what comes to follow that? A day where you feel fatigued, so then you are able to use that fatigued day to still accomplish good work, to still develop technique, to still get volume in, but to not be guessing at when it is going to occur.
There are two models of different undulating periodization strategies that I use. One we will refer to as the Alternating Model and the High/Medium/Low Model.

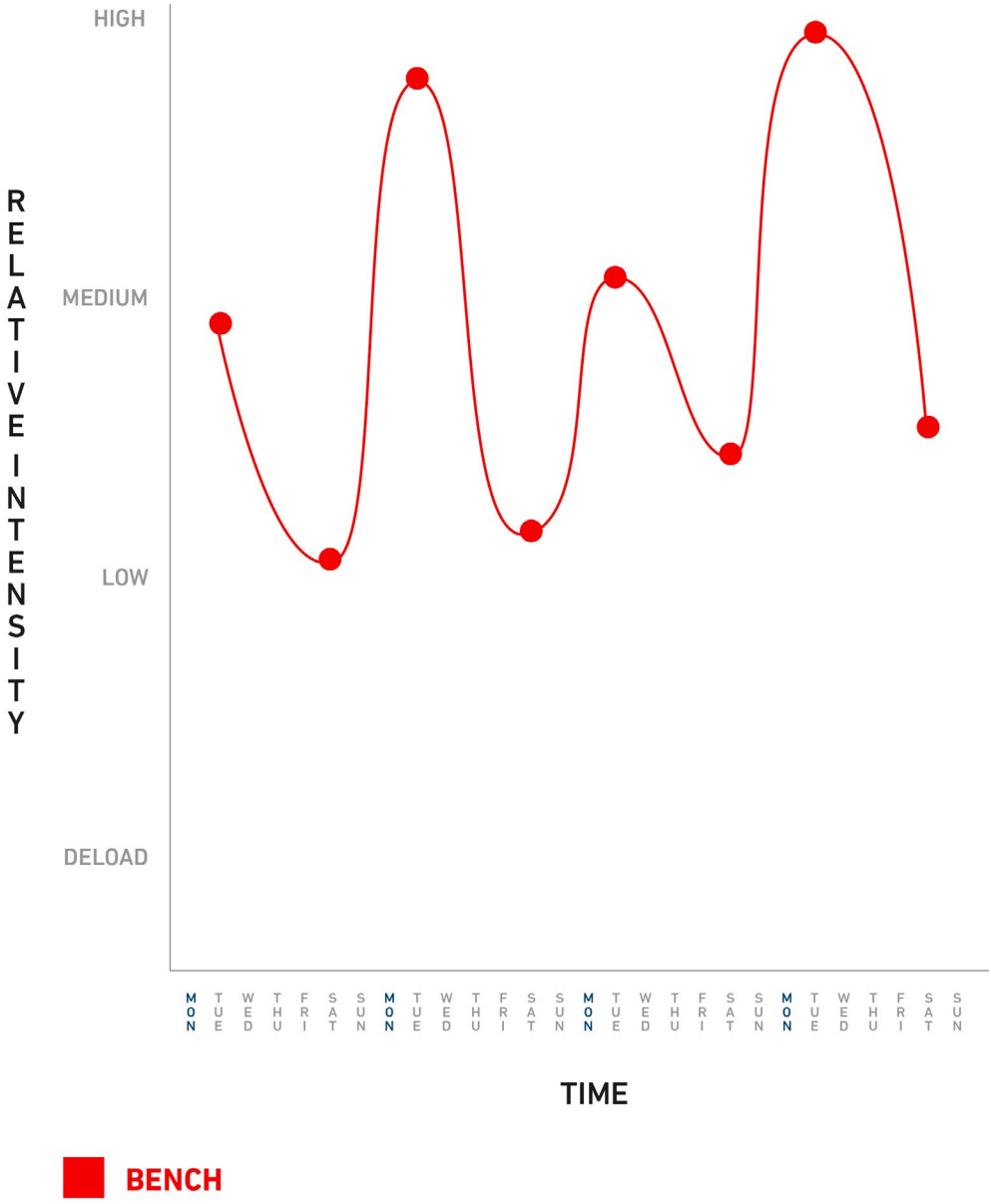
The alternating model for more intermediate athletes and advanced female athletes, this is how Marisa Inda's training is structured, this is how Kristen Dunsmore's training is structured. Advanced men's lifters, in a lower weight class, particularly shorter lifters as they are not able to generate as much fatigue, because the bar is not moving as far could also benefit from this strategy.

This is usually a 4 up, 1 down strategy that is going to alternate weeks between very hard deadlifting in weeks one and three or weeks two and four, and very hard squatting and benching in the alternating weeks, the opposite week of the deadlift.



When designing sessions to fit within the Alternating Model, typically the Heavier sessions will include working up to a top set/rep max, as well as back down sets, while the Lighter session is generally just straight across sets at a set weight. Further undulation can occur beyond just manipulating intensity/volume, through different exercise selection. For example, a heavier session in the Squat may be performed with Low Bar, while the lighter session is High Bar. This same strategy could be applied in a Hypertrophy phase with High Bar on the heavier session and Front Squat on the lighter week.

Alternating Bench Graph



As we bring in the bench workouts you can see following a very similar pattern to the squat workouts. I find that the squat and bench work harmoniously together, they tend to respond more similarly to training strategies. You will see an almost identical pattern through them.

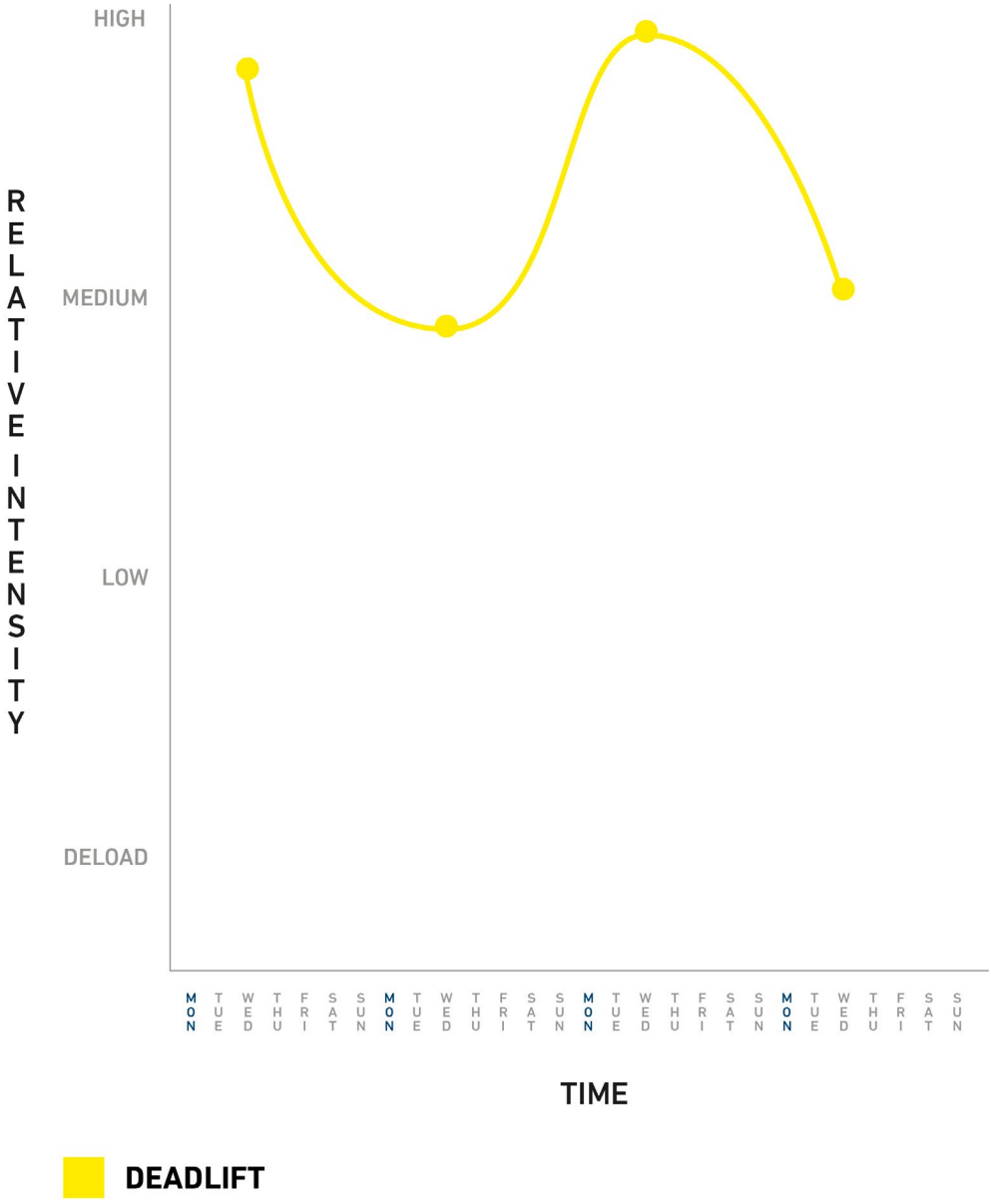
The only real change is that in the high week, the hard week for the squat and bench, we have the hard bench workout early in the week and the hard squat workout later in the week. That is done strategically, nothing we are doing here is by accident, we are doing that because a hard bench workout is much less likely to have a negative effect on the squat workout, where if we put a really hard squat workout on Monday and then tried to have a hard bench workout on Tuesday, the bench workout would suffer more from that squat workout, whether it is stress on the wrists and elbows from the low bar position or just general neural fatigue.

Now looking at the third lift, the deadlift. We're gonna have two different colors here for the deadlift. The yellow dot is our primary deadlift workout and then the green dot is a secondary deadlift workout. I distinguish between these two because they are periodized two different ways.

The yellow dot, our primary deadlift workout, is going to follow the same strategy as the squat and bench but flipped to the different week. Instead of going medium, high, medium, high, followed by a deload. Now the deadlift will go high, medium, high, medium. They are inverted from each other so week one can be very hard, stressful work for the deadlift, while week two, very hard, stressful work for the squat and bench, alternating throughout.

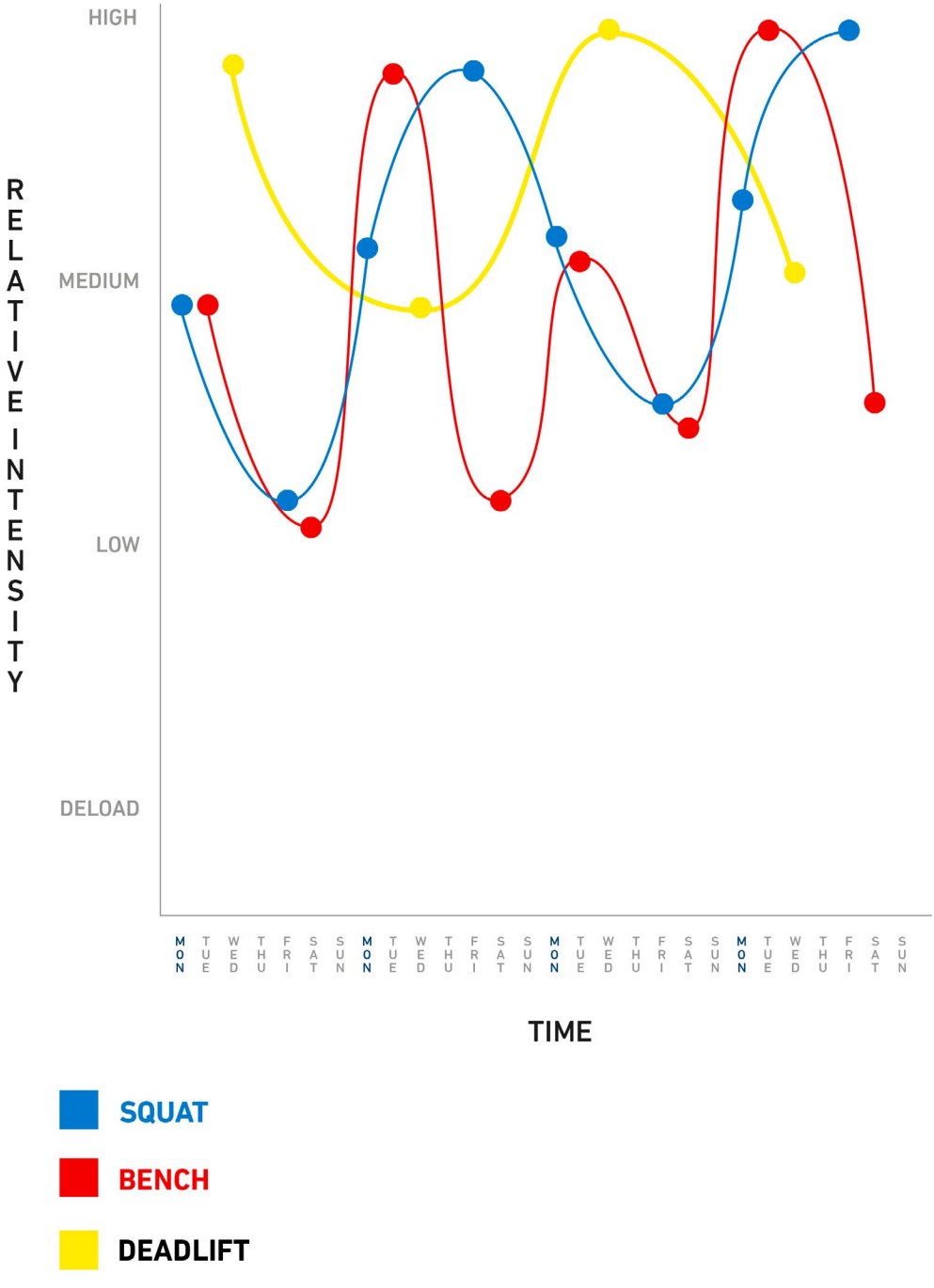
The secondary deadlift workout we treat a bit differently, just because the deadlift is so much more neurally taxing, that trying to put two workouts of it that are comparable difficulty the way we do in the squat and bench has tended to just not work out as well. And then we just have a more linear periodization strategy for that second deadlift workout, which is often just focused on technical development, trying to keep it much more low neural stress and is done after the second squat workout of each week.

Alternating Deadlift Graph



For example, the high deadlift workout in week one, depending on the phase, a strength phase for example for Marisa, might be up to five rep max in the deadlift of 160kg/352#, followed by back down sets, 320 for three sets of five. And then some accessory work like vert pull, back raises or GHRs and maybe some rows. While the second deadlift workout of the week, the secondary deadlift workout could either be three sets of five deadlifts at 275-295. If it's an earlier, more general phase, it might be RDLs or vert pulls at that point, just the deadlift movement to help stay in touch with technique to give those muscles some extra stimulus to respect idea of the technical SRA curve being shorter. The neural SRA curve for the deadlift is very long, that might make you inclined to say, "Well, I should only deadlift once a week," but in doing that strategy, technique tends to, at best stagnate, definitely not improve, and at worst start to decay.

Alternating Graph of all 3 Lifts

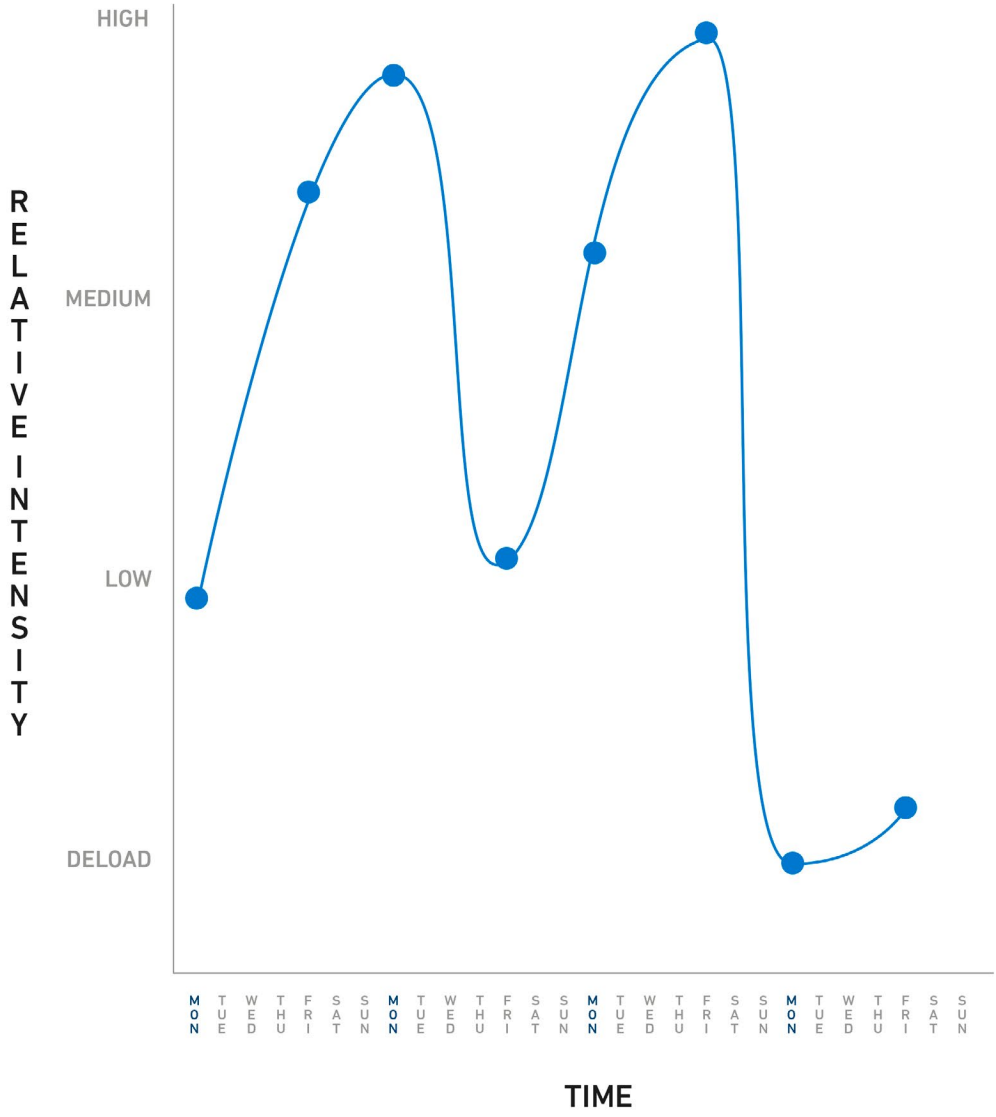


Week	Day 1	Day 2	Day 3	Day 4	Day 5	Day 6	Day 7
1	Medium SQ	Medium BP	Heavy DL	Off	Light SQ	Light BP	Off
2	Medium SQ	Heavy BP	Light DL	Off	Heavy SQ	Light BP	Off
3	Medium SQ	Medium BP	Heavy DL	Off	Light SQ	Light BP	Off
4	Medium SQ	Heavy BP	Light DL	Off	Heavy SQ	Light BP	Off

Now we're going to discuss the more advanced High/Medium/Low Model and this is what I would be using for myself, athletes like Steve Gentili, Andy Huang, Brandon Allen, Evan Mensing, guys who will put up these really big, 2,000+ pound totals. This strategy could also be appropriate for intermediate lifters who, for whatever reason, whether that is genetics, lifestyle factors like age, jobs, sleep, nutrition, have a hard time recovering.

This system builds more recovery into it and is going to work on the principle of these three different types of workouts: low workout, medium workout, high workout. Easy, medium, hard. Rotating through those to create fatigue management for each given lift and then also to work harmoniously together to create more predictable trends and better fatigue management of all the accumulated fatigue through the three different lifts.

Low/Med/High Squat Graph



SQUAT

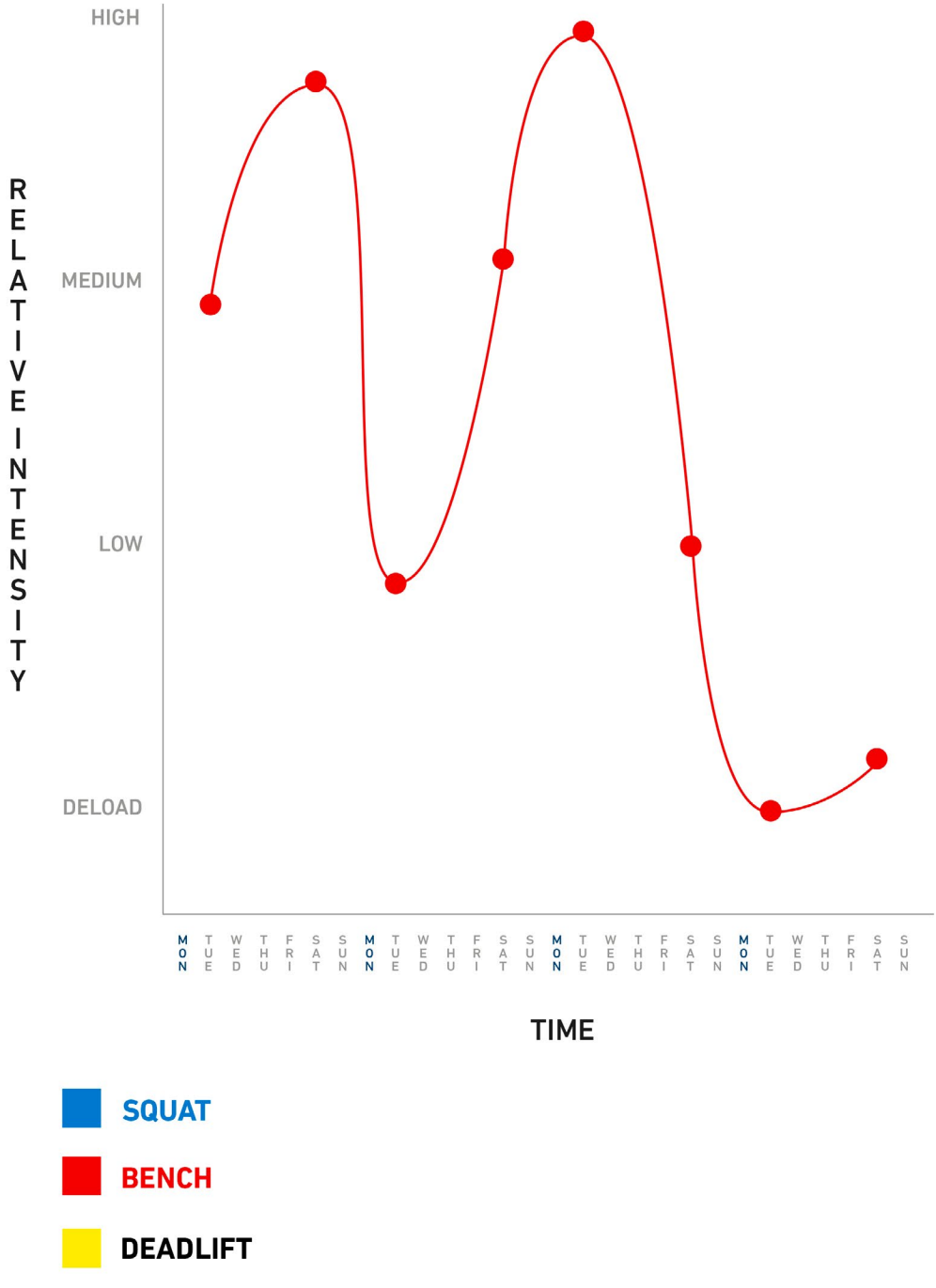
BENCH

DEADLIFT

Looking at the squat first, we start with a low squat on the first day, that would be Monday, Week 1, followed by a medium squat on Friday, Week 1. Then a high, or heavy squat on Monday, Week 2. A lot of fatigue is going to be accumulated from that, so what do we need to do to decay that fatigue? Bring it back down to a low workout. That low workout, because of the Principle of Overload, is going to have to be harder than the previous low workout and the same thing with the medium and the same thing with the high, but going through this low, medium, high, low, medium, high structure is, I found it to be extremely helpful to know that on the days that are supposed to be high, you feel good, you can lift and push the heavy weights on that one. Typically what follows that, a day where you feel really rundown, your joints hurt, weights that should feel light feel heavy, but that is fine because it is by design. We are programming lighter weights on those days, maybe still for a lot of volume and it is not going to be fun to do them, it is not going to be a great feeling workout, but you are going to get through it anyways, so it is going to add that level of predictability to it.

So we follow that strategy, low, medium, high, low, medium, high, and then a deload for the squat.

Low/Med/High Bench Graph



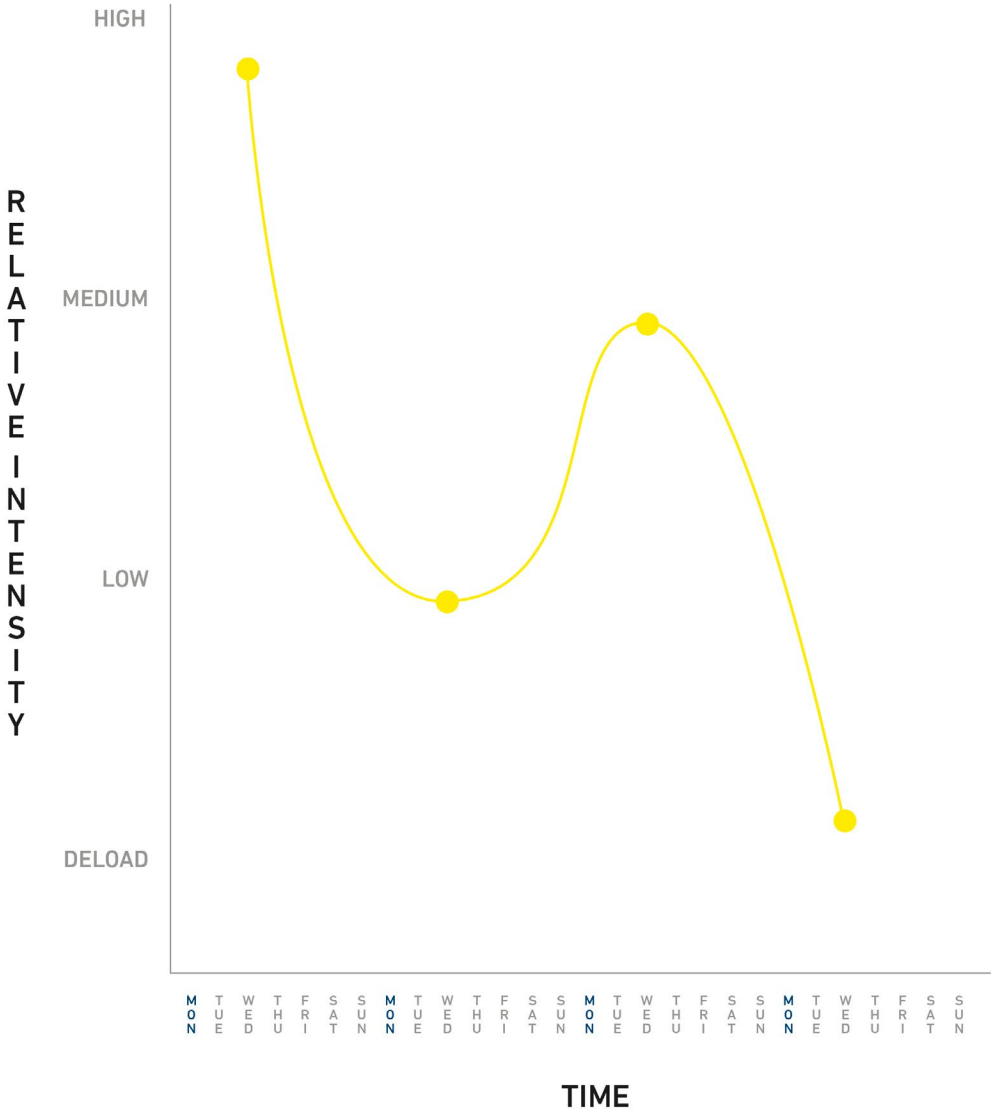
SQUAT

BENCH

DEADLIFT

Benching on Tuesday and Saturday, we are going to start out with the medium workout on Week 1, Tuesday and then a hard workout on Week 1 on Saturday and we want to get that hard bench workout in before we have the really hard squat workout, because if you are squatting low bar you can relate to this, after that hard squat workout, elbows hurt, maybe your shoulders hurt, the last thing you want to do is have a heavy bench workout. So we have the heavy bench workout on Saturday then the heavy squat workout on Monday and right now you are probably feeling very beat up and you do not want to have a hard bench workout, luckily it comes back down to a low bench workout on Tuesday, Week 2. Then back up to medium on Saturday, Week 2, again, before we get to the heavy squat workout, we are going to have the heavy bench workout on Week 3, Tuesday. Finally, after the heaviest squat workout of the whole cycle, on that Week 3, Friday, you'll still have a low bench workout. It might feel terrible, and that is fine, it feels terrible by design, we are expecting it to feel bad. That is why we programmed the lighter weights or manipulated exercise selection to allow you to still have a good workout, to still satisfy the Principle of Overload and do hard work, but without it compounding fatigue and pain on top of itself as it would if you had hard squat workout on Friday, hard bench workout on Saturday over and over.

Low/Med/High Deadlift Graph



SQUAT

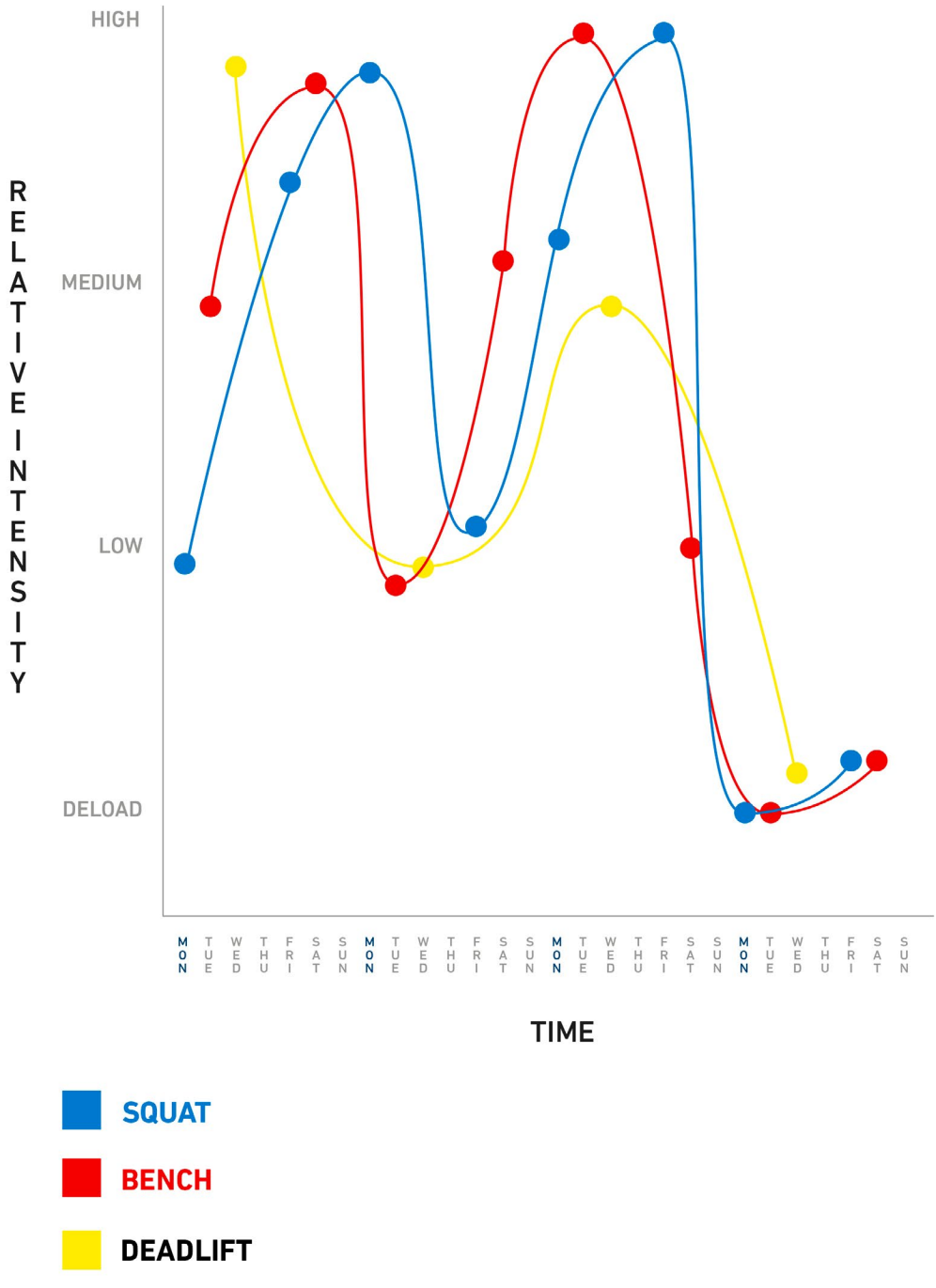
BENCH

DEADLIFT

Finally, as we put the deadlift workout in there, the squat, bench, and deadlift, are all working independently. Just as we had in the intermediate alternating model in the deadlift workout, we have two different deadlift workouts here. Our primary one, which follows a high, low, medium strategy and our secondary deadlift workout, which we do after our second squat workout in the week, that has its own periodization model, just a linear periodization of relatively easier workouts, 60-75% as difficult as the first workout of the week. You can see with the deadlift, Week 1, a heavy, hard workout. We start the cycle off with that and it is maybe not the most optimal thing when looking at just one training cycle, one 4 week block of 3 eeks up, 1 week down, but keeping in mind that the deadlift is the most fatiguing and you need to take your heavy deadlifts the farthest away from a meet, it ends up working out just fine.

We go heavy in week one, followed up by a light workout in Week 2. We always want to follow the heavy workout with the light workout, because the heavy workout is going to generate the most fatigue, which is going to make you feel the worst for the next subsequent workout. So we go low in Week 2, back up to medium in Week 3 and then, finally, in Week 4, a deload.

Low/Med/High Graph of all 3 Lifts



Though all of these lifts independently of each other through this high, medium, low structure or low, medium high structure, and it's important how they all work independently of each other, what is even more important is how they all work when you put them together.
Training does not happen in a vacuum. The squat workout you do on Monday affects the bench workout on Tuesday. The bench workout on Tuesday affects the deadlift workout on Wednesday. That affects Friday, that affects Saturday, that affects the next Monday. It all has to work together. Fatigue is being accumulated through all of the different lifts.

Where our bench workout can be negatively affected by a squat workout, vice versa, so that is why we use this alternating high, medium, low strategy. It works so well when you put them all together to create a trend of week one, and into early week two, is up in a lot of harder workouts, relatively harder workouts in that timeframe. You're going to have a hard deadlift, week one, Wednesday. Medium squat week one, Friday. Hard bench week one, Saturday. And then a hard squat on week two Monday. Those four workouts are going to generate a lot of fatigue, those are all going to be pretty challenging to very challenging workouts. So what do you need to do after that? You need to follow them up with lighter workouts, which is why on Tuesday, Wednesday, Thursday as an off day, and then Friday of week two we have low bench, low deadlift, low squat to decay that fatigue and now you're gonna start to feel better and better coming into week three, where we come with high bench on Tuesday, medium deadlift, which is still going to be a very challenging workout, and then our biggest squat workout of the whole cycle on that last Friday.

The Principle of Overload dictates training must become more difficult over time, the second low workout is always harder than the first low workout, it is always heavier or has more volume in it. The second medium workout, always heavier, has more volume than the first medium workout and the same thing with the high workouts. So our trend ends up being, if we're going to assign just a difficulty on a one to ten, not necessarily an RPE scale, our first week maybe is a 6 or 7. Our second week a 4 or 5, and then the third week, 8, 9, maybe even a 10, depending on how many consecutive phases you are going to pair one after another. It is that trend of pretty hard, easier, very hard, deload, that is going to allow us to create better fatigue management by strategically using Stimulus Recovery Adaptation and manipulating Overload so we can create longterm training predictability and better know these are the days where you are going to feel strong, where you can go heavy. All of this is trending towards the one day that matters most, where you are going to lift the most weight that you have ever lifted, the day of the competition.

Week	Day 1	Day 2	Day 3	Day 4	Day 5	Day 6	Day 7
1	Light SQ	Medium BP	Heavy DL	Off	Medium SQ	Heavy BP	Off
2	Heavy SQ	Light BP	Light DL	Off	Light SQ	Medium BP	Off
3	Medium SQ	Heavy BP	Medium DL	Off	Heavy SQ	Light BP	Off

The intermediate, Alternating model, and then the Advanced, High/Medium/Low Model, hopefully this gives you some better insight into potential ways to better structure your or your athlete's training. If they are struggling with fatigue management, you do not feel like they are able to get enough hard work in in whatever timeframes you're dealing with, this can be a great tool to use.

Selecting and Progressing Weights

While it seems like a simple, straightforward concept, the ability to accurately select which weights to use during training is one of the most critical aspects to a lifter's success, both within the context of a given training cycle and over their entire career. If you are failing to select the proper weights, it could mean that you are training is not challenging enough or that you are overreaching too soon, you could be running into fatigue management problems. Maybe you are missing lots of lifts in training and of course you do not become a stronger by missing lifts, you become stronger by making them, besides the psychological benefits of making as many of your lifts or all your lifts in training.

Starting slow is the first important concept to understand when selecting your training weights. The Principles of Overload dictates that training must become harder over time, so if you start a training cycle too heavy, there wont be enough room for you to increase weight over the weeks to satisfy this principle. We often write programs on a 3+1 paradigm, meaning 3 weeks of progressively harder training followed by a 1 week deload. Simply put, this will be training hard, harder, hardest and then deload, so depending on the lifter's qualification we will be aiming to make a PR in the 3rd week of this structure, possibly a very small PR on the 2nd week if the lifter is less experienced and still capable of progressing very quickly. Knowing and tracking your maxes for different rep ranges in different exercises is critical in being able to select weights effectively, this will allow you to use a rep max calculation to best pick your target weights throughout the program.

Here is how weight progressions in the Squat may look for Beginner, Intermediate and Advanced lifters over the course of a 3 week Hypertrophy Block using a linear progression.

Beginner Lifter. 1rm-300#, 8rm-240#.

Intermediate Lifter. 1rm-450#, 8rm-360#.

Advanced Lifter. 1rm-600#, 8rm-475#.

The (#) represents the projected 1rm from that set. This was found by using...

(Weights x Reps x .033) + Weight = Projected Max

Week	Beginner	Intermediate	Advanced
1	230x8 (290#)	340x8 (429#)	420x8 (530#)
2	240x8 (303#)	355x8 (448#)	450x8 (569#)
3	250x8 (316#)	370x8 (467#)	480x8 (606#)
Deload	220x8	320x8	400x8

A few important things to note when looking at these progressions are...

- They're just examples, you need to observe how you and your athletes respond to training and adjust based on that.
- Beginner athletes can and usually should start at a relatively higher % of their rep max because they aren't able to generate as much stimulus or fatigue as their more advanced counterparts.
- Beginner athletes can and usually should expect to make a PR by a greater amount in the final week of their mesocycle than advanced lifters should. They may even be able to match their existing PR or increase it by a fractional amount in an earlier week.

The next critical component in regards to weight selection, would be how to transition to the next block of training, in this scenario you have 2 options, performing another phase of the same type (2 consecutive Hypertrophy blocks) or moving from one phase to another (Going from Hypertrophy to Strength).

As you transition from one block to another, you should consider what you are basing your weights on, %s or RPE, this can be important in how you adjust your weights for the following block. If you are basing your weights on %s, there are 2 important things to remember: 1-What is it a % of? 2-%s are just a suggestion and should not be treated as a hard and fast rule.

In the above table for the Beginner lifter their lifts were at Week 1-77%, Week 2-80% and Week 3-83% based off of their 1rm of 300#. If they are performing another Hypertrophy Block of 8rm, then they could repeat the same %s but now base them off of 316#, their projected max from 250x8 in Week 3. If the athlete was to make a VERY big PR in a given block, for example if the athlete had done 265x8 in Week 3, a projected max of 334#, over a 10% increase to their established numbers, I would be wary to increase their numbers that significantly in the next block because beginner athletes are generally better at reps than singles so it likely isn't a true representation of how much their strength has increased and this may cause them to go too heavy, too soon in the next block. You could also look that their projected PR has increased by 16# and when repeating the Hypertrophy Block, increase all the weights by 16# (or 15# is likely more realistic).

If you are selecting your weights based on RPE, you should just use your previous block to inform your decisions and then adjust by feel on the day. I have found that with myself and my athletes, I do not often employ either %s or RPE when designing their programs, rather I just select the weights based on their previous training cycles so that we are setting up for PRs at the appropriate time and satisfying the Principle of Overload.

If you are transitioning from one block to another, such as moving from Hypertrophy to Strength, it is important to create a smooth transition to set the athlete up for success. During an extended Hypertrophy Block, particularly for more advanced athletes, it is reasonable that the lifter's force production capabilities may decline in the short term, so the heavier weights required in a Strength Block may feel shocking. Starting a Strength Block around the same weights that you completed the Hypertrophy Block will help smooth this transition from phase to phase.

An appropriate strategy could look like the following...

Week	Beginner	Intermediate	Advanced
Hypertrophy 3	250x8 (316#)	370x8 (467#)	480x8 (606#)
Deload	220x8	320x8	400x8
Strength 1	245x5 (285#)	365x5 (425#)	455x5 (530#)
Strength 2	260x5 (303#)	385x5 (448#)	490x5 (570#)
Strength 3	275x5 (320#)	405x5 (471#)	525x5 (611#)

Following a simple strategy like this and keeping track of your established and projected maxes will allow you and your athletes to make steady progress over the long term. It is important to keep in mind while moving through a progression like this, that some athletes, particularly less experienced ones, will have days where they feel particularly strong and the urge to chase a big number on that day can

be overwhelming, but I would encourage you to fight that urge, unless it is happening during a functional overreaching week before a planned deload. For example, if our Intermediate athlete above decided to do 395x5 in Week 2 rather than 385x5, it would generate more fatigue earlier in the training cycle and possibly preclude them from being able to complete the 405x5 during Week 3.

To summarize, when selecting weights for your or your athletes' training, it is important to know and track your maxes in the exercises you are using. Failure to track max singles, rep maxes and volume PRs will make it difficult to choose weights that allow you to drive Hypertrophic and Strength gains and will ultimately violate the Principle of Overload. Start slow, while it may be cliche, 'it does not matter where you start, just where you finish,' is very true when it comes to selecting weights during a mesocycle. Going too heavy, too early in a cycle will cause too much fatigue to accumulate early and make it difficult to satisfy the Principle of Overload because training must become more difficult over time. When choosing how aggressively to increase and athlete's weights over the course of a cycle, keep in mind their strength and experience, as these factors will largely influence their ability to generate stimulus and fatigue. When moving from one phase to another, it is important to create a smooth transition so progress can continue and injury can be avoided, so try to have the next phase start at, or even slightly below, where the previous phase finished. Let these words from the greatest Powerlifter of all-time, Ed Coan, help you during this critical aspect of your training.

"The key (for me) was, as is the case in most programs, to pick your numbers correctly. Every single rep, every single set, every single weight, every single exercise was already planned." **- Ed Coan**

PHASE POTENTIATION STRATEGY

Putting it all Together

Phase potentiation is really a fancy way for me to sound smarter than I really am, by talking about using one phase of training to increase the potential of the subsequent phase of training. The more official definition we are going to give it is a logical sequence of training phases to promote the best overall long-term outcomes. The way that we're going to apply phase potentiation for powerlifting in its simplest terms are we are going to grow bigger muscles, build more muscle, that is going to happen in the Hypertrophy Phase. Then we are going to teach those bigger muscles to produce more force in a Strength Phase. Finally, we are going to take those bigger, more force producing muscles and hone their technical prowess and improve the neural qualities of the one rep max in the peaking phase.

There are three main considerations to be made when figuring out how to properly phase potentiate a training program, specificity, sequence, and adaptive decay. Specificity, which creates the framework for all training decisions, as it pertains to phase potentiation we first need to look at directed adaptation. Specificity, as it pertains to Phase Potentiation, is going to essentially answer the questions, why is Phase Potentiation good? Why is it important? And the reason really is going to be that training a single modality at once with all of your efforts focused towards developing that one quality, is going to be superior to trying to train multiple qualities during the same time. This, in my opinion, is what differentiates and distinguishes block style periodization, or phasic structure periodization, as being superior to a concurrent style of periodization. Potentially, what people would call Daily Undulating Periodization can also fall into this trap though it is not necessarily inherent to that, a common error that people are making is that they are trying to train too varied of qualities within the same

training week, or within the same training month. In regards to directed adaptation, if you want to get better at the squat, that you need to continually load heavier and heavier weights onto the squat, week, after week, after week, session after session and without doing that, with changing it too frequently, you are going to fail to satisfy that idea of directed adaptation. We need to have focused directed adaptations, specific work directed at developing a quality session after session, week after week for a minimum of three weeks, to really see any adaptation occurring.

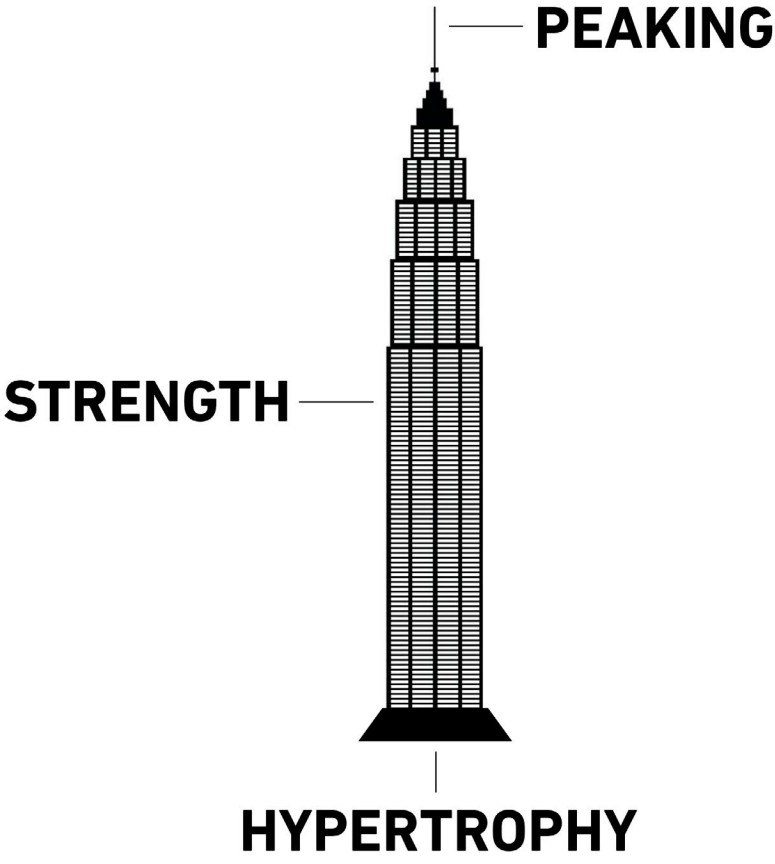
For the training sequence, we need to decide which phase precedes another and in doing so, we need to make logical, strategic decisions to make sure that potentiation is occurring. Potentiation, increasing the potential of the subsequent sequence. So we don't want to just randomly throw them together and say, well, Chad said we have to have phases to our training, no, the phases must have a strategic organization. Improve muscle size, strength, then technique and neural qualities. We want to be strategic and logical in the sequencing of our phases to make sure that this phase improves the potential of the next phase, and so on, and so on.

Let us use the analogy of a skyscraper. The base of our skyscraper is going to be Hypertrophy, it is going to be muscular size and work capacity. The bigger that you can make that base, the taller you will be able to build the skyscraper. It is important to consider that in the actual building of a skyscraper, the first thing they do is dig a foundation, it is actually going down. In properly phase potentiated training for powerlifting, you may have a decrease in the one rep max, particularly for more advanced lifters, to be able to lay a better foundation of hypertrophy.

You do not need to be ready to lift the heaviest weights all the time because your competitions are not coming up as a surprise. You know when you are going to compete, so you can take that step back in the short term, particularly for more advanced lifters, decrease the one rep max while you train for qualities of hypertrophy. Because while in the long run doing sets of 8, 10, 12 reps, setting 8, 10, 12 rep maxes are going to help you build more muscle. In the short term, it is going to detract from the neural qualities, and potentially a little bit from the technical qualities, especially depending on what you do from an exercise selection standpoint, but in the long term it is going to allow you to build more muscle, which is going to be a greater increase in the potential of the subsequent phases. We are going to dig the foundation and lay a big base, build as much muscle as possible with hypertrophy.

Then on top of that, you are going to build your main floors, that is going to be general strength. You are going to build that as tall as you can, as big as the hypertrophy you created will allow for. You could run into the issue of adaptive resistance there. You made your base as wide as you did, you can not infinitely build on top of that and just keep doing strength, strength, strength, strength, strength, and expect it to continually go up, Adaptive Resistance is going to stop that. The longer you continue to do something while you do the same thing, you are going to get some diminishing returns or slower gains with that. Once you have exhausted strength, you have pushed that to the limit of its adaptive resistance, you have built all the strength that you can.

Finally, we are going to throw a little antenna on top of our skyscraper and that is peaking. For an antenna to work well in the context of an actual skyscraper, it needs to be put up as high as possible, so the bigger hypertrophy base and the greater general strength main floors that you build, the higher than that peak is going to be able to go. If you were to put them out of order and the peaking training was done too early, out of order, you would be trying to balance general strength on top of that narrow base of the low volume of peaking. With that being said, that is why you cannot try and peak all the time, you can not put the antenna at the base. You can not always be doing singles, training low volume because you are not going to get the benefits of that big base, of that general strength, and putting the peak at the highest point of our training skyscraper.



The third principle that is going to make up this idea of Phase Potentiation, Adaptive Decay means that we need phases and we can not train everything at once. We have established that with the ideas of specificity and sequence the phases cannot be random, they need to build on each other logically because all of these physical qualities have a time that they will last. That is a very common criticism that conjugate and concurrent coaches will make of a more linear style programming, saying that you are always moving away from the qualities that you previously developed and that could be true if you are doing it wrong. As we moved from Hypertrophy to General Strength, there is going to be sufficient volume in there to retain the hypertrophic gains you made. Now you are not going to build more muscle by doing that, you are not going to build more muscle doing strength training than you would have with hypertrophy training, but you are not going to lose that muscle either. If you are doing sufficient volume during general strength, you are not going to have adaptive decay from the hypertrophy training that you were already doing.

Avoiding Adaptive Decay through Phase Potentiation is a matter of the proper application of Overloading and appropriate phase length and proportion. Let us take a look at how long your different phases can be and what proportion of your training they should make up.

Hypertrophy

Potential Phase Length: 3 Weeks to 4 Months
Potential Phase Proportion: 0-50%
Factors To Consider When Deciding on Hypertrophy Phase Length:

- Beginner lifters will lack muscle mass compared to their more experienced counterparts, so they will need relatively more Hypertrophy training.

- Advanced lifters, particularly if they are already cutting weight to make their class, may not need any Hypertrophy training. This is not to say that that higher volume phases can not be useful for them as it can help build work capacity but if they are not doing it to build muscle, Hypertrophy would become a misnomer.

- Female lifters will need relatively more and respond relatively better than their male counterparts, to Hypertrophy training.

- If an athlete was doing 12 weeks of directed training for a meet, I would expect that Beginner Lifters would spend 5-6 weeks in Hypertrophy, Intermediate Lifters would spend 4-5 weeks, Advanced Lifters who do not cut substantial weight for their class may spend 3-4 weeks and Very Advanced Lifters are likely cutting weight for their class or maximized their muscularity in that class and will be better served to include high volume phases preceding a 12 week meet prep, reserving those 12 weeks for Strength and Peaking work.

Strength

Potential Phase Length: 3 Weeks to 4 Months
Potential Phase Proportion: 20-70%
Factors To Consider When Deciding on Strength Phase Length:

- Strength Phases will be a more substantial part of Intermediate and Advanced Lifters Training.

- The stronger a lifter becomes, likely the longer they will need to dedicate to strength training be cause each overloading session will be more fatiguing and need to be spread out more for fatigue management purposes.

- If an athlete was doing 12 weeks of directed training for a meet, I would expect that Beginner Lifters would spend 3-4 weeks in Strength, Intermediate Lifters would spend 4-5 weeks, Advanced Lifters will need 4-6 weeks in Strength Phases and Very Advanced Lifters are may need from 5 weeks or potentially even up to 8 weeks of Strength work.

Peaking

Potential Phase Length: 2 Weeks to 2 Months
Potential Phase Proportion: 15-50%
Factors to Consider When Deciding on Peaking Phase Length:

- Beginner lifters, particularly low weight classes and females, can have very short peaking phases. Longer peaking phases for these athletes will result in Adaptive Decay of Hypertrophic qualities.

- The bigger and stronger an athlete is, the longer they will need for peaking. Bigger, more muscular athletes, will be able to retain Hypertrophy for longer. The stronger an athlete is, the more time they will need to work up to their peaking weights. For example, if an athlete finished their Strength Phase in the bench press with a 3rm of 185# and expect to finish peaking at 205#, it will only take a few sessions to build from 185-205#. Conversely, if an athlete finishes Strength with a 3rm of 485# and wants to finish Peaking at 530#, it will take them more sessions and longer time between those sessions, to build up from 485-530#.

- If an athlete was doing 12 weeks of directed training for a meet, I would expect that Beginner Lifters would spend 2-3 weeks in Peaking, Intermediate Lifters would spend 2-4 weeks, Advanced Lifters will need 3-5 weeks in Peaking Phases and Very Advanced Lifters are may need from 4 weeks or potentially even up to 8 weeks of Peaking work.

FAQ: Can I mix different phases together?

While this practice could run into issues of Directed Adaptation, I have found it to be useful to time phases differently between the 3 competition lifts. Typically, this will happen by having longer Hypertrophy Phases for the Bench Press and/or longer Peaking Phases for the Deadlift. The Bench Press responds the most to Hypertrophy training and because the weights of the Bench Press are relatively lightest of the three lifts, it will need the shortest time to build up during Peaking. The Deadlift typically has the heaviest weights being used and carries with it the greatest fatigue, so the lower volume of peaking and higher neural force production it can develop, may be beneficial. An example of how Phases can be strategically mixed together for long term success is as follows (each phase in this example is a 3+1 paradigm):

Lift	Phase 1	Phase 2	Phase 3	Phase 4
Squat	Hypertrophy	Strength	Strength	Peaking
Bench Press	Hypertrophy	Hypertrophy	Strength	Peaking
Deadlift	Hypertrophy	Strength	Peaking	Peaking

Putting It All Together

We have now discussed all critical aspects of program creation and now I want to help you create a program in a few strategic steps.

Step 1: Assess the Athlete

Answer all the questions from the Athlete Assessment section. These answers will guide our decisions throughout the rest of this process.

Here is an example athlete to help illustrate the next few steps:

Sex-Male	Weight-Middle	Height-Medium
Strength-High	Experience-Intermediate	Age-36 Years Old
Diet-Average	Sleep-Good	Stress-Average
Historical Workload-3	Historical Recovery Ability-3	Time to Meet-13 Weeks

Step 2: Establish the Athlete's Unique MEV and MRV

Finding the athlete's unique Minimum Effective Volume and Maximum Recoverable Volume will inform many of our training decisions. Primarily, we will know where to start our training and potentially where that training will eventually go to. The relationship between MEV and MRV will help us understand what phase lengths are appropriate for the athlete, as well as which periodization strategy will be most appropriate. Assuming the athlete will begin in a Hypertrophy Phase, we only need to find the MEV and MRV for Hypertrophy at this time.

Our example athlete from above have Volume Landmarks of:

MEV

Squat	Bench	Deadlift
8 Sets/Week	9.5 Sets/Week	6 Sets/Week

MRV

Squat	Bench	Deadlift
18 Sets/Week	21 Sets/Week	15 Sets/Week

Step 3: Find the Optimal Frequency for the Athlete

Now that we know how much training the athlete needs to be doing (MEV/MRV), we must decide how to best distribute that volume throughout the week. Finding a frequency that takes into account all varying SRA curve lengths to strike a balance between sufficient overloading sessions to drive adaptation and develop technique, without creating fatigue management issues regarding neural fatigue and tissue/tendon integrity is our goal.

Squat	Bench	Deadlift
2x/Week	3x/Week	1-2x/Week

There are several ways that this frequency could be distributed throughout a training week based on the athlete's schedule. Here are my preferred ways to structure a week based on this frequency.

3 Days/Week

Day 1	Day 2	Day 3	Day 4	Day 5	Day 6	Day 7
Squat	Off	Bench	Off	Deadlift	Off	Off
Bench				Squat		
Deadlift				Bench		

4 Days/Week

Day 1	Day 2	Day 3	Day 4	Day 5	Day 6	Day 7
Bench	Off	Squat	Off	Bench	Deadlift	Off
		Deadlift			Squat	
		Bench				

5 Days/Week A

Day 1	Day 2	Day 3	Day 4	Day 5	Day 6	Day 7
Squat	Bench	Deadlift	Off	Squat	Bench	Off
Bench				Deadlift		

5 Days/Week B

Day 1	Day 2	Day 3	Day 4	Day 5	Day 6	Day 7
Bench	Squat	Off	Deadlift	Bench	Squat	Off
	Bench				Deadlift	

Step 4: Establish the Athlete's Phasic Structure and Periodization Strategy

The relationship between MEV/MRV will help us understand more about how to best distribute the athlete's phasic structure and which periodization strategy will be most effective for them. In the case of our example athlete, they have a fairly wide gap between MEV/MRV so they could likely perform well within a linear structure. One consideration regarding this would be that because they have High Strength, they will be able to generate relatively high fatigue levels from a given session, so particularly during Strength and/or Peaking, it could be beneficial to move to an Alternating Periodization strategy to help manage Neural and Joint/Tissue Integrity Fatigue.

With 13 weeks until competition, an athlete at this level (Middle Weight, High Strength, Intermediate Experience) has likely not quite maximized their muscle mass at their given weight class so they will benefit from Hypertrophy Training but should place their focus most on General Strength. I would divide their 13 weeks of training into Hypertrophy 3+1 w/ Linear Periodization, Strength 4+1 w/ Alternating Periodization and Peaking 3+1 w/ Alternating Periodization.

Step 5: Distribute the Athlete's Volume Over Their Optimal Frequency and Select Exercises

When distributing an athlete's volume over their frequency we must consider how the week works together. Training does not happen in a vacuum, remember that what happens on Day 1 will affect Day 2 and so on. This means that we should modulate volume from one session to another in an effort to allow some sessions to be more primary, while others are secondary or tertiary. Further modulation of the relative fatigue a session will generate can be accomplished through exercise variation. For example, two sessions of Hypertrophy work for the Bench Press could both have 3 overloading sets but if one is done with Closegrip Bench Press and the other is done with DB Incline Bench, the Closegrip session would be relatively more fatiguing.

Let's use the 4 Day/Week training structure to distribute our MEV over: 8/9.5/6

Day 1	Day 2	Day 3	Day 4	Day 5	Day 6	Day 7
Bench 5 Sets	Off	Squat 5 Sets	Off	Bench 3 Sets	Deadlift 4 Sets	Off
		Deadlift 2 Sets			Squat 3 Sets	
		Bench 2 Sets				

The above just looks at the number of overloading sets the athlete will perform in a given session and you can see that some days are given more than others this will help allow the 'bigger' days to receive greater energy while 'smaller' sessions still contribute to weekly volume, develop technique and satisfy the Principle of SRA. The number of sets per session is referring to work done within the Overloading Parameters from our overview of the phases. For example, in Hypertrophy, we will count sets done with 55-75% intensity for sets of 6-12 reps in exercises that are stimulating enough to drive adaptation.

Now that we have established how much volume to perform in a given session we can decide what exercises to fill that volume with. When selecting exercises, we should consider an athlete's competition techniques, unique strengths and weaknesses, as well as what phase they are in. Deciding what an athlete's strength and weaknesses are should be done through observing the athlete, asking them about where they typically struggle with lifts, as well as looking at the relationship between exercise variations. For example, an athlete with a particularly strong low bar squat compared to high bar and/or front squat could be indicative of relatively weak quads; or someone who has a closegrip bench nearly equal to their moderate and wider grips, would be indicative of strong triceps but can also mean that pec and shoulder strength is lagging behind. For our example case, let's pretend that we have established the following weak points:

Squat	Bench	Deadlift
Low Bar	Medium Grip	Sumo
Weak Legs/Strong Back	Midrange	Off the Floor

The Principle of Variation will be also be significant when selecting the best exercises for them to utilize, particularly during Hypertrophy since it is furthest away from competition (meaning it can utilize the least specific training) and is a great time to help avoid Adaptive Resistance. Based on the above weak points and the athlete beginning in a Hypertrophy Block, we could structure training in the following manner:

Day 1	Day 2	Day 3	Day 4	Day 5	Day 6	Day 7
Spoto Press Up to 8rm at 8RPE, Drop 8-12%x4x8	Off	High Bar Squat Up to 8rm at 8RPE, Drop 8-12%x4x8	Off	Incline Bench 3x8-10 at 60-65%	Conventional Deadlift Up to 6rm at 8RPE, Drop 8-12%x3x6	Off
		Sumo Halting Deadlift 2x6 at 55-60%			Front Squat 3x8 at 60-65%	
		DB Military Press 2x10-12 at 55-60%				

The above would fulfill the athlete's MEV with exercises that will help address their unique weaknesses and are also a good fit for Hypertrophy based on how much fatigue they generate and how they will allow a movement over time to more specific exercises which will help avoid Adaptive Resistance.

You are probably wondering how to fit your accessory training into this, it can be fairly straightforward. On your 'bigger' bench press workouts, including some training for the lats, shoulders and triceps is appropriate. On the 'bigger' squat and deadlift workouts, including some training for the low back, quads, hamstrings and glutes is appropriate. Performing one movement for each body part, each session is without a doubt sufficient but you can perform more if you feel a certain muscle is lacking, or even perform less if you feel that it is interfering with your ability to perform and recover from your primary training.
For example, as lower body training tends to be more taxing, performing extra work for the Low Back and Quads after one session of primary training and extra work for the Hamstrings and Glutes after another session could be a good strategy. With assistance work added, the week may now look like this:

Day 1	Day 2	Day 3	Day 4	Day 5	Day 6	Day 7
Spoto Press Up to 8rm at 8RPE, Drop 8-12%x4x8	Off	High Bar Squat Up to 8rm at 8RPE, Drop 8-12%x4x8	Off	Incline Bench 3x8-10 at 60-65%	Conventional Deadlift Up to 6rm at 8RPE, Drop 8-12%x3x6	Off
Chest Supported DB Rows 5x10-12		Sumo Halting Deadlift 2x6 at 55-60%		Lat Pulldowns 5x10-12	Front Squat 3x8 at 60-65%	
DB Front Raises 2-3x12-15		DB Military Press 2x10-12 at 55-60%		DB Lateral Raises 2-3x12-15	Hamstring Curls 3x10-12	
DB Skullcrushers 2-3x12-15		Back Raises 3-4x10-12		Tricep Pushdowns 2-3x12-15	Single Leg Glute Bridge 3x10-12 each leg	
		Walking Lunges 2x30 Steps				

I do not count these smaller assistance exercises when considering MEV/MRV because they are not loaded significantly enough but if you start taking these exercises to failure often or trying to push for significant PRs on them, then you should include them but keep in mind that 3 sets of DB Front Raises should not be considered the equivalent of 3 sets of DB Military Press.

Progressing volume from one week to another can be a tough point of program design to understand. Our goal is to move from the athlete's MEV toward their MRV over the course of a Mesocyle for our example athlete that means we will aim to progress from:

Squat: 8 Sets/Week to 18 Sets/Week
Bench: 9.5 Sets/Week to 21 Sets/Week
Deadlift: 6 Sets/Week to 15 Sets/Week

Our above example shows the athlete's MEV spread out over the course of a week over their optimal frequency, below we will show how to move toward the athlete's MRV during his 3+1 Hypertrophy Block.

Day 1	Week 1	Week 2	Week 3	Deload
Spoto Press	Up to 8rm at 8RPE. Drop 8-12%x4x8	Up to 8rm at 9RPE. Drop 8-12%x5x8	Up to 8rm at 10RPE. Drop 8-12%x6x8	5x8 at 75% of 8rm from Week 3
Chest Supported DB Rows	5x10-12	5-6x10-12	6x10-12	4x10
DB Front Raises	2-3x12-15	3x12-15	3-4x12-15	2x12
DB Skullcrushers	2-3x12-15	3x12-15	3-4x12-15	2x12
Day 2				
Off				
Day 3				
High Bar Squat	Up to 8rm at 8RPE. Drop 8-12%x4x8	Up to 8rm at 9RPE. Drop 8-12%x5x8	Up to 8rm at 10RPE. Drop 8-12%x6x8	5x8 at 75% of 8rm from Week 3
Sumo Halting Deadlift	2x6 at 55-60%	3x6 at 57.5-62.5%	4x6 at 60-65%	2x6 at 50%
DB Military Press	2x10-12 at 55-60%	3x10-12 at 57.5-62.5%	4x10-12 at 60-65%	2x10 at 50%
Back Raises	3-4x10-12	4x10-12	4-5x10-12	3x10
Walking Lunges	2x30 Steps	2x40 Steps	2x50 Steps	2x20 Steps
Day 4				
Off				
Day 5				
Incline Bench	3x8-10 at 60-65%	4x8-10 at 62.5-67.5%	5x8-10 at 65-70%	3x8 at 55%
Lat Pulldowns	5x10-12	5-6x10-12	6x10-12	4x10
DB Lateral Raises	2-3x12-15	3x12-15	3-4x12-15	2x12
Tricep Pushdowns	2-3x12-15	3x12-15	3-4x12-15	2x12
Day 6				
Conventional Deadlift	Up to 6rm at 8RPE. Drop 8-12%x3x6	5x6 at 90% of 6rm from Week 1	Up to 6rm at 10RPE. Drop 8-12%x5x6	4x6 at 75% of 6rm from Week 3
Front Squat	3x8 at 60-65%	4x8 at 62.5-67.5%	5x8 at 65-70%	3x8 at 50%
Hamstring Curls	3x10-12	3-4x10-12	4x10-12 at 60-65%	3x10
Single Leg Glute Bridges	3x10-12 each leg	3-4x10-12 each leg	4x10-12 each leg	3x10 each leg
Day 7				
Off				
Total Overloading Squat Sets	8 Sets	10 Sets	12 Sets	8 Sets
Total Overloading Bench Sets	10 Sets	13 Sets	16 Sets	10 Sets
Total Overloading Deadlift Sets	6 Sets	8 Sets	10 Sets	6 Sets

You can see in this example that the athlete doesn't progress all the way to their MRV in Week 3 which is fine and somewhat expected because the training cycle is relatively short. If they were to perform 2 consecutive Hypertrophy Blocks, they would likely start Week 1 of Block 2 at the same volume as Week 2 of Block 1 and progress up to their MRV by Week 3. The rate at which volume is added to the program should be dependent on their feedback and how they are feeling from the training. The way that the accessory work is progressed above is one option, you could keep the number of sets flat throughout the training cycle and just progress the weights or only increase the sets and keep the weights the same. Focus on increasing the workload of primary exercises first and accessory work after that if your goal is being the best powerlifter.

Notice that in the Deload Week, there is a reduction to ~65% of the volume of the previous week at ~85% of the average intensity. Deload weeks should be at 50-70% of normal overloading volume and 80-90% of average overloading intensity. Let the athlete's fatigue dictate where in those ranges the work is done.

Now the athlete has completed their Hypertrophy Phase, it is time to transition to Strength. Key considerations now are the need to overload intensity over the course of the mesocycle and moving towards more specific exercise selections. Overloading intensity brings with it some Fatigue Management considerations, particularly in regards to Neural and Joint/Tendon Integrity. While Frequency can typically stay the same as during the Hypertrophy Phase, Strength may require greater modulation/undulation from session to session or week to week, for this reason, we will employ an Alternating Model over a 4+1 paradigm.

While MEV and MRV are important considerations for any phase, when moving from Hypertrophy to Strength, I tend to derive the starting volume of Strength by beginning at or slightly (10%) below where we finished Hypertrophy.
Here is how I would design a Strength Phase for our example athlete.

Day 1	Week 1	Week 2	Week 3	Week 4	Deload
Bench Press	6x6 at 70-75%	Up to 6rm at 9RPE. Drop 10-15%x4x6	5x4 at 75-80%	Up to 4rm at 10RPE. Drop 10-15%x3-4x4	3x4 at 80% of 4rm from Week 4
Chest Supported DB Rows	5x8-10	5x8-10	5x8-10	5x8-10	4x8
DB Front Raises	2-3x10-12	2-3x10-12	2-3x10-12	2-3x10-12	2x10
DB Skullcrushers	2-3x10-12	2-3x10-12	2-3x10-12	2-3x10-12	2x10
Day 2					
Off					
Day 3					
Low Bar Squat	6x5 at 70-75%	Up to 5rm at 9RPE. Drop 10-15%x4x5	5x3 at 75-80%	Up to 3rm at 10RPE. Drop 10-15%x3-4x3	3x3 at 80% of 3rm from Week 4
1" Deficit Sumo Deadlift	3x5 at 70-75%	3x5 at 72.5-77.5%	3x4 at 75-80%	3x3 at 77.5-82.5%	2x3 at 65%
Incline Bench	3x6 at 70-75%	3x6 at 72.5-77.5%	3x5 at 75-80%	3x4 at 77.5-82.5%	2x4 at 70%
Back Raises	3x8-10	3x8-10	3x8-10	3x8-10	2x8
Split Squats	3x8 each leg	3x8 each leg	3x8 each leg	3x8 each leg	2x8 each leg
Day 4					
Off					
Day 5					
Spoto Press	5x6 at 70-75%	5x6 at 72.5-77.5%	5x5 at 75-80%	5x4 at 77.5-82.5%	4x4 at 70%
Lat Pulldowns	5x8-10	5x8-10	5x8-10	5x8-10	4x8
DB Lateral Raises	2-3x10-12	2-3x10-12	2-3x10-12	2-3x10-12	2x10
Tricep Pushdowns	2-3x10-12	2-3x10-12	2-3x10-12	2-3x10-12	2x10
Day 6					
Sumo Deadlift	Up to 4rm at 9RPE. Drop 10-15%x3-4x4	4x4 at 75-80%	Up to 3rm at 10RPE. Drop 10-15%x3x3	4x3 at 80-85%	3x3 at 80% of 3rm from Week 3
High Bar Squat	4x6 at 70-75%	4x6 at 72.5-77.5%	4x5 at 75-80%	4x4 at 77.5-82.5%	3x4 at 65%
Hamstring Curls	3x8-10	3x8-10	3x8-10	3x8-10	2x8
Single Leg Glute Bridges	3x8 each leg	3x8 each leg	3x8 each leg	3x8 each leg	2x8 each leg
Day 7					
Off					

Notice a few key components of the above program.

Alternating Periodization is Applied

- The primary work for the Squat and Bench are heavy on the opposite week that the heavy deadlift is being trained. This is done to help more energy to be available for heavy training compared to if all 3 were done heavy during the same training week.

Specificity of Exercise Selection Increased

- The athlete is know performing their competition techniques weekly. Also, exercises like Front Squat and DB Military Press have been traded out for slightly more transferrable variations, High Bar Squat and Incline Bench.

- Variation is still present during this phase though as the athlete has unique weak points to address which strategically selected variations can help with. Also, variations that manage output (ie. Most people can't go as heavy on Spoto Press as they can Competition Bench Press) will allow the athlete to still train hard without accumulating as much fatigue.

Accessory Work Is Lower Priority

- Accessory exercises are inherently general, so as the program moves to more specificity closer to competition, they should occupy less of your total training.

- Accessory work is important to maintaining Hypertrophy, so don't drop it out of your training entirely.

- Overloading the accessory work from week to week is preferable but isn't a must, particularly if you feel it is interfering with your energy to train the bigger movements or your ability to recover.

Day 1	Week 1	Week 2	Week 3	Taper
Bench Press	Up to 2rm at 9.5RPE. Drop 10-15%x4x2	5x3 at 80-85%	Up to 1rm at 9.5RPE. Drop 10-15%x3x1-2	Squat 6x1-2 at 70%
Chest Supported DB Rows	5x6-8	5x6-8	4x6-8	Bench 8x1-2 at 75%
DB Front Raises	2-3x8-10	2-3x8-10	2x8-10	Deadlift 5x1 at 65%
DB Skullcrushers	2-3x8-10	2-3x8-10	2x8-10	
Day 2				
Off				Off
Day 3				
Low Bar Squat	Up to 2rm at 9.5RPE. Drop 10-15%x3x2	4x3 at 80-85%	Up to 1rm at 9.5RPE. Drop 10-15%x2x1-2	Squat 6x1-2 at 60%
Sumo Deadlift	6x1 at 60%	5x1 at 60%	4x1 at 60%	Bench Up to 1 at 8RPE. 6x1-2 at 70%
Bench Press	6x1-2 at 70%	5x1-2 at 70%	4x1-2 at 70%	Deadlift 5x1 at 55%
Back Raises	2-3x8-10	2-3x8-10	2x8-10	
Day 4				
Off				Off
Day 5				
Spoto Press	5x5 at 75-80%	5x4 at 77.5-82.5%	5x3 at 80-85%	Squat 5x1-3 at 20-50%
Lat Pulldowns	5x6-8	5x6-8	4x6-8	Bench 6x1-3 at 30-60%
DB Lateral Raises	2-3x8-10	2-3x8-10	2x8-10	Deadlift 4x1 at 20-50%
Tricep Pushdowns	2-3x8-10	2-3x8-10	2x8-10	
Day 6				
Sumo Deadlift	4x2 at 85%	Up to 1rm at 9.5RPE. Drop 10-15%x2-3x1	From 3” Blocks-3x1 at 95% of 1rm from Week 2	Compete
Low Bar Squat	6x1-2 at 65%	5x1-2 at 65%	4x1-2 at 65%	
Split Squats	2-3x6 each leg	2-3x6 each leg	2x6 each leg	
Day 7				
Off				

Key points of this phase of the program to note are:

- Specificity is very high with almost all focus on the competition exercises.

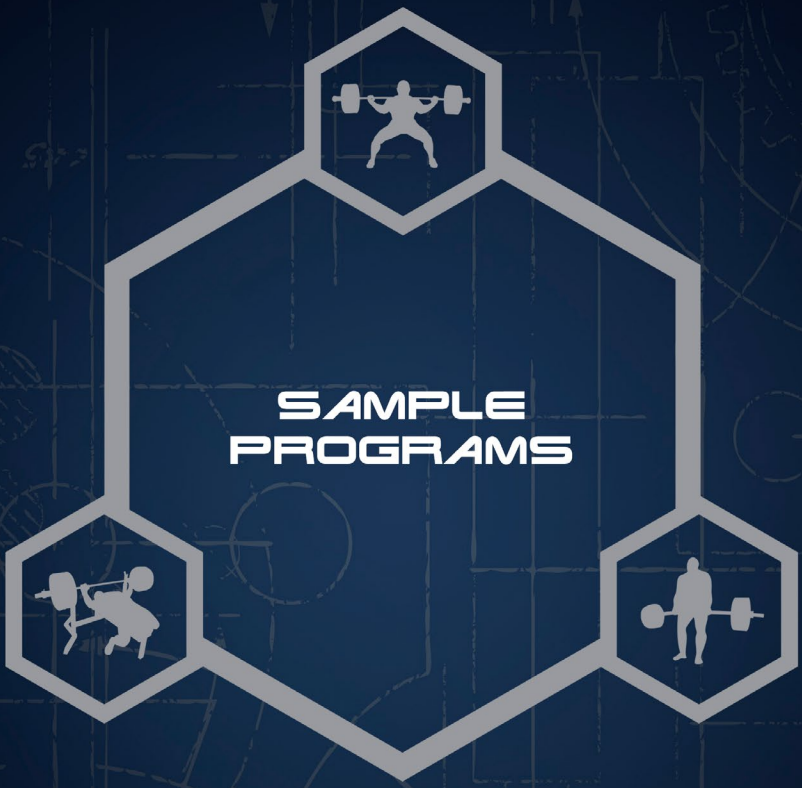
- Bench Press training still includes the Spoto Press to address the specific weakness of Midrange Strength and is done so in slightly higher volume
because the Bench Press can tolerate higher volume and it is helpful to avoid losing hypertrophy.

- Accessory work is further reduced in priority.

- The final Sumo Deadlift workout is done from 3" blocks to allow for Neural adaptations to still occur by handling heavy weight but at a slightly lower fatigue cost because of the shorter ROM.

- Technical sessions are utilized for each lift. These will help improve technique, as well improving recovery through improved blood flow to the specific muscles.

Thank you for taking the time to read and absorb all of this information. I'm looking forward to creating continuing education opportunities around this book to help you improve your understanding and application of these concepts. Here are some more sample scenarios and programs for you to look over and understand how I practically apply the ideas of this text into actual programs. Good luck!



SAMPLE PROGRAMS

Scenario #1

Athlete Assessment:

Sex-Male	Weight-Middle	Height-Medium
Strength-Low	Experience-Beginner	Age-24 Years Old
Diet-Average	Sleep-Good	Stress-Low
Historical Workload-2	Historical Recovery Ability-3	Time to Meet-20 Weeks

Competition Technique/Weaknesses:

Squat	Bench	Deadlift
Low Bar	Medium Grip	Sumo
Weak Legs/Strong Back	Off the Chest	Off the Floor

MEVs:

Squat	Bench	Deadlift
5 Sets/Week	7.5 Sets/Week	3 Sets/Week

MRVs:

Squat	Bench	Deadlift
19 Sets/Week	22 Sets/Week	16 Sets/Week

Frequency:

Training Days	Squat	Bench	Deadlift
5 Days/Week	2-3x/Week	3-4x/Week	2x/Week

Phasic Structure:

Considering this athlete's Size, Strength, Experience and Age, along with the wide gap between his MEV and MRV, a Linear Periodization would be most appropriate for them. Also, because of the big difference in MEV and MRV, a longer mesocycle length should be utilized. With 20 weeks until competition, this athlete would benefit most from 10 Weeks of Hypertrophy, 6 Weeks of Strength and 4 Weeks of Peaking. That will break down as two 4+1 Hypertrophy Blocks, one 5+1 Strength Block and a 3+1 Peaking Block.

Hypertrophy Phase 1

Day 1	Week 1	Week 2	Week 3	Week 4	Deload
High Bar Squat	Up to 8rm at 7RPE, Drop 8-12%x8	Up to 8rm at 8RPE, Drop 8-12%x2x8	Up to 8rm at 9RPE, Drop 8-12%x3x8	Up to 8rm at 10RPE, Drop 8-12%x4x8	3x8 at 75% of 8rm from Week 4
Feet Up Bench	1x8-10 at 60%	2x8-10 at 63%	3x8-10 at 65%	4x8-10 at 68%	3x8 at 55%
Walking Lunges	2x20 Steps	2x30 Steps	3x30 Steps	3x40 Steps	2x20 Steps
Day 2					
Widegrip Bench	Up to 8rm at 7RPE, Drop 8-12%x2x8	Up to 8rm at 8RPE, Drop 8-12%x3x8	Up to 8rm at 9RPE, Drop 8-12%x4x8	Up to 8rm at 10RPE, Drop 8-12%x5x8	4x8 at 75% of 8rm from Week 4
Lat Pulldown or DB Rows	3x10-12	3-4x10-12	4x10-12	4-5x10-12	4x10
DB Flies	2-3x12-15	2-3x12-15	3x12-15	3-4x12-15	2-3x12
DB Front Raises	2-3x12-15	2-3x12-15	3x12-15	3-4x12-15	2-3x12
DB Skullcrushers	2-3x12-15	2-3x12-15	3x12-15	3-4x12-15	2-3x12
Day 3					
Conventional Deadlift	Up to 6rm at 7RPE, Drop 8-12%x6	Up to 6rm at 8RPE, Drop 8-12%x2x6	Up to 6rm at 9RPE, Drop 8-12%x3x6	Up to 6rm at 10RPE, Drop 8-12%x4x6	2x6 at 75% of 6rm from Week 4
Belt Squat	1x10-12 at 6RPE	2x10-12 at 6-7RPE	3x10-12 at 7RPE	4x10-12 at 8RPE	3x10-12 at 6RPE
Back Raises	3x10-12	3-4x10-12	4x10-12	4-5x10-12	3-4x10
Day 4					
Off					
Day 5					
High Bar Squat	2x8 at 85% of 8rm from Day 1	3x8 at 85% of 8rm from Day 1	4x8 at 85% of 8rm from Day 1	5x8 at 85% of 8rm from Day 1	3x8 at 75% of 8rm from Week 4
1" Deficit Dead	1x6 at 85% of 6rm from Day 3	2x6 at 85% of 6rm from Day 3	3x6 at 85% of 6rm from Day 3	4x6 at 85% of 6rm from Day 3	3x6 at 75% of 6rm from Week 4
DB Bench	1x10 at 6RPE	2x10 at 6-7RPE	3x10 at 7RPE	4x10 at 8RPE	3x10 at 6RPE
Split Squats	2x10 each leg	2x10 each leg	3x10 each leg	3x10 each leg	2x10 each leg
Day 6					
Widegrip Bench	3x8 at 85% of 8rm from Day 2	4x8 at 85% of 8rm from Day 2	5x8 at 85% of 8rm from Day 2	6x8 at 85% of 8rm from Day 2	4x8 at 75% of 8rm from Week 4
Lat Pulldown or DB Rows	3x10-12	3-4x10-12	4x10-12	4-5x10-12	4x10
Pushups	2-3x12-15	2-3x12-15	3x12-15	3-4x12-15	2-3x12
DB Lateral Raises	2-3x12-15	2-3x12-15	3x12-15	3-4x12-15	2-3x12
Tricep Pushdowns	2-3x12-15	2-3x12-15	3x12-15	3-4x12-15	2-3x12
Day 7					
Off					

Hypertrophy Phase 2

Day 1	Week 1	Week 2	Week 3	Week 4	Deload
High Bar Squat	Up to 8rm at 7RPE, Drop 8-12%x2x8	Up to 8rm at 8RPE, Drop 8-12%x3x8	Up to 8rm at 9RPE, Drop 8-12%x4x8	Up to 8rm at 10RPE, Drop 8-12%x5x8	4x8 at 75% of 8rm from Week 4
Feet Up Bench	2x8-10 at 63%	3x8-10 at 65%	4x8-10 at 68%	5x8-10 at 70%	4x8 at 55%
Walking Lunges	2x30 Steps	3x30 Steps	3x40 Steps	4x40 Steps	2x20 Steps
Day 2					
Widegrip Bench	Up to 8rm at 7RPE, Drop 8-12%x3x8	Up to 8rm at 8RPE, Drop 8-12%x4x8	Up to 8rm at 9RPE, Drop 8-12%x5x8	Up to 8rm at 10RPE, Drop 8-12%x6x8	5x8 at 75% of 8rm from Week 4
Lat Pulldown or DB Rows	4x10-12	4-5x10-12	5x10-12	5-6x10-12	5x10
DB Flies	2-3x12-15	3x12-15	3-4x12-15	4x12-15	2-3x12
DB Front Raises	2-3x12-15	3x12-15	3-4x12-15	4x12-15	2-3x12
DB Skullcrushers	2-3x12-15	3x12-15	3-4x12-15	4x12-15	2-3x12

Day 3					
Conventional Deadlift	Up to 6rm at 7RPE, Drop 8-12%x2x6	Up to 6rm at 8RPE, Drop 8-12%x3x6	Up to 6rm at 9RPE, Drop 8-12%x4x6	Up to 6rm at 10RPE, Drop 8-12%x5x6	3x6 at 75% of 6rm from Week 4
Belt Squat	2x10-12 at 6RPE	3x10-12 at 6-7RPE	4x10-12 at 7RPE	5x10-12 at 8RPE	3x10-12 at 6RPE
Back Raises	3-4x10-12	4x10-12	4-5x10-12	5x10-12	3-4x10
Day 4					
Off					
Day 5					
High Bar Squat	3x8 at 85% of 8rm from Day 1	4x8 at 85% of 8rm from Day 1	5x8 at 85% of 8rm from Day 1	6x8 at 85% of 8rm from Day 1	4x8 at 75% of 8rm from Week 4
1" Deficit Dead	2x6 at 85% of 6rm from Day 3	3x6 at 85% of 6rm from Day 3	4x6 at 85% of 6rm from Day 3	5x6 at 85% of 6rm from Day 3	3x6 at 75% of 6rm from Week 4
DB Bench	2x10 at 6RPE	3x10 at 6-7RPE	4x10 at 7RPE	5x10 at 8RPE	3x10 at 6RPE
Split Squats	3x10 each leg	3x10 each leg	4x10 each leg	4x10 each leg	2x10 each leg
Day 6					
Widegrip Bench	4x8 at 85% of 8rm from Day 2	5x8 at 85% of 8rm from Day 2	6x8 at 85% of 8rm from Day 2	7x8 at 85% of 8rm from Day 2	4x8 at 75% of 8rm from Week 4
Lat Pulldown or DB Rows	4x10-12	4-5x10-12	5x10-12	5-6x10-12	5x10
Pushups	2-3x12-15	3x12-15	3-4x12-15	4x12-15	2-3x12
DB Lateral Raises	2-3x12-15	3x12-15	3-4x12-15	4x12-15	2-3x12
Tricep Pushdowns	2-3x12-15	3x12-15	3-4x12-15	4x12-15	2-3x12
Day 7					
Off					

Strength Phase

Day 1	Week 1	Week 2	Week 3	Week 4	Week 5	Deload
Low Bar Squat	Up to 5rm at 6RPE, Drop 8-12%x6x5	Up to 5rm at 7RPE, Drop 8-12%x5x5	Up to 5rm at 8RPE, Drop 8-12%x4-5x5	Up to 5rm at 9RPE, Drop 8-12%x4x5	Up to 5rm at 10RPE, Drop 8-12%x3-4x5	3x5 at 75% of 5rm from Week 5
Feet Up Bench	4x6 at 70%	4x6 at 73%	4x5 at 75%	3-4x5 at 78%	3x4 at 80%	2x4 at 70%
Walking Lunges	3x20 Steps	3x20 Steps	3x20 Steps	3x20 Steps	3x20 Steps	2x20 Steps
Day 2						
Bench Press	Up to 5rm at 6RPE, Drop 8-12%x7x5	Up to 5rm at 7RPE, Drop 8-12%x6x5	Up to 5rm at 8RPE, Drop 8-12%x5-6x5	Up to 5rm at 9RPE, Drop 8-12%x5x5	Up to 5rm at 10RPE, Drop 8-12%x4-5x5	4x5 at 75% of 5rm from Week 5
Lat Pulldown or DB Rows	5x8-10	5x8-10	5x8-10	5x8-10	5x8-10	4x8
DB Flies	3x10-12	3x10-12	3x10-12	3x10-12	3x10-12	2x10
DB Front Raises	3x10-12	3x10-12	3x10-12	3x10-12	3x10-12	2x10
DB Skullcrushers	3x10-12	3x10-12	3x10-12	3x10-12	3x10-12	2x10
Day 3						
Sumo Deadlift	Up to 3rm at 6RPE, Drop 8-12%x5x3	Up to 3rm at 7RPE, Drop 8-12%x4-5x3	Up to 3rm at 8RPE, Drop 8-12%x4x3	Up to 3rm at 9RPE, Drop 8-12%x3-4x3	Up to 3rm at 10RPE, Drop 8-12%x3x3	3x3 at 75% f 3rm from Week 5
Belt Squat	3x10-12 at 6RPE	3x10-12 at 6-7RPE	3x10-12 at 7RPE	3x10-12 at 7-8RPE	3x10-12 at 8RPE	2x10 at 6RPE
Back Raises	4x8-10	4x8-10	4x8-10	4x8-10	4x8-10	3x8
Day 4						
Off						
Day 5						
High Bar Squat	5x6 at 70%	5x6 at 73%	5x5 at 75%	4-5x5 at 78%	4x4 at 80%	3x4 at 70%
1" Conv Deficit Deadlift	3x5 at 70%	3x5 at 73%	3x4 at 75%	3x4 at 78%	3x3 at 80%	2x3 at 70%
Widegrip Bench	3x6 at 70%	3x6 at 73%	3x5 at 75%	3x5 at 78%	3x4 at 80%	2x4 at 70%

Split Squats	2x8 each leg	2x8 each leg	2x8 each leg	2x8 each leg	2x8 each leg	2x8 each leg

Day 6						
Bench Press	5x5 at 85% of 5rm from Day 2	4-5x5 at 85% of 5rm from Day 2	4x5 at 85% of 5rm from Day 2	3-4x5 at 85% of 5rm from Day 2	3x5 at 85% of 5rm from Day 2	3x5 at 75% of 5rm from Week 5
Lat Pulldown or DB Rows	5x8-10	5x8-10	5x8-10	5x8-10	5x8-10	4x8
DB Bench	3x10-12 at 6RPE	3x10-12 at 6-7RPE	3x10-12 at 7RPE	3x10-12 at 7-8RPE	3x10-12 at 8RPE	2x10 at 6RPE
DB Lateral Raises	3x10-12	3x10-12	3x10-12	3x10-12	3x10-12	2x10
Tricep Pushdowns	3x10-12	3x10-12	3x10-12	3x10-12	3x10-12	2x10
Day 7						
Off						

Peaking Phase

Day 1	Week 1	Week 2	Week 3	Taper
Low Bar Squat	Up to 3rm at 9RPE, Drop 10-15%x4x3	Up to 2rm at 9RPE, Drop 10-15%x3-4x2	Up to 1rm at 9RPE, Drop 10-15%x3x1	Low Bar Squat 4x2 at 70%
Bench Press	5x2 at 70%	4x2 at 70%	3x2 at 70%	Bench Press 5x2 at 75%
Walking Lunges	2x16 Steps	2x14 Steps	2x12 Steps	Deadlift 4x1 at 65%
Day 2				
Bench Press	Up to 3rm at 9RPE, Drop 10-15%x5x3	Up to 2rm at 9RPE, Drop 10-15%x5x2	Up to 1rm at 9RPE, Drop 10-15%x5x1	Off
Lat Pulldown or DB Rows	5x6-8	5x6-8	5x6-8	
DB Flies	2-3x8-10	2-3x8-10	2-3x8-10	
DB Front Raises	2-3x8-10	2-3x8-10	2-3x8-10	
DB Skullcrushers	2-3x8-10	2-3x8-10	2-3x8-10	
Day 3				
Sumo Deadlift	Up to 2rm at 9RPE, Drop 10-15%x3x2	4x2 at 90-100% of Week 1-From 3" Blocks	Up to 1rm at 9RPE, Drop 10-15%x2-3x1	Low Bar Squat 4x2 at 65%
Low Bar Squat	5x2 at 70%	4x2 at 70%	3x2 at 70%	Bench Press Up to 1 at 90%, 4x2 at 70%
Back Raises	3x6-8	3x6-8	3x6-8	Deadlift 4x1 at 60%
Day 4				
Off				Off
Day 5				
Low Bar Squat	5x3 at 85%	4x2-3 at 88%	3x2 at 90%	Low Bar Squat 4x1 at 50-60%
Sumo Deadlift	5x1 at 70%	4x1 at 70%	3x1 at 70%	Bench Press 5x1 at 60-70%
Split Squats	2x8 each leg	2x6 each leg	2x5 each leg	Deadlift 3x1 at 40-50%
Day 6				
Widegrip Bench	5x5 at 78%	5x4 at 80%	5x3 at 83%	Compete
Lat Pulldown or DB Rows	5x6-8	5x6-8	5x6-8	
DB Bench	2-3x8-10	2-3x8-10	2-3x8-10	
DB Lateral Raises	2-3x8-10	2-3x8-10	2-3x8-10	
Tricep Pushdowns	2-3x8-10	2-3x8-10	2-3x8-10	
Day 7				
Off				

Scenario #2

Athlete Assessment:

Sex-Female	Weight-Light	Height-Short
Strength-Moderate	Experience-Intermediate	Age-31 Years Old
Diet-Average	Sleep-Good	Stress-Average
Historical Workload-3	Historical Recovery Ability-3	Time to Meet-16 Weeks

Competition Technique/Weaknesses:

Squat	Bench	Deadlift
Low Bar	Wide Grip	Conventionals
Out of the Hole/Balanced	Midrange	Lockout

MEVs:

Squat	Bench	Deadlift
11.5 Sets/Week	13 Sets/Week	9.5 Sets/Week

MRVs:

Squat	Bench	Deadlift
25 Sets/Week	28 Sets/Week	19 Sets/Week

Frequency:

Training Days	Squat	Bench	Deadlift
4 Days/Week	3x/Week	4x/Week	2x/Week

Phasic Structure:

Considering this athlete's Size, Strength, Experience and Age, along with the wide gap between her MEV and MRV, a Linear Periodization would be most appropriate for them. Also, because of the big difference in MEV and MRV, a longer mesocycle length should be utilized. With 16 weeks until competition, this athlete would benefit most from 6 Weeks of Hypertrophy, 6 Weeks of Strength and 4 Weeks of Peaking. That will break down as one 5+1 Hypertrophy Block, one 5+1 Strength Block and a 3+1 Peaking Block.

Hypertrophy Phase

Day 1	Week 1	Week 2	Week 3	Week 4	Week 5	Deload
High Bar Squat	Up to 8rm at 6RPE, Drop 8-12%x4x8	Up to 8rm at 7RPE, Drop 8-12%x5x8	Up to 8rm at 8RPE, Drop 8-12%x5-6x8	Up to 8rm at 9RPE, Drop 8-12%x6x8	Up to 8rm at 10RPE, Drop 8-12%x6-7x8	5x8 at 75% of 8rm from Week 5
Spoto Press	2x8 at 60%	3x8 at 63%	3-4x8 at 65%	4x8 at 68%	4-5x8 at 70%	3-4x8 at 60%
RDL	3x8 at 6RPE	3-4x8 at 6-7RPE	4x8 at 7RPE	4-5x8 at 7-8RPE	5x8 at 8RPE	4x8 at 6RPE
Walking Lunges	2x20 Steps	2x30 Steps	3x30 Steps	3x40 Steps	4x40 Steps	3x20 Steps
Day 2						
Off						
Day 3						
Bench Press	Up to 8rm at 6RPE, Drop 8-12%x5x8	Up to 8rm at 7RPE, Drop 8-12%x5-6x8	Up to 8rm at 8RPE, Drop 8-12%x6x8	Up to 8rm at 9RPE, Drop 8-12%x6-7x8	Up to 8rm at 10RPE, Drop 8-12%x7x8	6x8 at 75% of 8rm from Week 5
Pause HB Squat	2x8 at 6RPE	2-3x8 at 6-7RPE	3x8 at 7RPE	3-4x8 at 7-8RPE	4x8 at 8RPE	3x8 at 6RPE
Bentover Rows	2x8 at 6RPE	3x8 at 6-7RPE	3-4x8 at 7RPE	4x8 at 7-8RPE	4-5x8 at 8RPE	3-4x8 at 6RPE
Lat Pulldown or DB Rows	4x10-12	4x10-12	5x10-12	5x10-12	6x10-12	5x10
DB Front Raises	2-3x12-15	2-3x12-15	3x12-15	3x12-15	3-4x12-15	3x12
DB Skullcrushers	2-3x12-15	2-3x12-15	3x12-15	3x12-15	3-4x12-15	3x12
Day 4						
Off						
Day 5						
2” Block Pull	Up to 6rm at 6RPE, Drop 8-12%x3x6	Up to 6rm at 7RPE, Drop 8-12%x3-4x6	Up to 6rm at 8RPE, Drop 8-12%x4x6	Up to 6rm at 9RPE, Drop 8-12%x4-5x6	Up to 6rm at 10RPE, Drop 8-12%x5x6	4x6 at 75% of 6rm from Week 5

High Bar Squat	2x8 at 85% of 8rm from Day 1	3x8 at 85% of 8rm from Day 1	3x8 at 85% of 8rm from Day 1	4x8 at 85% of 8rm from Day 1	4x8 at 85% of 8rm from Day 1	3x8 at 75% of 8rm from Week 5
Pause HB Squat	2x8 at 6RPE	2x8 at 6-7RPE	3x8 at 7RPE	3x8 at 7-8RPE	4x8 at 8RPE	2x8 at 6RPE
DB Military Press	2x8-10 at 6RPE	2-3x8-10 at 6-7RPE	3x8-10 at 7RPE	3-4x8-10 at 7-8RPE	4x8-10 at 8RPE	3x8 at 6RPE
Back Raises	3x10-12	3-4x10-12	4x10-12	4-5x10-12	5x10-12	4x10
Day 6						
Incline Bench	3x8 at 6RPE	3-4x8 at 6-7RPE	4x8 at 7RPE	4-5x8 at 7-8RPE	5x8 at 8RPE	3-4x8 at 6RPE
Lat Pulldown or DB Rows	4x10-12	4x10-12	5x10-12	5x10-12	6x10-12	5x10
DB Lateral Raises	2-3x12-15	2-3x12-15	3x12-15	3x12-15	3-4x12-15	3x12
Tricep Pushdowns	2-3x12-15	2-3x12-15	3x12-15	3x12-15	3-4x12-15	3x12
Day 7						
Off						

Strength Phase

Day 1	Week 1	Week 2	Week 3	Week 4	Week 5	Deload
Low Bar Squat	Up to 5rm at 6RPE, Drop 8-12%x7x5	Up to 5rm at 7RPE, Drop 8-12%x6x5	Up to 5rm at 8RPE, Drop 8-12%x5-6x5	Up to 5rm at 9RPE, Drop 8-12%x5x5	Up to 5rm at 10RPE, Drop 8-12%x4-5x5	4x5 at 75% of 5rm from Week 5
Spoto Press	4x6 at 70%	4x6 at 73%	3-4x6 at 75%	3x5 at 78%	3x5 at 80%	2x5 at 65%
2" Block Pull	4-5x5 at 70%	4x5 at 73%	3-4x5 at 75%	3x4 at 78%	3x3 at 80%	2x3 at 65%
Walking Lunges	3x20 Steps	3x20 Steps	3x20 Steps	3x20 Steps	3x20 Steps	2x20 Steps
Day 2						
Off						
Day 3						
Bench Press	Up to 5rm at 6RPE, Drop 8-12%x8x5	Up to 5rm at 7RPE, Drop 8-12%x7x5	Up to 5rm at 8RPE, Drop 8-12%x6-7x5	Up to 5rm at 9RPE, Drop 8-12%x6x5	Up to 5rm at 10RPE, Drop 8-12%x5-6x5	5x5 at 75% of 5rm from Week 5
Dead Squat	8x1 at 70%-45 sec rest	8x1 at 73%-45 sec rest	8x1 at 75%-45 sec rest	6x1 at 78%-60 sec rest	5x1 at 80%-75 sec rest	4x1 at 70%-75 sec rest
Bentover Rows	5x6 at 6RPE	5x6 at 6-7RPE	4-5x6 at 7RPE	4x6 at 7-8RPE	3-4x6 at 8RPE	3x6 at 6RPE
Lat Pulldown or DB Rows	5x8-10	5x8-10	5x8-10	5x8-10	5x8-10	4x8
DB Front Raises	3x10-12	3x10-12	3x10-12	3x10-12	3x10-12	2x10
DB Skullcrushers	3x10-12	3x10-12	3x10-12	3x10-12	3x10-12	2x10
Day 4						
Off						
Day 5						
Deadlift	Up to 3rm at 6RPE, Drop 8-12%x6x3	Up to 3rm at 7RPE, Drop 8-12%x5-6x3	Up to 3rm at 8RPE, Drop 8-12%x5x3	Up to 3rm at 9RPE, Drop 8-12%x4-5x3	Up to 3rm at 10RPE, Drop 8-12%x4x3	4x3 at 75% of 3rm from Week 5
Low Bar Squat	4x5 at 85% of 5rm from Day 1	4x5 at 85% of 5rm from Day 1	3-4x5 at 85% of 5rm from Day 1	3-4x5 at 85% of 5rm from Day 1	3x5 at 85% of 5rm from Day 1	
Pause LB Squat	3x5 at 75% of 5rm from Day 1	3x5 at 75% of 5rm from Day 1	3x5 at 75% of 5rm from Day 1	2-3x5 at 75% of 5rm from Day 1	2x5 at 75% of 5rm from Day 1	
DB Military Press	4x6-8 at 6RPE	3-4x6-8 at 6-7RPE	3x6-8 at 7RPE	2-3x6-8 at 7-8RPE	2x6-8 at 8RPE	2x8 at 6RPE
Back Raises	4x8-10	4x8-10	3-4x8-10	3-4x8-10	3x8-10	2-3x8
Day 6						
Incline Bench	5x6 at 6RPE	4-5x6 at 6-7RPE	4x6 at 7RPE	3-4x6 at 7-8RPE	3x6 at 8RPE	2-3x6 at 6RPE
Lat Pulldown or DB Rows	5x8-10	5x8-10	5x8-10	5x8-10	5x8-10	4x8
DB Lateral Raises	3x10-12	3x10-12	3x10-12	3x10-12	3x10-12	2x10

Tricep Pushdowns	3x10-12	3x10-12	3x10-12	3x10-12	3x10-12	2x10
Day 7						
Off						

Peaking Phase

Day 1	**Week 1**	**Week 2**	**Week 3**	**Taper**
Low Bar Squat	Up to 3rm at 9RPE, Drop 8-12%x6x3	Up to 2rm at 9RPE, Drop 8-12%x5x2	Up to 1rm at 9RPE, Drop 8-12%x4x1	Low Bar Squat 5x2 at 75%
Bench Press	5x3 at 80%	4-5x3 at 83%	4x3 at 85%	Bench Press 6x2 at 80%
Deadlift	8x1 at 70%	7x1 at 70%	6x1 at 70%	Deadlift 5x1 at 70%
Walking Lunges	2x16 Steps	2x14 Steps	2x12 Steps	
Day 2				
Off				
Day 3				
Bench Press	Up to 3rm at 9RPE, Drop 8-12%x6x3	Up to 2rm at 9RPE, Drop 8-12%x5x2	Up to 1rm at 9RPE, Drop 8-12%x4x1	Low Bar Squat 5x2 at 70%
Slingshot Bench	2x3 at 105% of 3rm-Before Drop Sets	None	2x1 at 105% of 1rm-Before Drop Sets	Bench Press Up to 1 at 90%, 5x2 at 75%
Pause LB Squat	4x3 at 80%	3-4x3 at 83%	3x3 at 85%	Deadlift 5x1 at 60%
Lat Pulldown or DB Rows	6x6-8	5-6x6-8	5x6-8	
DB Front Raises	2-3x8-10	2-3x8-10	2-3x8-10	
DB Skullcrushers	2-3x8-10	2-3x8-10	2-3x8-10	
Day 4				
Off				Off
Day 5				
Deadlift	Up to 2rm at 9RPE, Drop 10-15%x5x2	6x2 at 85-95% of Week 1-From 2" Blocks	Up to 1rm at 9RPE, Drop 10-15%x3-4x1	Low Bar Squat 4x1 at 50-60%

Low Bar Squat	6x2 at 90% of 3rm from Day 1	5x1 at 90% of 2rm from Day 1	3x1 at 85% of 1rm from Day 1	Bench Press 5x1 at 60-70%
Bench Press	6x2 at 75%	5x2 at 70%	4x2 at 70%	Deadlift 3x1 at 40-50%
Back Raises	4x6-8	3-4x6-8	3x6-8	
Day 6				
Spoto Press	6x5 at 78%	6x4 at 80%	6x3 at 83%	Compete
Lat Pulldown or DB Rows	6x6-8	5-6x6-8	5x6-8	
DB Lateral Raises	2-3x8-10	2-3x8-10	2-3x8-10	
Tricep Pushdowns	2-3x8-10	2-3x8-10	2-3x8-10	
Day 7				
Off				

Scenario #3

Athlete Assessment:

Sex-Male	Weight-Heavy	Height-Tall
Strength-High	Experience-Intermediate	Age-36 Years Old
Diet-Average	Sleep-Good	Stress-High
Historical Workload-3	Historical Recovery Ability-2	Time to Meet-13 Weeks

Competition Technique/Weaknesses:

Squat	Bench	Deadlift
High Bar	Close Grip	Conventional
Strong Legs/Weak Back	Off the Chest	Midrange

MEVs:

Squat	Bench	Deadlift
6 Sets/Week	7.5 Sets/Week	4 Sets/Week

MRVs:

Squat	Bench	Deadlift
11 Sets/Week	14 Sets/Week	8 Sets/Week

Frequency:

Training Days	Squat	Bench	Deadlift
5 Days/Week	1-2x/Week	2-3x/Week	1-2x/Week

Phasic Structure:

Considering this athlete's Size, Strength, Experience and Age, along with the more narrow gap between his MEV and MRV, an Alternating Periodization would be most appropriate for them. The athlete's high strength levels and large size will allow them to generate higher fatigue levels from each training session so they will need shorter mesocycles. With 13 weeks until competition, this athlete would benefit most from 4 Weeks of Hypertrophy, 5 Weeks of Strength and 4 Weeks of Peaking. That will break down as one 3+1 Hypertrophy Block, one 4+1 Strength Block and a 3+1 Peaking Block.

Hypertrophy Phase

Day 1	Week 1	Week 2	Week 3	Deload
High Bar Squat	Up to 8rm at 8RPE, Drop 8-12%x3x8	4x8 at 90% of 8rm from Week 1	Up to 8rm at 10RPE, Drop 8-12%x4x8	3x8 at 75% of 8rm from Week 3
Good Mornings	3x8 at 6RPE	3-4x8 at 7RPE	4x8 at 8RPE	3x8 at 6RPE
Back Raises	2-3x10-12	3x10-12	3-4x10-12	3x10
Day 2				
Widegrip Bench	Up to 8rm at 8RPE, Drop 8-12%x3x8	5x8 at 90% of 8rm from Week 1	Up to 8rm at 10RPE, Drop 8-12%x5x8	4x8 at 75% of 8rm from Week 3
Lat Pulldown or DB Rows	4x10-12	4-5x10-12	5x10-12	4x10
DB Front Raises	2x12-15	2-3x12-15	3x12-15	2x12
DB Skullcrushers	2x12-15	2-3x12-15	3x12-15	2x12
Day 3				
3” Block Pull	3x6 at 7RPE	Up to 6rm at 9RPE, Drop 8-12%x3x6	5x6 at 7-8RPE	2x6 at 75% of 6rm from Week 2
Bentover Rows	2x10 at 6RPE	2-3x10 at 7RPE	3x10 at 8RPE	2x10 at 6RPE
Back Raises	2-3x10-12	3x10-12	3-4x10-12	3x10
DB Flies	2x10-12	2-3x10-12	3x10-12	2x10 at 6RPE
Day 4				
Off				
Day 5				
Front Squat	2x8 at 6RPE	3x8 at 6-7RPE	4x8 at 7RPE	3x8 at 6RPE
RDLs	3x8 at 6RPE	4x8 at 6-7RPE	5x8 at 7RPE	3x8 at 6RPE
Weighted Planks	3x30 sec	3x45 sec	3x60 sec	3x40 sec
Day 6				
DB Bench	2x8-10 at 7RPE	3x8-10 at 7-8RPE	4x8-10 at 8RPE	3x8 at 6RPE
Lat Pulldown or DB Rows	4x10-12	4-5x10-12	5x10-12	4x10
DB Flies	2x12-15	2-3x12-15	3x12-15	2x12
Tricep Pushdowns	2x12-15	2-3x12-15	3x12-15	2x12
Day 7				
Off				

Strength Phase

Day 1	Week 1	Week 2	Week 3	Week 4	Deload
High Bar Squat	5x5 at 75-80%	Up to 5rm at 9RPE, Drop 8-12%x3-4x5	5x3 at 80-85%	Up to 3rm at 10RPE, Drop 8-12%x3x3	4x3 at 70%
Good Mornings	4x6 at 6RPE	4x6 at 6-7RPE	3-4x6 at 7RPE	3x6 at 7-8RPE	3x6 at 6RPE
Back Raises	3x8-10	3x8-10	3x8-10	3x8-10	2x8
Day 2					
Bench Press	6x5 at 75-80%	Up to 5rm at 9RPE, Drop 8-12%x4-5x5	6x3 at 80-85%	Up to 3rm at 10RPE, Drop 8-12%x4x3	5x3 at 70%
Lat Pulldown or DB Rows	5x8-10	5x8-10	5x8-10	5x8-10	4x8
DB Front Raises	2-3x10-12	2-3x10-12	2-3x10-12	2-3x10-12	2x10
DB Skullcrushers	2-3x10-12	2-3x10-12	2-3x10-12	2-3x10-12	2x10
Day 3					
Deadlift	Up to 4rm at 9RPE, Drop 8-12%x2-3x4	4x4 at 75-80%	Up to 3rm at 10RPE, Drop 8-12%x2x3	4x3 at 80-85%	3x3 at 70%
Bentover Rows	4x8 at 6RPE	4x8 at 6-7RPE	3-4x8 at 7RPE	3x8 at 7-8RPE	2x8 at 6RPE
Back Raises	3x8-10	3x8-10	3x8-10	3x8-10	2x8
Day 4					
Off					
Day 5					
High Bar Squat	3x5 at 70-75%	3x5 at 72.5-77.5%	3x4 at 75-80%	3x4 at 77.5-82.5%	2x4 at 70%
Front Squat	2x5 at 6RPE	2x5 at 6-7RPE	2x4 at 7RPE	2x4 at 7-8RPE	2x4 at 6RPE
Pause at the Knees Deadlift	3x5 at 6RPE	3x5 at 6-7RPE	3x4 at 7RPE	3x4 at 7-8RPE	2x4 at 6RPE
Weighted Planks	5x30 seconds	5x30 seconds	5x30 seconds	5x30 seconds	3x30 seconds
Day 6					
Widegrip Bench	5x6 at 75-80%	5x6 at 77.5-82.5%	5x4 at 80-85%	4-5x4 at 82.5-87.5%	4x4 at 70%
DB Bench	2-3x8-10 at 6RPE	2-3x8-10 at 6-7RPE	2x8-10 at 7RPE	2x8-10 at 7-8RPE	2x8 at 6RPE
Lat Pulldown or DB Rows	5x8-10	5x8-10	5x8-10	5x8-10	4x8
DB Flies	2-3x10-12	2-3x10-12	2-3x10-12	2-3x10-12	2x10
Tricep Pushdowns	2-3x10-12	2-3x10-12	2-3x10-12	2-3x10-12	2x10
Day 7					
Off					

Peaking Phase

Day 1	Week 1	Week 2	Week 3	Taper
High Bar Squat	Up to 2rm at 9RPE, Drop 10-15%x2-3x2	4x3 at 80-85%	Up to 1rm at 9RPE, Drop 10-15%x2x1	Squat 4-6x2 at 50-70%
Deadlift	6x1 at 60%	5x1 at 60%	4x1 at 60%	Bench 5-8x2-3 at 55-75%
Bench Press	6x2 at 70%	5x2 at 70%	4x2 at 70%	Deadlift 3-5x1 at 50-70%
Good Mornings	3x5 at 6RPE	3x5 at 6-7RPE	3x5 at 7RPE	
Day 2				
Off				Off
Day 3				
Bench Press	Up to 2rm at 9RPE, Drop 10-15%x3-4x2	6x3 at 80-85%	Up to 1rm at 9RPE, Drop 10-15%x2-3x1-2	Squat 4-6x2 at 50-70%
Slingshot Bench	2x2 at 105% of 2rm-Before Drop Sets	None	2x1 at 105% of 1rm-Before Drop Sets	Bench Up to 80%x1, 55-75%x5-8x2
Lat Pulldown or DB Rows	5x6-8	5x6-8	5x6-8	Deadlift 3-5x1 at 40-60%
DB Flies	2-3x8-10	2-3x8-10	2-3x8-10	
DB Skullcrushers	2-3x8-10	2-3x8-10	2-3x8-10	
Day 4				
Off				Off
Day 5				
Deadlift	4x2 at 80-85%	Up to 1rm at 9RPE, Drop 10-15%x2x1-2	6x1 at 75%	Squat 4-6x1 at 30-50%
Pause at the Knees Deadlift	3x3 at 6RPE	3x3 at 6-7RPE	2-3x3 at 7RPE	Bench 5-8x1 at 40-60%
Squat	6x2 at 65%	5x2 at 65%	4x2 at 65%	Deadlift 3-5x1 at 20-40%
Back Raises	3x6-8	3x6-8	3x6-8	
Day 6				
Widegrip Bench	4x5 at 75-80%	4x4 at 77.5-82.5%	4x3 at 80-85%	Compete
Lat Pulldown or DB Rows	5x6-8	5x6-8	5x6-8	
DB Front Raises	2-3x8-10	2-3x8-10	2-3x8-10	
DB Skullcrushers	2-3x8-10	2-3x8-10	2-3x8-10	
Day 7				
Off				

Scenario #4

Athlete Assessment:

Sex-Female	Weight-Light	Height-Short
Strength-Very High	Experience-Advanced	Age-40 Years Old
Diet-Good	Sleep-Good	Stress-Average
Historical Workload-4	Historical Recovery Ability-4	Time to Meet-10 Weeks

Competition Technique/Weaknesses:

Squat	Bench	Deadlift
Low Bar	Medium Grip	Conventional
Weak Legs/Strong Back	Lockout	Off the Floor

MEVs (Strength):

Squat	Bench	Deadlift
12 Sets/Week	14.5 Sets/Week	11 Sets/Week

MRVs (Strength):

Squat	Bench	Deadlift
15 Sets/Week	17 Sets/Week	13 Sets/Week

Frequency:

Training Days	Squat	Bench	Deadlift
6 Days/Week	3x/Week	4x/Week	2x/Week

Phasic Structure:

Considering this athlete's Size, Strength, Experience and Age, along with the more narrow gap between her MEV and MRV, an Alternating Periodization would be most appropriate for them. The athlete's very high strength levels and advanced experience will cause them to generate relatively high fatigue in a given session, despite her small stature, so shorter-average length mesocycles will be appropriate. With 10 weeks until competition, this athlete will forgo a true Hypertrophy Phase, given their strength and experience, they've likely maximized their bodyweight in their weight class and in this short time period need to focus on Force Production and Technical Prowess. Our case study athlete would benefit most from 6 Weeks of Strength

and 4 Weeks of Peaking. That will break down as one 5+1 Strength Block and a 3+1 Peaking Block.

Day 1	Week 1	Week 2	Week 3	Week 4	Week 5	Deload
High Bar Squat	6x6 at 70-75%	5-6x6 at 72.5-77.5%	5x5 at 80-85%	4-5x5 at 82.5-87.5%	4x4 at 85-90%	3x4 at 70%
BB Skullcrushers	3x8 at 6RPE	3x8 at 6-7RPE	3x8 at 7RPE	3x8 at 7-8RPE	3x8 at 8RPE	2x8 at 6RPE
Walking Lunges	3x30 Steps	2-3x30 Steps	3x20 Steps	2-3x20 Steps	2x20 Steps	2x20 Steps
Day 2						
Bench Press	Up to 5rm at 9RPE, Drop 8-12%x4x5	6x5 at 72.5-77.5%	Up to 4rm at 9.5RPE, Drop 8-12%x3-4x4	5x4 at 80-85%	Up to 3rm at 10RPE, Drop 8-12%x3x3	4x3 at 75% of 3rm from Week 5
Lat Pulldown or DB Rows	5x8-10	5x8-10	5x8-10	5x8-10	5x8-10	4x8
DB Front Raises	3x10-12	3x10-12	3x10-12	3x10-12	3x10-12	2x10
DB Skullcrushers	3x10-12	3x10-12	3x10-12	3x10-12	3x10-12	2x10
Day 3						
Deadlift	6x5 at 70-75%	Up to 5rm at 9RPE, Drop 8-12%x4x5	5x4 at 75-80%	Up to 3rm at 10RPE, Drop 8-12%x3-4x3	5x3 at 80-85%	3x3 at 75% of 3rm from Week 4
Pendlay Rows	4x8 at 6RPE	4x8 at 6-7RPE	3-4x8 at 7RPE	3x8 at 7-8RPE	2-3x8 at 8RPE	2x8 at 6RPE
Belt Squat	2-3x8-10 at 6RPE	2-3x8-10 at 6-7RPE	2-3x8-10 at 7RPE	2-3x8-10 at 7-8RPE	2-3x8-10 at 8RPE	2x8 at 6RPE
Back Raises	3x8-10	3x8-10	3x8-10	3x8-10	3x8-10	2x8
Day 4						
Dips	3x8 at 6RPE	3x8 at 6-7RPE	3x6 at 7RPE	3x6 at 7-8RPE	3x5 at 8RPE	2x5 at 6RPE
Lat Pulldown or DB Rows	4x8-10	4x8-10	4x8-10	4x8-10	4x8-10	3x8
Reverse Flies	2-3x10-12	2-3x10-12	2-3x10-12	2-3x10-12	2-3x10-12	2x10
DB Curls	2-3x10-12	2-3x10-12	2-3x10-12	2-3x10-12	2-3x10-12	2x10
Day 5						
Low Bar Squat	Up to 5rm at 9RPE, Drop 8-12%x3x5	5x5 at 72.5-77.5%	Up to 4rm at 9.5RPE, Drop 8-12%x3x4	4x4 at 80-85%	Up to 3rm at 10RPE, Drop 8-12%x2x3	4x3 at 75% of 3rm from Week 5
Front Squat	2x6 at 6RPE	2x6 at 6-7RPE	2x5 at 7RPE	2x5 at 7-8RPE	2x4 at 8RPE	2x4 at 6RPE
Deficit Deadlift	3x6 at 6RPE	3x6 at 6-7RPE	3x5 at 7RPE	3x5 at 7-8RPE	3x4 at 8RPE	2x4 at 6RPE
Goblet Squat	2x8-10	2x8-10	2x8-10	2x8-10	2x8-10	2x8

Day 6						
Closegrip Bench	5x6 at 70-75%	4-5x6 at 72.5-77.5%	4x5 at 80-85%	3-4x5 at 82.5-87.5%	3x4 at 85-90%	2x4 at 70%
Lat Pulldown or DB Rows	5x8-10	5x8-10	5x8-10	5x8-10	5x8-10	4x8
DB Lateral Raises	3x10-12	3x10-12	3x10-12	3x10-12	3x10-12	2x10
Tricep Pushdowns	3x10-12	3x10-12	3x10-12	3x10-12	3x10-12	2x10
Day 7						
Off						

Peaking Phase

Day 1	Week 1	Week 2	Week 3	Taper
Low Bar Squat	Up to 2rm at 9RPE, Drop 10-15%x3x2	5x3 at 80-85%	Up to 1rm at 9RPE, Drop 10-15%x2-3x1-2	Squat 4-6x2 at 60-80%
Bench Press	Up to 1rm at 8RPE, Drop 10%x3x3	Up to 1rm at 8RPE, Drop 10%x2x3	Up to 1rm at 8RPE, Drop 10%x3	Bench 5-8x2-3 at 65-85%
Deadlift	6x1 at 70%	6x1 at 70%	6x1 at 70%	Deadlift 3-5x1 at 60-80%
Belt Squat	2-3x6-8 at 6RPE	2-3x6-8 at 6RPE	2-3x6-8 at 6RPE	

Day 2				
Off				Off
Day 3				
Bench Press	Up to 1rm at 9RPE, Drop 7%x4x2	Up to 1rm at 9.5RPE, Drop 7%x3x2	Up to 1rm at 10RPE, Drop 7%x2x2	Squat 4-6x2 at 60-80%
Slingshot Bench	2x2 at 105% of 1rm-Before Drop Sets	None	2x1 at 105% of 1rm-Before Drop Sets	Bench Up to 92.5%x1, 60-80%x5-8x2
Squat	5x2 at 75%	5x2 at 75%	5x2 at 75%	Deadlift 3-5x1 at 50-70%
Lat Pulldown or DB Rows	5x6-8	5x6-8	5x6-8	
DB Front Raises	2-3x8-10	2-3x8-10	2-3x8-10	
DB Skullcrushers	2-3x8-10	2-3x8-10	2-3x8-10	
Day 4				
Off				Off
Day 5				
Deadlift	4x2 at 80-85%	Up to 1rm at 9RPE, Drop 10-15%x2x1-2	8x1 at 75%	Squat 4-6x1 at 30-50%
Deficit Deadlift	2x5 at 6RPE	2x4 at 6RPE	2x3 at 6RPE	Bench 5-8x1 at 40-60%
Low Bar Squat	4x3 at 80-85%	4x2 at 82.5-87.5%	4x1 at 85-90%	Deadlift 3-5x1 at 20-40%
Bench Press	3-4x2 at 75%	3-4x2 at 75%	3-4x2 at 75%	
Pendlay Row	3x5 at 6RPE	3x5 at 6RPE	3x5 at 6RPE	

Day 6				
Closegrip Bench	4x5 at 75-80%	4x4 at 77.5-82.5%	4x3 at 80-85%	Compete
Lat Pulldown or DB Rows	5x6-8	5x6-8	5x6-8	
DB Lateral Raises	2-3x8-10	2-3x8-10	2-3x8-10	
Tricep Pushdowns	2-3x8-10	2-3x8-10	2-3x8-10	
Day 7				
Off				

Scenario #5

Athlete Assessment:

Sex-Female	Weight-Super	Height-Tall
Strength-High	Experience-Intermediate	Age-44 Years Old
Diet-Good	Sleep-Good	Stress-Average
Historical Workload-2	Historical Recovery Ability-3	Time to Meet-12 Weeks

Competition Technique/Weaknesses:

Squat	Bench	Deadlift
Low Bar	Medium Grip	Conventional
Strong Legs/Weak Back	Midrange	Lockout

MEVs:

Squat	Bench	Deadlift
6 Sets/Week	7.5 Sets/Week	4 Sets/Week

MRVs:

Squat	Bench	Deadlift
14 Sets/Week	17 Sets/Week	11 Sets/Week

Frequency:

Training Days	Squat	Bench	Deadlift
5 Days/Week	2x/Week	3x/Week	2x/Week

Phasic Structure:

Considering this athlete's Size, Strength, Experience and Age, along with the more narrow gap between her MEV and MRV, an Undulating Periodization would be most appropriate for them. The athlete's high strength levels and size will cause them to generate high fatigue in a given session, so shorter length mesocycles will be appropriate. With 12 weeks until competition, this athlete could forgo a true Hypertrophy Phase, given their strength but would likely benefit from the increased work capacity benefits of higher volume training. Our case study athlete would benefit most from 4 Weeks of Hypertrophy, 4 Weeks of Strength and 4 Weeks of Peaking. That will break down as one 3+1 Hypertrophy Block, one 3+1 Strength Block and a 3+1 Peaking Block.

Day 1	Week 1	Day 1	Week 2
High Bar Squat	3x10 at 55-60%	High Bar Squat	Up to 6rm at 8RPE, Drop 8-12%x2x6
Good Morning	3x10-12 at 6RPE	RDL	3x10-12 at 6RPE
Back Raises	3x10-12	Back Raises	3x10-12
Day 2		**Day 2**	
Incline Bench	3x8 at 60-65%	Seated BB Military Press	3x10 at 6RPE
Lat Pulldown or DB Rows	5x10-12	Lat Pulldown or DB Rows	5x10-12
DB Front Raises	2-3x12-15	DB Flies	2-3x12-15
DB Skullcrushers	2-3x12-15	DB Curls	2-3x12-15
Day 3		**Day 3**	
3” Block Pull	Up to 6rm at 9RPE, Drop 8-12%x2x6	3” Block Pull	3-4x10 at 55-60%
BB Shrugs	3x10-12 at 6RPE	BB Shrugs	3x10-12 at 7RPE
Back Raises	3x10-12	Back Raises	3x10-12
Day 4		**Day 4**	
Off		Off	
Day 5		**Day 5**	
High Bar Squat	3x8 at 60-65%	High Bar Squat	4x10 at 57.5-62.5%
BB Hip Thrusts	2x8 at 6RPE	Good Morning	3x10-12 at 7RPE
DB Military Press	2x8-10 at 6RPE	BB Hip Thrusts	2-3x8 at 7RPE
Back Raises	3x10-12	DB Military Press	2-3x8-10 at 7RPE
Day 6		Back Raises	3x10-12
Spoto Press	Up to 6rm at 8RPE, Drop 8-12%x2x6	**Day 6**	
Lat Pulldown or DB Rows	5x10-12	Incline Bench	4x8 at 62.5-67.5%
DB Lateral Raises	2-3x12-15	Lat Pulldown or DB Rows	5x10-12
Tricep Pushdowns	2-3x12-15	DB Front Raises	2-3x12-15
Day 7		DB Skullcrushers	2-3x12-15
Off		**Day 7**	
		Off	

Day 1	Week 3	Day 1	Deload
High Bar Squat	4x8 at 62.5-67.5%	Low Bar Squat	3x6 at 80% of 6rm from Week 3
Back Raises	3x10-12	BB Hip Thrust	2x10 at 6RPE

Day 2		Back Raises	2x10
Spoto Press	Up to 6rm at 10RPE, Drop 8-12%x3x6	**Day 2**	
Lat Pulldown or DB Rows	5x10-12	Bench Press	4x6 at 80% of 6rm from Week 3
DB Lateral Raises	2-3x12-15	Lat Pulldown or DB Rows	4x10
Tricep Pushdowns	2-3x12-15	DB Front Raises	2x12
Day 3		DB Skullcrushers	2x12
3” Block Pull	4x8 at 60-65%	**Day 3**	
BB Shrugs	3x10-12 at 8RPE	Deadlift	2x6 at 80% of 6rm from Week 1
Back Raises	3x10-12	BB Shrugs	2x10 at 6RPE
Day 4		Back Raises	2x10
Off		**Day 4**	
Day 5		Off	
High Bar Squat	Up to 6rm at 10RPE, Drop 8-12%x2x6	**Day 5**	
RDL	3x10-12 at 7RPE	High Bar Squat	3x6 at 80% of 6rm from Week 3
BB Hip Thrusts	3x8 at 8RPE	BB Hip Thrusts	2x8 at 6RPE
DB Military Press	3x8-10 at 8RPE	DB Military Press	2x8 at 6RPE

Back Raises	3x10-12	Back Raises	2x10
Day 6		**Day 6**	
Seated BB Military Press	4x10 at 7RPE	Spoto Press	4x6 at 80% of 6rm from Week 3
Lat Pulldown or DB Rows	5x10-12	Lat Pulldown or DB Rows	4x10
DB Flies	2-3x12-15	DB Lateral Raises	2x12
DB Curls	2-3x12-15	Tricep Pushdowns	2x12
Day 7		**Day 7**	
Off		Off	

Strength Phase

Day 1	Week 1	Day 1	Week 2
Low Bar Squat	4x6 at 70-75%	Low Bar Squat	Up to 4rm at 9RPE, Drop 8-12%x2-3x4
Good Mornings	3x8 at 7RPE	RDL	3x8 at 7RPE
Back Raises	3x8-10	Back Raises	3x8-10
Day 2		**Day 2**	
Bench Press	5x5 at 75-80%	Spoto Press	5x6 at 70-75%
Lat Pulldown or DB Rows	5x8-10	Lat Pulldown or DB Rows	5x8-10
DB Front Raises	2-3x10-12	DB Flies	2-3x10-12
DB Skullcrushers	2-3x10-12	DB Curls	2-3x10-12
Day 3		**Day 3**	
Deadlift	Up to 3rm at 10RPE, Drop 8-12%x2-3x3	3” Block Pull	4x5 at 70-80%
BB Hip Thrust	3x8-10 at 7RPE	BB Hip Thrust	3x8-10 at 7-8RPE
Back Raises	3x8-10	Back Raises	3x8-10
Day 4		**Day 4**	
Off		Off	
Day 5		**Day 5**	
High Bar Squat	3-4x5 at 75-80%	Low Bar Squat	4x5 at 72.5-77.5%
3” Block Pull	3x5 at 7RPE	3” Block Pull	3x5 at 7-8RPE
Back Raises	3x8-10	Good Mornings	3x8 at 7-8RPE
Day 6		Back Raises	3x8-10
Bench Press	Up to 4rm at 9RPE, Drop 8-12%x3x4	**Day 6**	
Lat Pulldown or DB Rows	5x8-10	Bench Press	5x4 at 77.5-82.5%
DB Lateral Raises	2-3x10-12	Lat Pulldown or DB Rows	5x8-10
Tricep Pushdowns	2-3x10-12	DB Front Raises	2-3x10-12
Day 7		DB Skullcrushers	2-3x10-12
Off		**Day 7**	
		Off	

Day 1	Week 3	Day 1	Deload
High Bar Squat	3-4x4 at 77.5-82.5%	Low Bar Squat	3x3 at 80% of 3rm from Week 3
Back Raises	3x8-10	RDL	2x8 at 6RPE

Day 2		Back Raises	2x8
Bench Press	Up to 3rm at 10RPE, Drop 8-12%x3x3	**Day 2**	
Lat Pulldown or DB Rows	5x8-10	Bench Press	4x3 at 80% of 3rm from Week 3
DB Lateral Raises	2-3x10-12	Lat Pulldown or DB Rows	4x8
Tricep Pushdowns	2-3x10-12	DB Front Raises	2x10
Day 3		DB Skullcrushers	2x10
Deadlift	4x4 at 75-85%	**Day 3**	
BB Hip Thrust	3x8-10 at 8RPE	Deadlift	2x3 at 80% of 6rm from Week 1
Back Raises	3x8-10	BB Hip Thrust	2x8 at 6RPE
Day 4		Back Raises	2x8
Off		**Day 4**	
Day 5		Off	
Low Bar Squat	Up to 3rm at 10RPE, Drop 8-12%x2x3	**Day 5**	
3" Block Pull	3x5 at 8RPE	High Bar Squat	3x5 at 65%
RDL	3x8 at 8RPE	3" Block Pull	2x5 at 6RPE
Back Raises	3x8-10	Back Raises	2x8
Day 6		**Day 6**	
Spoto Press	5x5 at 72.5-77.5%	Spoto Press	4x3 at 70%
Lat Pulldown or DB Rows	5x8-10	Lat Pulldown or DB Rows	4x8
DB Flies	2-3x10-12	DB Lateral Raises	2x10
DB Curls	2-3x10-12	Tricep Pushdowns	2x10
Day 7		**Day 7**	
Off		Off	

Peaking Phase

Day 1	Week 1	Week 2	Week 3	Taper
Deadlift	Up to 1rm at 9RPE, Drop 10-15%x2x1-2	4x3 at 80-85% from 3" Blocks	4x1-2 at 85-90%	Squat 4-6x1 at 70-80%
BB Hip Thrust	3x5 at 6RPE	3x5 at 6RPE	3x5 at 6RPE	Bench Up to 90%x1, 3-5x3 at 75%
Squat	5x2 at 70%	4x2 at 70%	3x2 at 70%	Deadlift 5x1 at this 60-70%
Bench Press	6x2 at 75%	5x2 at 75%	4x2 at 75%	
Day 2				
Off				Off
Day 3				
Bench Press	Up to 1rm at 8RPE, 4x3 at 80-85%	Up to 1rm at 9RPE, 3x2 at 85-90%	Up to 1rm at 10RPE, 2x1 at 90-95%	Squat 4-6x1 at 60-70%

Slingshot Bench	2x1 at 105% of 1rm-Before Drop Sets	None	2x1 at 105% of 1rm-Before Drop Sets	Bench Up to 85%x1, 3-5x2 at 70%
Lat Pulldown or DB Rows	5x6-8	5x6-8	5x6-8	Deadlift 5x1 at 50-60%
DB Front Raises	2-3x8-10	2-3x8-10	2-3x8-10	
DB Skullcrushers	2-3x8-10	2-3x8-10	2-3x8-10	
Day 4				
Off				Off
Day 5				
Low Bar Squat	4x2 at 85-90%	Up to 1rm at 9.5RPE, Drop 10-15%x2x2	4x3 at 80-85%	Squat 4-6x1-2 at 30-50%
Deadlift	5x1 at 70%	4x1 at 70%	3x1 at 70%	Bench 4-6x2-3 at 40-60%
Good Mornings	3x5 at 6RPE	3x5 at 6RPE	3x5 at 6RPE	Deadlift 5x1 at 30-50%
Back Raises	3x6-8	3x6-8	3x6-8	
Day 6				
Bench Press	2x5 at 80-85%	2x4 at 82.5-87.5%	2x3 at 85-90%	COMPETE
Spoto Press	2x5 at 80-85%	2x4 at 82.5-87.5%	2x3 at 85-90%	
Lat Pulldown or DB Rows	5x6-8	5x6-8	5x6-8	
DB Flies	2-3x8-10	2-3x8-10	2-3x8-10	
Tricep Pushdowns	2-3x8-10	2-3x8-10	2-3x8-10	
Day 7				
Off				

Scenario #6

Athlete Assessment:

Sex-Male	Weight-Heavy	Height-Tall
Strength-Very High	Experience-Advanced	Age-34 Years Old
Diet-Good	Sleep-Average	Stress-Low
Historical Workload-2	Historical Recovery Ability-4	Time to Meet-13 Weeks

Competition Technique/Weaknesses:

Squat	Bench	Deadlift
Low Bar	Medium Grip	Sumo
Out of the Hole/Balanced	Lockout	Off the Floor

MEVs:

Squat	Bench	Deadlift
7 Sets/Week	8.5 Sets/Week	5 Sets/Week

MRVs:

Squat	Bench	Deadlift
9 Sets/Week	12 Sets/Week	6 Sets/Week

Frequency:

Training Days	Squat	Bench	Deadlift
4 Days/Week	1-2x/Week	2x/Week	1-2x/Week

Phasic Structure:

An athlete of this type, very muscular, strong and advanced male lifter, with a very small gap between their MEV and MRV presents a unique and exciting challenge to train. This is the type of challenge I faced in my own training, as well as working with many athletes of this type like Brandon Allen, Andy Huang, Steve Gentili and others and to solve it, developed the Undulating Periodization strategy which I employ. The athlete's high strength levels and size will cause them to generate extremely high fatigue in a given session, so shorter length mesocycles will be appropriate and the undulations are necessary to help manage fatigue. The stronger an athlete becomes the bigger the variance between High and Low sessions will need to be.

With 13 weeks until competition, this athlete could forgo a true Hypertrophy Phase, given their strength but could likely benefit from the increased work capacity benefits of higher volume training, this type of training can also help avoid Adaptive Resistance and mitigate stress on the joints from extremely heavy training. Our case study athlete would benefit most from 4 Weeks of Hypertrophy, 4 Weeks of Strength and 4 Weeks of Peaking. That will break down as one 3+1 Hypertrophy Block, one 3+1 Strength Block and a 3+2 Peaking Block, because of this athlete's qualification, they will use a 2 week taper, hence the 3+2.

Day 1	**Week 1**	**Day 1**	**Week 2**
High Bar Squat	4x6 at 65-70%	High Bar Squat	4-5x6 at 70-75%
Conventional 1" Deficit Dead	3x6 at 6RPE	Conventional 1" Deficit Dead	3x6 at 6-7RPE
BB Hip Thrusts	3x8-10 at 6RPE	BB Hip Thrusts	3x8-10 at 6-7RPE
Day 2		**Day 2**	
Off		Off	
Day 3		**Day 3**	
Closegrip Bench	5x6 at 70-75%	Closegrip Bench	Up to 6rm at 9RPE, Drop 8-12%x3-4x6
Lat Pulldowns or DB Rows	5x10-12	Lat Pulldowns or DB Rows	5x10-12
DB Flies	2-3x12-15	DB Flies	2-3x12-15
DB Front Raises	2-3x12-15	DB Front Raises	2-3x12-15
DB Skullcrushers	2-3x12-15	DB Skullcrushers	2-3x12-15
Day 4		**Day 4**	
Off		Off	
Day 5		**Day 5**	
Halting Sumo Deadlifts	4x5-Building from 6 to 9RPE	Halting Sumo Deadlifts	5x5 at 80% of 9RPE from Week 1
Dead Squats	60%x6x1-30 sec rest	Dead Squats	60%x8x1-30 sec rest
Back Raises	3x8-10	Back Raises	3x8-10
Day 6		**Day 6**	
2 Board Press	3x6 at 6RPE	2 Board Press	3x6 at 6-7RPE
BB Skullcrushers	2x8 at 6RPE	BB Skullcrushers	3x8 at 6-7RPE
Lat Pulldowns or DB Rows	5x10-12	Lat Pulldowns or DB Rows	5x10-12

DB Lateral Raises	2-3x12-15	DB Lateral Raises	2-3x12-15
Tricep Pushdowns	2-3x12-15	Tricep Pushdowns	2-3x12-15
Day 7		**Day 7**	
Off		Off	

Day 1	**Week 3**	**Day 1**	**Deload**
High Bar Squat	Up to 6rm at 9RPE, Drop 8-12%x3-4x6	High Bar Squat	3x6 at 70% of 6rm from Week 3
Conventional 1" Deficit Dead	3-4x6 at 7RPE	Conventional 1" Deficit Dead	2-3x6 at 6RPE
BB Hip Thrusts	3x8-10 at 7RPE	BB Hip Thrusts	2x8-10 at 6RPE
Day 2		**Day 2**	
Off		Off	
Day 3		**Day 3**	
Closegrip Bench	5x6 at 72.5-77.5%, Drop 10%xAMRAP	Closegrip Bench	4x6 at 75% of 6rm from Week 2
Lat Pulldowns or DB Rows	5x10-12	Lat Pulldowns or DB Rows	4x10
DB Flies	2-3x12-15	DB Flies	2x12
DB Front Raises	2-3x12-15	DB Front Raises	2x12
DB Skullcrushers	2-3x12-15	DB Skullcrushers	2x12
Day 4		**Day 4**	
Off		Off	
Day 5		**Day 5**	
Halting Sumo Deadlifts	5x5 at 90% of 9RPE from Week 1	Halting Sumo Deadlifts	3x5 at 65% of 9RPE from Week 1
Dead Squats	60%x10x1-30 sec rest	Dead Squats	60%x5x1-30 sec rest
Back Raises	3x8-10	Back Raises	2x8
Day 6		**Day 6**	
2 Board Press	4x6 at 7RPE	2 Board Press	3x6 at 6RPE
BB Skullcrushers	3x8 at 7RPE	BB Skullcrushers	2x6 at 6RPE

Lat Pulldowns or DB Rows	5x10-12	Lat Pulldowns or DB Rows	4x10
DB Lateral Raises	2-3x12-15	DB Lateral Raises	2x12
Tricep Pushdowns	2-3x12-15	Tricep Pushdowns	2x12
Day 7		**Day 7**	
Off		Off	

Strength Phase

Day 1	Week 1	Day 1	Week 2
Low Bar Squat	4-5x5 at 70-75%	Low Bar Squat	4-5x4 at 75-80%
Conventional 1" Deficit Dead	3x4 at 7RPE	Conventional 1" Deficit Dead	3x4 at 7-8RPE
BB Hip Thrusts	3x6-8 at 7RPE	BB Hip Thrusts	3x6-8 at 7-8RPE
Day 2		**Day 2**	
Off		Off	
Day 3		**Day 3**	
Bench Press	5x4 at 75-80%	Bench Press	Up to 3rm at 9.5RPE. Drop 8-12%x3-4x3
Lat Pulldowns or DB Rows	5x8-10	Lat Pulldowns or DB Rows	5x8-10
DB Flies	2-3x10-12	DB Flies	2-3x10-12
DB Front Raises	2-3x10-12	DB Front Raises	2-3x10-12
DB Skullcrushers	2-3x10-12	DB Skullcrushers	2-3x10-12
Day 4		**Day 4**	
Off		Off	
Day 5		**Day 5**	
Sumo Deadlift	Up to 3rm at 9.5RPE. Drop 8-12%x2-3x3	Halting Sumo Deadlift	5x5 at 70% of 3rm from Week 1
Pause LB Squat	3x5 at 6-7RPE	Pause LB Squat	3x5 at 7-8RPE
Back Raises	3x6-8	Back Raises	3x6-8
Day 6		**Day 6**	
Closegrip 2 Board Press	4x5 at 7RPE	Closegrip 2 Board Press	4x5 at 7-8RPE
Lat Pulldowns or DB Rows	5x8-10	Lat Pulldowns or DB Rows	5x8-10
DB Lateral Raises	2-3x10-12	DB Lateral Raises	2-3x10-12
Tricep Pushdowns	2-3x10-12	Tricep Pushdowns	2-3x10-12
Day 7		**Day 7**	
Off		Off	

Day 1	Week 3	Day 1	Deload
Low Bar Squat	Up to 3rm at 9.5RPE. Drop 8-12%x3x3	Low Bar Squat	3-4x3 at 70% of 3rm from Week 3
Conventional 1" Deficit Dead	3x4 at 8-9RPE	Conventional 1" Deficit Dead	2x4 at 6RPE
BB Hip Thrusts	3x6-8 at 8RPE	BB Hip Thrusts	2x6 at 6RPE
Day 2		**Day 2**	
Off		Off	

Day 3		**Day 3**	
Bench Press	5x5 at 70-75%	Bench Press	4x3 at 75% of 3rm from Week 2
Lat Pulldowns or DB Rows	5x8-10	Lat Pulldowns or DB Rows	4x8
DB Flies	2-3x10-12	DB Flies	2x10
DB Front Raises	2-3x10-12	DB Front Raises	2x10
DB Skullcrushers	2-3x10-12	DB Skullcrushers	2x10
Day 4		**Day 4**	
Off		Off	
Day 5		**Day 5**	
Sumo Deadlift	4x4 at 85% of 3rm from Week 1	Sumo Deadlift	3x3 at 65% of 3rm from Week 3
Pause LB Squat	3x5 at 8-9RPE	Pause LB Squat	2x5 at 6RPE
Back Raises	3x6-8	Back Raises	2x6
Day 6		**Day 6**	
Closegrip 2 Board Press	4x5 at 8RPE	Closegrip 2 Board Press	3x5 at 6RPE
Lat Pulldowns or DB Rows	5x8-10	Lat Pulldowns or DB Rows	4x10
DB Lateral Raises	2-3x10-12	DB Lateral Raises	2x10
Tricep Pushdowns	2-3x10-12	Tricep Pushdowns	2x10
Day 7		**Day 7**	
Off		Off	

Peaking Phase

Day 1	Week 1	Day 1	Week 2
Sumo Deadlift	Up to 1rm at 9.5RPE, Drop 10-20%x2x2	Halting Deadlift	4x3 at 6RPE
Halting Deadlift	2x3 at 6RPE	Low Bar Squat	4x2 at 60%
Low Bar Squat	5x2 at 60%	Bench Press	5x2 at 70%
Bench Press	6x2 at 70%	**Day 2**	
Day 2		Off	
Off		**Day 3**	
Day 3		Bench Press	5x2 from 6 to 9RPE
Bench Press	5x3 from 6 to 9RPE	Lat Pulldowns or DB Rows	5x6-8
Lat Pulldowns or DB Rows	5x6-8	DB Flies	2-3x8-10
DB Flies	2-3x8-10	DB Front Raises	2-3x8-10
DB Front Raises	2-3x8-10	DB Skullcrushers	2-3x8-10
DB Skullcrushers	2-3x8-10	**Day 4**	
Day 4		Off	
Off		**Day 5**	
Day 5		Low Bar Squat	Up to 1rm at 9.5RPE, Drop 10-20%x2-3x2
Low Bar Squat	4x2 from 6 to 8RPE	Pause LB Squat	2x3 at 6RPE
Pause LB Squat	2x3 at 6RPE	Sumo Deadlift	4x1 at 50%
Sumo Deadlift	5x1 at 50%	Back Raises	2x6-8
Back Raises	2x6-8	**Day 6**	
Day 6		Closegrip 2 Board Press	3x4 at 6-7RPE
Closegrip 2 Board Press	3x5 at 6-7RPE	Lat Pulldowns or DB Rows	5x6-8
Lat Pulldowns or DB Rows	5x6-8	DB Lateral Raises	2-3x8-10
DB Lateral Raises	2-3x8-10	Tricep Pushdowns	2-3x8-10
Tricep Pushdowns	2-3x8-10	**Day 7**	
Day 7		Off	
Off			

Day 1	Week 3	Taper 1	Taper 2
Sumo Deadlift	4x1-2 at 80-85%	Squat 5x2-3 at 50%	Off
Low Bar Squat	3x2 at 60%	Bench 6x2-3 at 60%	**Day 2**
Bench Press	4x2 at 70%	Deadlift 4x1-2 at 40%	Squat 5x1-2 at 60%
Day 2		**Day 2**	Bench 6x1-2 at 70%
Off		Off	Deadlift 4x1 at 50%

Day 3		**Day 3**	**Day 3**
Bench Press	Up to 1rm at 9.5RPE, Drop 10-20%x3-4x2	Squat 5x2-3 at 60%	Off
Lat Pulldowns or DB Rows	5x6-8	Bench 6x2-3 at 70%	**Day 4**
DB Flies	2-3x8-10	Deadlift 4x1-2 at 50%	Squat 5x1 at 40-50%
DB Front Raises	2-3x8-10	**Day 4**	Bench 6x1 at 50-60%
DB Skullcrushers	2-3x8-10	Off	Deadlift 4x1 at 30-40%
Day 4		**Day 5**	**Day 5**
Off		Squat 5x2-3 at 50%	Off
Day 5		Bench 6x2-3 at 60%	**Day 6**
Low Bar Squat	4x3 at 6-8RPE	Deadlift 4x1-2 at 40%	COMPETE

Pause LB Squat	2x3 at 6RPE	**Day 6**	
Sumo Deadlift	3x1 at 50%	Off	
Back Raises	2x6-8	**Day 7**	
Day 6		Squat Up to 1 at 80%, 50-60%x2x3	
Closegrip 2 Board Press	3x3 at 6-7RPE	Bench Up to 1 at 90%, 60-70x2-3x3-5	
Lat Pulldowns or DB Rows	5x6-8	Deadlift 4x1-2 at 50%	
DB Lateral Raises	2-3x8-10		
Tricep Pushdowns	2-3x8-10		
Day 7			
Off			